

LA-UR-26-24464

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Title: MultiPEM Toolbox: User Manual

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Intended for: Report

Issued: 2026-05-28



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MultiPEM Toolbox: User Manual

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May 16, 2026

1 Introduction

This document explains use of the **Multi-Phenomenology Explosion Monitoring** (Multi-PEM) Toolbox, a collection of R scripts for estimating the unknown device parameters of a new event with uncertainty quantification. The methodology and application used for illustration in this user manual are published in *Geophysical Journal International*¹ and fully documented in a Los Alamos National Laboratory technical report² hereafter designated “WPA” for reference. Additional details on the application are found in a recent journal article³. Two assessment types are available: *rapid* and *complete*.

Rapid assessments are conducted in two stages, as described in Section 2. In the first stage, benchmark data are used to estimate forward and error model parameters (WPA, §4.1) and (if relevant) errors-in-variables yield values for benchmark sources (WPA, §2, Equation (3)). Forward model parameters can be of two types: specific to signature within phenomenology and even emplacement condition, or global across more than one signature within phenomenology or across phenomenologies. In the second stage, new event data are used to estimate the unknown new event device parameters (WPA, §4.2) with uncertainty quantification.

Two options for treating the inferred first stage parameters in second stage Bayesian analysis are available: fixing them at their maximum likelihood estimate (default), or multiple imputation⁴. Multiple imputation involves utilizing several posterior samples (imputations) of the first stage parameters as fixed values in the second stage posterior sampling of the new event device parameters. Second stage sampling is conducted across imputations in parallel to improve computational efficiency. This method produces improved uncertainty quantification of the new event device parameters compared with the default treatment of the first stage parameters, at the expense of additional computation.

Complete assessments are conducted in a single stage, as described in Section 3. Benchmark

¹Williams, B.J., Picard, R., & Anderson, D.N., Uncertainty estimation for multiphenomenology explosion monitoring, *Geophys. J. Int.* **245**(3), June 2026, ggag125, <https://doi.org/10.1093/gji/ggag125>.

²Williams, B.J., Picard, R.R., & Anderson, D.N. (2025). Multi-phenomenology Yield Characterization. Los Alamos National Laboratory Technical Report LA-UR-23-21950 (rev.4).

³Ford, S.R., Bulaevskaya, V., Ramirez, A., Johannesson, G., & Rodgers, A.J. (2021). Joint Bayesian inference for near-surface explosion yield and height-of-burst, *J Geophys Res Solid Earth* 126:e2020JB020968.

⁴Plummer, M. (2015). Cuts in Bayesian graphical models, *Stat Comput* 25:37-43.

and (if relevant) new event data are used simultaneously to estimate all forward model, error model, and (if relevant) errors-in-variables yields and new event device parameters with uncertainty quantification on the latter.

As the name suggests, rapid assessments generally run substantially faster than complete assessments (even with multiple imputation), because the results of first stage analysis can be stored and incorporated into estimating a relatively low-dimensional space of new event device parameters whenever relevant new event data becomes available. On the other hand, complete assessments must be run on the full set of model and device parameters with benchmark and new event data every time the latter becomes available.

1.1 Running An Application in MultiPEM Toolbox

MultiPEM Toolbox applications can be run directly in R, or through Docker. The latter is useful if a common run environment is desired for multiple users. Details are provided here for the former, followed by brief remarks on the latter in the next subsection.

Begin with the following initial steps:

- Install the latest version of R
- Install the following auxillary R packages
 - `Matrix`
 - `numDeriv`
 - `doFuture`
 - `adaptMCMC`
 - `FME`
 - `abind` (optional for Sequential Monte Carlo (SMC) sampling)
 - `ramcmc` (optional for SMC sampling)

For example, to install the `Matrix` package, run the command

```
> install.packages("Matrix")
```

inside an R session. The `FME` package requires R version 4.0 or higher (through its dependency on the `MASS` package) and its function `modMCMC` is used as the default posterior sampler in the run files associated with the second stage of the rapid (`runMPEM_0.r`) assessments. The run files associated with the first stage of the rapid (`runMPEM.r`) and complete (`runMPEM.r`) assessments use the `MCMC` function of the `adaptMCMC` package for posterior sampling, due to the extra stability provided for higher-dimensional parameter spaces. If an older version of R is used, the `iMCMC` command in these run files should be changed from `"FME"` to `"RAM"`.

- The `future` package places a limit of 500 MiB on the size of global variables that can be exported to parallel processes. This can be overridden by placing the following

command

```
options(future.globals.maxSize = Inf)
```

in the `.Rprofile` file located in the user's home directory, or by adding it to the preprocessing component of the `runMPEM.r` and `runMPEM_0.r` files.

- Clone the open source MultiPEM Toolbox repository from GitHub:

```
% git clone https://github.com/lanl/MultiPEM_Toolbox.git
```

There are three MultiPEM analyses contained in the illustrative application (WPA, §5), named `2-Phen-oc`, `2-Phen-sa`, and `4-Phen`. The first estimates new event device parameters based on fusion of data from the *optical* and *surface effects* phenomenologies; the second fuses data from the *seismic* and *acoustic* phenomenologies; and finally the last performs this same task but with data fusion across all four phenomenologies.

1.1.1 Rapid Assessments

The following steps may be invoked in sequential order to run the MultiPEM Toolbox on the illustrative application for rapid assessments. A similar workflow will pertain to any application. This application uses multiple cores for likelihood/posterior maximization and posterior sampling.

1. Create a symbolic link to the global code directory

```
% cd ./MultiPEM_Toolbox/Runfiles
% # link to global code (used by all applications)
% ln -s ../Code/ Code
```

2. Create symbolic links to the application (“IYDT-gsrp”) data and code directories

```
% cd IYDT-gsrp
% # link to application (IYDT-gsrp) data files
% ln -s ../../Applications/Data/IYDT-gsrp/ Data
% # link to application code (used by IYDT-gsrp application)
% ln -s ../../Applications/Code/IYDT-gsrp/ Code
```

3. Run the first stage analysis for each single phenomenology

- Seismic (phenomenology 1)

```
% cd Seismic
% # link to phenomenology code (used by seismic
% # phenomenology to specify prior distribution
% # of the forward model coefficients (and its gradient))
% ln -s ../../Applications/Code/IYDT-gsrp/Phenomenology/ Code
% cd I-SUGAR-hob-0
% R CMD BATCH runMPEM.r runMPEM.out &
% # check status
% tail runMPEM.out
```

```

% # upon completion of run (~2.1 hours), copy
% # .RData file for use in all relevant future
% # seismic second stage rapid assessments
% # a completed run will show the results of proc.time()
% # at the end of the runMPEM.out file
% cp .RData .RData-s

```

- Acoustic (2)

```

% cd ../../Acoustic/I-SUGAR-hob-0
% R CMD BATCH runMPEM.r runMPEM.out &
% # check status
% tail runMPEM.out
% # upon completion of run (~1.4 hours), copy
% # .RData file for use in all relevant future
% # acoustic second stage rapid assessments
% # a completed run will show the results of proc.time()
% # at the end of the runMPEM.out file
% cp .RData .RData-a

```
- Optical (3)

```

% cd ../../Optical
% # link to phenomenology code (used by optical
% # phenomenology to specify prior distribution
% # of the forward model coefficients (and its gradient))
% ln -s ../../Applications/Code/IYDT-gsrp/Phenomenology/ Code
% cd I-EIV-SUGAR-hob-0
% R CMD BATCH runMPEM.r runMPEM.out &
% # check status
% tail runMPEM.out
% # upon completion of run (~46 minutes), copy
% # .RData file for use in all relevant future
% # optical second stage rapid assessments
% # a completed run will show the results of proc.time()
% # at the end of the runMPEM.out file
% cp .RData .RData-o

```
- Surface Effects/Crater (4)

```

% cd ../../Crater/I-EIV-SUGAR-0
% R CMD BATCH runMPEM.r runMPEM.out &
% # check status
% tail runMPEM.out
% # upon completion of run (~10 minutes), copy
% # .RData file for use in all relevant future
% # crater second stage rapid assessments
% # a completed run will show the results of proc.time()

```

```
% # at the end of the runMPEM.out file
% cp .RData .RData-c
```

The `runMPEM.r` and `runMPEM.out` files are described in Section 2.1. The status of running code is checked by issuing the following command in the run directory,

```
% tail runMPEM.out
```

Successfully completed runs show the results of the `proc.time()` command at the end of the `runMPEM.out` file. Symbolic links are created to a phenomenology-specific code directory for **Seismic** and **Optical** prior to conducting the runs (see comments in above code). The `.RData` files resulting from each completed analysis should be copied and stored for use in relevant future second stage analyses (see comments in above code). Upon completion of the maximum likelihood estimation component of these runs (typically much earlier than the entire run), copy each resulting `opt.RData` file to the `Opt` directory in each MultiPEM analysis, to be used as starting values for MultiPEM log-likelihood maximization.

```
% cd ../../2-Phen-oc/Opt
% cp ../../Optical/I-EIV-SUGAR-hob-0/opt.RData opt_1_eiv_0.RData
% cp ../../Crater/I-EIV-SUGAR-0/opt.RData opt_2_eiv_0.RData
% cd ../../2-Phen-sa/Opt
% cp ../../Seismic/I-SUGAR-hob-0/opt.RData opt_1_0.RData
% cp ../../Acoustic/I-SUGAR-hob-0/opt.RData opt_2_0.RData
% cd ../../4-Phen/Opt
% cp ../../Seismic/I-SUGAR-hob-0/opt.RData opt_1_0.RData
% cp ../../Acoustic/I-SUGAR-hob-0/opt.RData opt_2_0.RData
% cp ../../Optical/I-EIV-SUGAR-hob-0/opt.RData opt_3_eiv_0.RData
% cp ../../Crater/I-EIV-SUGAR-0/opt.RData opt_4_eiv_0.RData
```

4. Run the second stage analysis for each single phenomenology, copying the `.RData` file from the first stage analysis into the run directories if necessary. For example, the directory

```
./MultiPEM_Toolbox/Runfiles/IYDT-gsrp/Seismic/I-SUGAR-hob-0
```

contains a MultiPEM analysis for the new event device parameters, assuming a “flat” prior on these parameters for the Bayesian analysis (WPA, §5.6, p. 21). Alternatively, the user could perform an analysis that assumes an “informative” prior on these parameters. Both analyses would use the same first stage results. To perform the second stage run for an informative prior, the first stage `.RData` file would be copied into a new run directory created for this analysis (say, `I-SUGAR-hob-0-pi`),

```
% cd ./MultiPEM_Toolbox/Runfiles/IYDT-gsrp/Seismic/I-SUGAR-hob-0-pi
% cp ../I-SUGAR-hob-0/.RData-s .RData
```

and the analysis would be run in the `I-SUGAR-hob-0-pi` directory analogous to the run shown below for the “flat” prior.

- Seismic (phenomenology 1)

```
% # noninformative prior distribution on new event
% # device parameters
% cd ../../Seismic/I-SUGAR-hob-0
% # if necessary, change iMCMC to "RAM" in runMPEM_0.r
% R CMD BATCH runMPEM_0.r runMPEM_0.out &
% # a completed run will show the results of proc.time()
% # at the end of the runMPEM_0.out file
```

- Acoustic (2)

```
% # noninformative prior distribution on new event
% # device parameters
% cd ../../Acoustic/I-SUGAR-hob-0
% # if necessary, change iMCMC to "RAM" in runMPEM_0.r
% R CMD BATCH runMPEM_0.r runMPEM_0.out &
% # a completed run will show the results of proc.time()
% # at the end of the runMPEM_0.out file
```

- Optical (3)

```
% # noninformative prior distribution on new event
% # device parameters
% cd ../../Optical/I-EIV-SUGAR-hob-0
% # if necessary, change iMCMC to "RAM" in runMPEM_0.r
% R CMD BATCH runMPEM_0.r runMPEM_0.out &
% # a completed run will show the results of proc.time()
% # at the end of the runMPEM_0.out file
```

- Surface Effects/Crater (4)

```
% # noninformative prior distribution on new event
% # device parameters
% cd ../../Crater/I-EIV-SUGAR-0
% # if necessary, change iMCMC to "RAM" in runMPEM_0.r
% R CMD BATCH runMPEM_0.r runMPEM_0.out &
% # a completed run will show the results of proc.time()
% # at the end of the runMPEM_0.out file
```

The runMPEM_0.r and runMPEM_0.out files are described in Section 2.2.

5. Run the first stage MultiPEM analysis (illustrated here for 4-Phen)

```
% cd ../../4-Phen
% # link to phenomenology code (used by seismic and optical
% # phenomenologies to specify prior distributions of their
% # respective forward model coefficients (and their gradients))
% ln -s ../../Applications/Code/IYDT-gsrp/Phenomenology/ Code
% cd I-EIV-SUGAR-hob-0
% R CMD BATCH runMPEM.r runMPEM.out &
```



```
% # upon completion of run (~7.4 hours), copy
% # .RData file for use in all relevant future
% # multiPEM second stage rapid assessments
% # a completed run will show the results of proc.time()
% # at the end of the runMPEM.out file
% cp .RData .RData-4
```

The `runMPEM.r` and `runMPEM.out` files are described in Section 2.1. A symbolic link is created to a phenomenology-specific code directory prior to conducting the run (see comments in above code). The `.RData` file resulting from the completed analysis should be copied and stored for use in future second stage analyses (see comments in above code).

6. Run the second stage MultiPEM analyses, copying the `.RData` file from the first stage analysis into the run directories if necessary.

```
% # noninformative prior distribution on new event
% # device parameters
% # if necessary, change iMCMC to "RAM" in runMPEM_0.r
% R CMD BATCH runMPEM_0.r runMPEM_0.out &
% # a completed run will show the results of proc.time()
% # at the end of the runMPEM_0.out file
```

The `runMPEM_0.r` and `runMPEM_0.out` files are described in Section 2.2.

1.1.2 Complete Assessments

The following steps may be invoked in sequential order to run the MultiPEM Toolbox on the illustrative application for complete assessments. A similar workflow will pertain to any application. This application uses multiple cores for likelihood/posterior maximization and posterior sampling.

1. Create a symbolic link to the global code directory

```
% cd ./MultiPEM_Toolbox/Runfiles
% # link to global code (used by all applications)
% # NOT REQUIRED IF LINK CREATED PREVIOUSLY
% ln -s ../Code/ Code
```

2. Create symbolic links to the application (“IYDT-gsrp”) data and code directories

```
% cd IYDT-gsrp
% # link to application (IYDT-gsrp) data files
% # NOT REQUIRED IF LINK CREATED PREVIOUSLY
% ln -s ../../Applications/Data/IYDT-gsrp/ Data
% # link to application code (used by IYDT-gsrp application)
% # NOT REQUIRED IF LINK CREATED PREVIOUSLY
% ln -s ../../Applications/Code/IYDT-gsrp/ Code
```

3. Run the complete analysis for each single phenomenology

- Seismic (phenomenology 1)

```
% cd Seismic
% # link to phenomenology code (used by seismic
% # phenomenology to specify prior distribution
% # of the forward model coefficients (and its gradient))
% # NOT REQUIRED IF LINK CREATED PREVIOUSLY
% ln -s ../../Applications/Code/IYDT-gsrp/Phenomenology/ Code
% cd I-SUGAR-hob
% # if necessary, change iMCMC to "RAM" in runMPEM.r
% R CMD BATCH runMPEM.r runMPEM.out &
% # check status
% tail runMPEM.out
% # a completed run (~2.7 hours) will show the results of
% # proc.time() at the end of the runMPEM.out file
```

- Acoustic (2)

```
% cd ../../Acoustic/I-SUGAR-hob
% # if necessary, change iMCMC to "RAM" in runMPEM.r
% R CMD BATCH runMPEM.r runMPEM.out &
% # check status
% tail runMPEM.out
% # a completed run (~1.5 hours) will show the results of
% # proc.time() at the end of the runMPEM.out file
```

- Optical (3)

```
% cd ../../Optical
% # link to phenomenology code (used by optical
% # phenomenology to specify prior distribution
% # of the forward model coefficients (and its gradient))
% # NOT REQUIRED IF LINK CREATED PREVIOUSLY
% ln -s ../../Applications/Code/IYDT-gsrp/Phenomenology/ Code
% cd I-EIV-SUGAR-hob
% # if necessary, change iMCMC to "RAM" in runMPEM.r
% R CMD BATCH runMPEM.r runMPEM.out &
% # check status
% tail runMPEM.out
% # a completed run (~48 minutes) will show the results of
% # proc.time() at the end of the runMPEM.out file
```

- Surface Effects/Crater (4)

```
% cd ../../Crater/I-EIV-SUGAR
% # if necessary, change iMCMC to "RAM" in runMPEM.r
% R CMD BATCH runMPEM.r runMPEM.out &
% # check status
```

```
% tail runMPEM.out
% # a completed run (~12 minutes) will show the results of
% # proc.time() at the end of the runMPEM.out file
```

The `runMPEM.r` and `runMPEM.out` files are described in Section 3. The status of running code is checked by issuing the following command in the run directory,

```
% tail runMPEM.out
```

Successfully completed runs show the results of the `proc.time()` command at the end of the `runMPEM.out` file. Symbolic links are created to a phenomenology-specific code directory for **Seismic** and **Optical** prior to conducting the runs (see comments in above code; not required to recreate links if they were created previously, e.g. for rapid assessments). Upon completion of the maximum likelihood estimation component of these runs (typically much earlier than the entire run), copy each resulting `opt.RData` file to the `Opt` directory in each MultiPEM analysis, to be used as starting values for MultiPEM log-likelihood maximization.

```
% cd ../../4-Phen/Opt
% cp ../../Seismic/I-SUGAR-hob/opt.RData opt_1.RData
% cp ../../Acoustic/I-SUGAR-hob/opt.RData opt_2.RData
% cp ../../Optical/I-EIV-SUGAR-hob/opt.RData opt_3_eiv.RData
% cp ../../Crater/I-EIV-SUGAR/opt.RData opt_4_eiv.RData
```

4. Run the complete MultiPEM analysis (illustrated here for 4-Phen)

```
% cd ../../4-Phen
% # link to phenomenology code (used by seismic and optical
% # phenomenologies to specify prior distributions of their
% # respective forward model coefficients (and their gradients))
% # NOT REQUIRED IF LINK CREATED PREVIOUSLY
% ln -s ../../Applications/Code/IYDT-gsrp/Phenomenology/ Code
% # noninformative prior distribution on new event
% # device parameters
% cd I-EIV-SUGAR-hob
% # if necessary, change iMCMC to "RAM" in runMPEM.r
% R CMD BATCH runMPEM.r runMPEM.out &
% # a completed run (~7.7 hours) will show the results of
% # proc.time() at the end of the runMPEM.out file
```

The `runMPEM.r` and `runMPEM.out` files are described in Section 3. A symbolic link is created to a phenomenology-specific code directory prior to conducting the run (see comments in above code; not required to recreate links if they were created previously, e.g. for rapid assessments).

1.2 Running MultiPEM Toolbox Through Docker

Applications can be run in the MultiPEM Toolbox through Docker, assuming Docker has been installed on the user's system. The basic steps are stated in the following `README` file,

```
% less ./Runfiles-Docker/README
```

First, a global Docker image is built. This installs the desired version of `R` with the required supporting packages, and incorporates the global subroutines. Second, an application-specific Docker image is built on top of the global image. This incorporates all application relevant subroutines and data. Third, single phenomenology or MultiPEM analysis-specific Docker images are built on top of the application image. These incorporate all run files and (if relevant) `R` data objects containing a starting value for optimization. Finally – for each single phenomenology or MultiPEM analysis – a Docker image is built on top of the analysis image and a Docker container is started to conduct the run, for each use case. Details are provided in the following `README` files for both rapid and complete assessments,

```
% less ./Runfiles-Docker/IYDT-gsrp/Seismic/README
% less ./Runfiles-Docker/IYDT-gsrp/Acoustic/README
% less ./Runfiles-Docker/IYDT-gsrp/Optical/README
% less ./Runfiles-Docker/IYDT-gsrp/Crater/README
% less ./Runfiles-Docker/IYDT-gsrp/4-Phen/README
```

As with the analyses of Section 1.1, all single phenomenology runs are conducted first, and (if needed) all optimization results are copied to the MultiPEM `Opt` directories prior to conducting the subsequent MultiPEM runs.

2 Rapid Assessment

Rapid assessments will be illustrated by examining the run files associated with a multi-phenomenology analysis in which signals from four phenomenologies are combined to infer the log-yield and height-of-burst (HOB) of a near-surface nuclear explosion (WPA, §5).

```
% cd ./Runfiles/IYDT-gsrp/4-Phen/I-EIV-SUGAR-hob-0
```

Rapid assessments consist of two stages. In the first stage, benchmark data are employed to estimate forward model parameters (e.g. regression coefficients) and error model parameters (e.g. source bias, path bias, observational error covariance), and (if relevant) errors-in-variables yield values of benchmark sources. This stage may be run for one or multiple scenarios of interest upon identification of relevant historical data for each scenario, and the resulting `.RData` file(s) stored for later use in processing new event data.

In the second stage, new event data are processed to infer unknown device parameters (e.g. yield, HOB/DOB, geolocation, event time) with uncertainty quantification. In Bayesian analysis, forward and error model parameters, and (if relevant) benchmark source errors-in-variables yields, may be treated in two ways:

- Fixed at values obtained from the first stage, or
- Imputed using posterior samples from the first stage⁵.

Either approach results in rapid assessments being executed with far less compute time than complete assessments. The first approach has the potential consequence of underestimating uncertainty in the unknown device parameters of interest for the new event, which is avoided by selecting the second approach with the expense of additional compute time.

2.1 First Stage

The first stage analysis is defined in the `runMPEM.r` file, provided in the first three sections of Appendix A with line numbers referred to in the ensuing discussion. Appendix A.1 provides the preprocessing component of the first stage, Appendix A.2 provides the code employed to maximize the likelihood function of the benchmark data with respect to the parameters of the forward and error models, while Appendix A.3 provides the code employed to optionally sample the posterior distribution of these parameters.

The first stage analysis is run in batch mode as follows,

```
% R CMD BATCH runMPEM.r runMPEM.out &
```

The main features of the output file `runMPEM.out` are provided in Appendix A.4.

2.1.1 Preprocessing

The preprocessing component of the first stage analysis in Appendix A.1 is primarily responsible for describing features of the benchmark data, and the parameters of the forward and

⁵Plummer, M. (2015). Cuts in Bayesian graphical models, *Stat Comput* 25:37-43.

error models.

- Line 25+: Load all R packages utilized by multiple supporting subroutines, most notably log-likelihood and log-prior calculations and their associated gradients.
- Line 36: Specify directory location (relative to run directory) of all global (application independent) subroutines.
- Line 39: Read in code performing first stage preprocessing of benchmark data.
- Line 42: Specify directory location (relative to run directory) of all application-specific subroutines.
- Line 45: Specify root directory (relative to run directory) containing all application-specific benchmark data files.
- Lines 48-51: A scalar or vector specifying the names of benchmark data files for each phenomenology, utilizing an ordering of the phenomenologies (for MultiPEM analysis) that is maintained throughout the input deck (as indicated here in Lines 54-57). Data files are text files (CSV formatted) containing all measured signatures (in the first column(s)) and input covariates (in succeeding column(s)) including all those required in forward and error model calculations. Directories specifying the exact locations of these files relative to the root data directory (Line 45) may also be included in the filenames.
- Line 60: Indicate if forward model parameters global to multiple signatures within phenomenology or across phenomenologies will be modeled. If `TRUE`, nominal values for these parameters may optionally be placed in the benchmark data file(s). If a subset of sources are to be assigned default values for some or all of these parameters, the value `NA` should be assigned to these parameters in the benchmark data file(s) for these sources, and the default values provided in the relevant forward model(s).
- Lines 62-65: If `calp` is `TRUE` (Line 60), provide a string vector of names for each of the global forward model parameters (Line 64).
- Line 68: A scalar or vector specifying the number of observed signatures for each phenomenology; in this example, 2 for each phenomenology.
- Lines 72-79: Specify the number of *common* forward model parameters within each phenomenology (WPA, §4.1, first paragraph). For a given forward model, common parameters maintain the same constant value within signature for every log-likelihood calculation. The `pbeta` object is initialized as a null list with elements for each phenomenology in the proper order (Line 72), initialized to zero vectors of length equal to the number of observed signatures (Line 73). If relevant, subsequent lines specify the number of common forward model parameters for each signature within each phenomenology. For example, the *acoustic* forward model for each signature contains 2 common forward model parameters (Line 75).
- Lines 82-85: Specify if the forward model(s) for any phenomenology depend on event emplacement conditions (Line 82), followed by (if relevant) a vector indicating the

number of distinct emplacement conditions considered for each phenomenology in the proper order (Line 84). This specification allows distinct forward model parameters to be associated with different emplacement conditions (as specified subsequently). If **Th** is **TRUE** (Line 82), a factor named **Type** must be present in the benchmark (and new event) data file for each relevant phenomenology, indicating the emplacement condition pertaining to each entry. In this example, the *seismic* and *acoustic* forward model parameters may vary for 3 distinct emplacements (“soft”, “hard”, and “wet” rock types), while the *optical* and *surface effects* forward models are independent of emplacement condition.

- Lines 89-104: Specify the number of *emplacement* dependent forward model parameters within each phenomenology if relevant (WPA, §4.1, first paragraph). For a given forward model, emplacement parameters remain constant within signature for log-likelihood calculations with a given emplacement condition, but may be modified within signature for each distinct emplacement. The **pbetat** object is initialized as a null list with elements for each phenomenology in the proper order (Line 90), initialized as null lists with elements for each emplacement condition (Line 92) if multiple emplacements are present. Subsequent lines specify the number of forward model parameters for each signature within each emplacement condition for each phenomenology. For example, the *seismic* forward model for each signature within each emplacement contains 5 forward model parameters (Line 97) allowed to vary across emplacements. **pbetat** must be specified if multiple emplacements are present for any phenomenology (at least one element of **Th** is greater than 1).
- Lines 108-123: Specify the location of *common* forward model parameters within the full parameter vector, for phenomenologies possessing both common and emplacement dependent parameters. The **ibetar** object is initialized as a null list with elements for each phenomenology in the proper order (Line 109), initialized as null lists with elements for each signature within each emplacement condition (Line 114) if multiple emplacements are present. Subsequent lines specify the position of common parameters in the full forward model parameter vector, for phenomenologies possessing both common and emplacement dependent forward model parameters. For example, the *acoustic* forward model parameter vector takes common parameter values in its first two positions for each signature within each emplacement condition (Line 120).
- Line 126: Indicate if errors-in-variables yield values for benchmark events will be modeled (WPA, §2, Equation (3); §A.4). If **TRUE**, this allows uncertain yields for benchmark events (often assumed known with certainty) to vary within user-specified guidelines.
- Lines 129-145: If relevant, specify details of errors-in-variables yield models for benchmark events.
 - Line 132: Specify phenomenologies for application of errors-in-variables yield models to benchmark events
 - Lines 136-137: Provide the sources subject to errors-in-variables yield models for each phenomenology. The **seiv** object is initialized as a null list with elements for each phenomenology in the proper order (Line 136), with vectors indicating

the relevant sources for each relevant phenomenology (Line 137). The "ALL" designation indicates that every source in the benchmark data set for the indicated phenomenology will be modeled with an errors-in-variables yield. `seiv` must be specified if `ieiv` is provided.

- Line 140: The standard deviation of the errors-in-variables Gaussian distribution for each benchmark event log-yield. For each event, the mean of this distribution is taken to be its provided (design or measured) log-yield. In this example, a “total” error (3 standard deviations) of 30% in each provided yield is allowed. Note that this error is relative because yields are treated on a logarithmic scale. `eiv_w_sd` must be specified if `ieiv` is provided.
- Lines 149-158: Specify if *source* random effects (WPA, §2, Equation (2); §3; §A.5) should be included in the error model (Line 149). If so, the `pvc_1` object is initialized as a null list with elements for each phenomenology in the proper order (Line 152), initialized to zero vectors of length equal to the number of observed signatures (Line 153). Subsequent lines specify the number of source random effects for each signature within each phenomenology. For example, the *seismic* error model for each signature contains a single source bias term (Line 155). If `pvc_1` is `TRUE` (Line 149), a factor named `Source` may be provided in the benchmark (and new event) data file for each relevant phenomenology, identifying the source pertaining to each entry. This factor must be present if there is more than one data entry for any source. In order to include source random effects in the error model for an observed signature, the benchmark data must contain more than one source, with at least one source containing more than one observation. A warning message will be printed to the log file if one of these conditions is violated.
- Lines 161-177: Specify if *path* random effects (WPA, §2, Equation (2); §3; §A.5), also referred to as *station* random effects, should be included in the error model (Line 161). If so, the `pvc_2` object is initialized as a null list with elements for each phenomenology in the proper order (Line 164), initialized to zero vectors of length equal to the number of observed signatures (Line 165). Subsequent lines specify the number of path random effects for each signature within each phenomenology. For example, the *seismic* error model for each signature contains a single path bias term (Line 167). The type of path random effect desired is specified by the `pctype` object, initialized as a null list with elements for each phenomenology in the proper order (Line 172). In this application, both the seismic and acoustic error models contain crossed path random effects (Lines 174 and 176). If `pvc_2` is `TRUE` (Line 161), a factor named `Path` must be provided in the benchmark (and new event) data file for each relevant phenomenology, identifying the source-to-sensor path pertaining to each entry. In order to include path random effects in the error model for an observed signature, the benchmark data must contain more than one path, with at least one path containing more than one observation. Additionally, specification of crossed path effects (`pctype` is "Crossed") requires the signature to be observed from at least one common path for two or more sources, while specification of nested (within source) path effects requires more than one path for at least one source, with more than one observation for at least one of those paths. A

warning message will be printed to the log file if one of these conditions is violated.

- Line 181: Indicate if the user is providing code to compute coefficient matrices for *source* or *path* random effects (WPA, §4.1). If **FALSE**, the functions `calc_zmat.r` and `calc_zmat_0.r` located in the global code directory,

MultiPEM_Toolbox/Code

compute default coefficient matrices for the benchmark and new event data, respectively. If **TRUE**, then two user-provided functions of the same names must be placed in the application code directory; in this example,

MultiPEM_Toolbox/Applications/Code/IYDT-gsrp

Table 1 shows notional data for two *seismic* benchmark sources.

Table 1: Data for seismic benchmark sources HRI-1 and HRI-2.

Y1	Y2	Source	Path	Type	lRange	W	HOB
-15.091	-9.252	HRI-1	Path_1	1	6.932	6.291	5
-15.089	-9.180	HRI-1	Path_1	1	6.932	6.291	5
-15.836	-10.218	HRI-1	Path_2	1	7.570	6.291	5
-15.892	-10.180	HRI-1	Path_2	1	7.570	6.291	5
-16.176	-10.557	HRI-1	Path_2	1	7.800	6.291	5
-16.907	-11.366	HRI-1	Path_2	1	8.371	6.291	5
-16.931	-11.338	HRI-1	Path_2	1	8.371	6.291	5
-14.835	-9.199	HRI-2	Path_1	1	6.930	6.291	3
-14.860	-9.184	HRI-2	Path_1	1	6.930	6.291	3
-15.674	-10.089	HRI-2	Path_1	1	7.568	6.291	3
-15.754	-10.197	HRI-2	Path_1	1	7.568	6.291	3
-16.002	-10.530	HRI-2	Path_2	1	7.802	6.291	3
-16.060	-10.605	HRI-2	Path_2	1	7.802	6.291	3
-16.534	-11.115	HRI-2	Path_2	1	8.239	6.291	3
-16.741	-11.230	HRI-2	Path_3	1	8.373	6.291	3
-16.737	-11.288	HRI-2	Path_3	1	8.373	6.291	3
-17.208	-11.656	HRI-2	Path_3	1	8.738	6.291	3

If source and nested (within source) path random effects are included in the error model, the source and path bias vectors (WPA, §4.1, p. 7) associated with these sources (with HRI-1 and HRI-2 designated as 1 and 2) are given by

$$\begin{aligned}
\mathbf{E}_{S,11r} &= \begin{pmatrix} \mathbf{Z}_{11r,1} \\ \mathbf{Z}_{11r,2} \end{pmatrix} b_{1r,1}^{(S)} & \mathbf{E}_{P,11r} &= \begin{pmatrix} \mathbf{E}_{P,111r} \\ \mathbf{E}_{P,112r} \end{pmatrix} = \begin{bmatrix} \mathbf{Z}_{111r} & \mathbf{0}_2 \\ \mathbf{0}_5 & \mathbf{Z}_{112r} \end{bmatrix} \begin{pmatrix} b_{1r,11}^{(P)} \\ b_{1r,12}^{(P)} \end{pmatrix} \\
\mathbf{E}_{S,12r} &= \begin{pmatrix} \mathbf{Z}_{12r,1} \\ \mathbf{Z}_{12r,2} \\ \mathbf{Z}_{12r,3} \end{pmatrix} b_{1r,2}^{(S)} & \mathbf{E}_{P,12r} &= \begin{pmatrix} \mathbf{E}_{P,121r} \\ \mathbf{E}_{P,122r} \\ \mathbf{E}_{P,123r} \end{pmatrix} = \begin{bmatrix} \mathbf{Z}_{121r} & \mathbf{0}_4 & \mathbf{0}_4 \\ \mathbf{0}_3 & \mathbf{Z}_{122r} & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{Z}_{123r} \end{bmatrix} \begin{pmatrix} b_{1r,21}^{(P)} \\ b_{1r,22}^{(P)} \\ b_{1r,23}^{(P)} \end{pmatrix},
\end{aligned}$$

where the default coefficient matrices are given by

$$\begin{array}{ll}
\mathbf{Z}_{11r,1} = \mathbf{1}_2 & \mathbf{Z}_{111r} = \mathbf{1}_2 \\
\mathbf{Z}_{11r,2} = \mathbf{1}_5 & \mathbf{Z}_{112r} = \mathbf{1}_5 \\
\mathbf{Z}_{12r,1} = \mathbf{1}_4 & \mathbf{Z}_{121r} = \mathbf{1}_4 \\
\mathbf{Z}_{12r,2} = \mathbf{1}_3 & \mathbf{Z}_{122r} = \mathbf{1}_3 \\
\mathbf{Z}_{12r,3} = \mathbf{1}_3 & \mathbf{Z}_{123r} = \mathbf{1}_3
\end{array}$$

for $\mathbf{1}_q$ and $\mathbf{0}_q$ the q -vectors of ones and zeros, respectively. The source and path random effects $\{b_{1r,1}^{(S)}, b_{1r,2}^{(S)}\}$ and $\{b_{1r,11}^{(P)}, b_{1r,12}^{(P)}, b_{1r,21}^{(P)}, b_{1r,22}^{(P)}, b_{1r,23}^{(P)}\}$ are mutually independent realizations of their respective random effects distributions (WPA, §4.1, Equation (5), p. 7). For each signature, this structure indicates that there is a single source bias effect applied to every observation within each source, while observations from each path are adjusted by distinct (and independently distributed) path bias effects (signatures are collected from two and three paths respectively for HRI-1 and HRI-2).

If instead a signature is observed from one or more common source-to-sensor paths across two or more sources, referred to as “crossed paths” (`ptype` is “Crossed” (Line 172)) – assuming the source-to-sensor paths observed for the sources HRI-1 and HRI-2 are not present for any other source – the path bias vectors ($\mathbf{E}_{P,11r}, \mathbf{E}_{P,12r}$) corresponding to the two sources above are replaced by the single path bias vector

$$\mathbf{E}_{P,1\{1,2\}r} = \begin{bmatrix} \mathbf{Z}_{111r} & \mathbf{0}_2 & \mathbf{0}_2 \\ \mathbf{0}_5 & \mathbf{Z}_{112r} & \mathbf{0}_5 \\ \mathbf{Z}_{121r} & \mathbf{0}_4 & \mathbf{0}_4 \\ \mathbf{0}_3 & \mathbf{Z}_{122r} & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{Z}_{123r} \end{bmatrix} \begin{pmatrix} b_{1r,\{1,2\}1}^{(P)} \\ b_{1r,\{1,2\}2}^{(P)} \\ b_{1r,\{1,2\}3}^{(P)} \end{pmatrix}.$$

The entry $\{1, 2\}$ for the source index indicates that sources 1 (HRI-1) and 2 (HRI-2) must be considered jointly as a group, due to covariance between their observed signatures induced by the common source-to-sensor propagation paths `Path_1` and `Path_2`. The path random effects $\{b_{1r,\{1,2\}1}^{(P)}, b_{1r,\{1,2\}2}^{(P)}, b_{1r,\{1,2\}3}^{(P)}\}$ are mutually independent realizations of the path random effect distribution (WPA, §4.1, Equation (5), p. 7).

- Lines 184-187: Calls the preprocessing function `prepro_cal` for the benchmark data. Table 2 describes all inputs to this function with default values. Only inputs with no default values must be provided.

2.1.2 Maximum Likelihood Estimation

The maximum likelihood estimation component of the first stage analysis in Appendix A.2 is responsible for utilizing benchmark data to estimate the parameters of the forward and error models, and possibly the yield of each benchmark source for phenomenologies adopting the errors-in-variables yield model (WPA, §A.4). The resulting estimates are supplied to all relevant second stage analyses.

Table 2: Inputs to `prepro_cal` function.

Input	Default	Brief Description
<code>gdir</code>	<code>none</code>	directory location of global subroutines
<code>adir</code>	<code>none</code>	directory location of application subroutines
<code>rdir</code>	<code>none</code>	root directory location of data files
<code>cdir</code>	<code>none</code>	directory locations (if relevant) and names of benchmark data files under <code>rdir</code>
<code>Rh</code>	<code>none</code>	vector with number of signatures for each phenomenology
<code>pbeta</code>	<code>none</code>	list containing empirical model common parameter counts by phenomenology
<code>izmat</code>	<code>FALSE</code>	user-provided code for computing variance component coefficient matrices
<code>ieiv</code>	<code>NULL</code>	numerical identifier of phenomenologies utilizing errors-in-variables yields in analysis of benchmark data
<code>seiv</code>	<code>NULL</code>	list containing identifiers of benchmark sources assigned errors-in-variables yields by phenomenology (<code>ALL</code> – every source)
<code>ewsd</code>	<code>NULL</code>	standard deviation of errors-in-variables Gaussian likelihood
<code>Th</code>	<code>NULL</code>	number of emplacement conditions for each phenomenology
<code>pbetat</code>	<code>NULL</code>	list containing empirical model emplacement-dependent parameter counts by phenomenology
<code>ibetar</code>	<code>NULL</code>	list containing locations of empirical model common parameters in full parameter vector by phenomenology
<code>pvc_1</code>	<code>NULL</code>	list containing source variance component parameter counts by phenomenology
<code>pvc_2</code>	<code>NULL</code>	list containing path variance component parameter counts by phenomenology
<code>ptype</code>	<code>NULL</code>	list indicating treatment of path variance component parameter by phenomenology (<code>Crossed</code> – common paths present across sources)
<code>cnames</code>	<code>NULL</code>	names of global forward model parameters

- Line 6: Read in code performing first stage maximum likelihood estimation of forward and error model parameters, and (if relevant) benchmark source errors-in-variables yields, based on benchmark data.
- Line 9: User specified seed to ensure repeatability of maximum likelihood estimation.
- Lines 13-17: Provide names of forward models for each signature by phenomenology (WPA, §5.1-5.4). The `fm` object is initialized as a null list with elements for each phenomenology in the proper order (Line 13). Subsequent lines specify the function names as vectors of strings having length equal to the number of signatures for each phenomenology (Lines 14-17). The code for all forward models from each phenomenology is concatenated into a single file named `forward.r` and placed in the application code directory; in this example,

`MultiPEM_Toolbox/Applications/Code/IYDT-gsrp`

Note that these forward models accept the parameters to be calibrated as their main argument. In this example, the *seismic* forward model $f_{sr}(\cdot)$ as a function of the parameters β_{sr} is given as follows (WPA, §5.2, pp. 11-12),

$$\begin{aligned}\log(\tilde{d}_{sr}(\beta_{sr})) &= \beta_{sr,1} + \beta_{sr,2} \log(\tilde{\delta}_s) + \beta_{sr,3} \text{logistic}(\beta_{sr,4} \tilde{h}_s + \beta_{sr,5}) \\ f_{sr}(\beta_{sr}) &= \log(d_{sr}(\beta_{sr}))\end{aligned}\tag{1}$$

for

$$\text{logistic}(x) = \frac{1}{1 + \exp(-x)}.$$

The scaled signatures and covariates of this forward model are given by

$$\begin{aligned}\tilde{d}_{s1} &= d_{s1} \exp(-w/3) & \tilde{d}_{s2} &= d_{s2} \\ \tilde{\delta}_s &= \delta \exp(-w/3) & \tilde{h}_s &= h \exp(-w/3),\end{aligned}$$

where d_{s1} and d_{s2} are P-wave displacement and maximum velocity, and the covariates $v = (w, h, \delta)$ are log-yield, HOB/DOB, and range. The function `f_s` returns a vector of forward model calculations evaluated for the supplied value of β_{sr} , each element corresponding to each row of a matrix of covariates (having columns (w, h, δ)).

- Line 20: Indicate if forward model Jacobian matrices are provided for efficient log-likelihood maximization.
- Lines 22-30: If `igrad` is `TRUE`, names of forward model Jacobian functions must be provided for each signature by phenomenology. The `gfm` object is initialized as a null list with elements for each phenomenology in the proper order (Line 25). Subsequent lines specify the Jacobian function names as vectors of strings having length equal to the number of signatures for each phenomenology (Lines 26-29). The code for all forward model Jacobian functions from each phenomenology is concatenated into a single file named `jacobian.r` and placed in the application code directory; in this example,

`MultiPEM_Toolbox/Applications/Code/IYDT-gsrp`

Note that these Jacobian functions accept the parameters to be calibrated as their main argument. In this example, the gradient vector of the *seismic* forward model of Equation (1) is computed from the partial derivatives of $f_{sr}(\cdot)$ for each parameter as

follows,

$$\begin{aligned}
\frac{\partial f_{sr}}{\partial \beta_{sr,1}} &= 1 \\
\frac{\partial f_{sr}}{\partial \beta_{sr,2}} &= \log(\tilde{\delta}_s) \\
\frac{\partial f_{sr}}{\partial \beta_{sr,3}} &= \text{logistic}(\beta_{sr,4}\tilde{h}_s + \beta_{sr,5}) \\
\frac{\partial f_{sr}}{\partial \beta_{sr,4}} &= \beta_{sr,3}\tilde{h}_s \times \text{logistic}(\beta_{sr,4}\tilde{h}_s + \beta_{sr,5}) \times \text{logistic}(-\beta_{sr,4}\tilde{h}_s - \beta_{sr,5}) \\
\frac{\partial f_{sr}}{\partial \beta_{sr,5}} &= \beta_{sr,3} \times \text{logistic}(\beta_{sr,4}\tilde{h}_s + \beta_{sr,5}) \times \text{logistic}(-\beta_{sr,4}\tilde{h}_s - \beta_{sr,5})
\end{aligned}$$

The function `g_s` returns a Jacobian matrix (`jbeta`) of forward model gradients for the parameters, evaluated at the supplied value of β_{sr} , with rows corresponding to the rows of a matrix of covariates (having columns (w, h, δ)). If `calp` is `TRUE` (Line 60 of Appendix A.1), partial derivatives of $f_{sr}(\cdot)$ with respect to the global forward model parameters must also be calculated analogously to `jbeta` and returned by `g_s` as the object `jcalp`. If `eiv` is `TRUE` (Line 126 of Appendix A.1), the partial derivative of $f_{sr}(\cdot)$ for log-yield w is also required,

$$\begin{aligned}
\frac{\partial f_{sr}}{\partial w} &= -\frac{1}{3} \left(\beta_{sr,2} + \beta_{sr,3}\beta_{sr,4}\tilde{h}_s \times \text{logistic}(\beta_{sr,4}\tilde{h}_s + \beta_{sr,5}) \times \text{logistic}(-\beta_{sr,4}\tilde{h}_s - \beta_{sr,5}) \right) \\
&\quad + \frac{1}{3} \delta_1(r)
\end{aligned} \tag{2}$$

for $\delta_A(x)$ the indicator function of set A . The function `g_s` will also return a Jacobian vector (`jtheta`) of forward model partial derivatives for log-yield, evaluated at the supplied value of β_{sr} , each element corresponding to each row of the same covariate matrix used in the calculation of `jbeta`.

- Line 35: Indicate if the same forward model function is used to compute multiple signatures, and signature-specific code within this function is required.
- Lines 37-41: If `iResponse` is `TRUE` (Line 35), initialize `iResponse` to a null list with elements for each phenomenology in the proper order (Line 38). For each relevant phenomenology, subsequent lines provide vectors of length equal to the number of signatures, each element of which is a tag identifying code specific to the corresponding signature. This mechanism is utilized for the *seismic* (Line 39) and *acoustic* (Line 40) phenomenologies.
- Line 44: Indicate if fixed inputs are to be provided to the forward models for at least one phenomenology.
- Lines 46-53: If `fPars` is `TRUE` (Line 44), initialize `fPars` to a null list with elements for each phenomenology in the proper order (Line 47). For each relevant phenomenology,

subsequent lines provide the value(s) of all fixed inputs. For example, the *acoustic* forward model requires fixed values for `yield_scaling` (Line 49), `pressure_scaling` (Line 50), and `temp_scaling` (Line 51).

- Line 56: Specify the number of starting parameter vectors for the log-likelihood maximization routine.
- Line 59: Specify the number of cores to use for parallel optimization (across distinct starting values) of the benchmark data log-likelihood function.
- Line 62: Specify if the Broyden-Fletcher-Goldfarb-Shanno (BFGS) algorithm is to be used for maximization of the log-likelihood function. If `TRUE`, functions to compute forward model Jacobian matrices analytically must be provided, or numerical gradients will be utilized (generally increasing compute time). If `FALSE`, the gradient-free Nelder-Mead simplex algorithm will be utilized for optimization, which is generally much slower computationally than BFGS with analytical gradients.
- Lines 66-69: If relevant, specifies the location (relative to the run directory) of parameter values or estimates to be used as the first starting value for log-likelihood maximization. These values are stored in a `.RData` object as a list, with elements corresponding to within signature forward model (e.g. common parameters `beta`, emplacement-dependent parameters `tbeta`) and error model (e.g. source variance components `vc_1`, path variance components `vc_2`, observation error parameters `eps`) quantities of interest. If relevant, global forward model parameter estimates (`calp`) and benchmark source errors-in-variables yield estimates (`w_eiv`) are also provided. For multi-phenomenology analyses, values or estimates from individual phenomenologies may be input in the proper order, and they will be concatenated appropriately.
- Line 73: If desired, name of output `.RData` file to store optimization results from this run. The elements of the list to be written are described in the previous item.
- Lines 75-89: If `calp` is `TRUE` (Line 60 of Appendix A.1), provides specifications for maximum likelihood estimation of global forward model parameters.
 - Line 77: If `cst` is `TRUE`, specifies an initial starting value for the global forward model parameters (Lines 80-81) for log-likelihood maximization. This value supersedes the value read in from the first file provided in the string vector `opt_files_in` (Line 66), if the `calp` list element is provided.
 - Line 85: Specifies the level of confidence intervals for the true values of each global forward model parameter from the maximum likelihood estimate and the estimated Fisher information matrix.
- Line 93: Indicate if phenomenology specific code is required in the postprocessing function.
- Lines 95-97: If `Phen` is `TRUE` (Line 93), specifies a matrix in which the first column provides the numerical phenomenology indicator (see Lines 54-57 of the preprocessing code in Appendix A.1), and the second column provides the phenomenology name

in string format. In this example, specific code is required to process results for the *seismic* phenomenology (Line 96).

- Line 100: Indicate if gradient verification is to be conducted on the log-likelihood function. If `TRUE` and `igrad` is `TRUE` (Line 20), analytical and numerical gradients at the optimal parameter value, and other randomly sampled parameter values, are compared for consistency.
- Line 103: Specify the strategy for running parallel jobs using the `future` package in R. The available options are given by starting an R session and issuing the following commands,

```
% R
> require(future)
> help(plan)
```

- Lines 106-111: Calls the log-likelihood maximization function `calc_mle_cal` for the benchmark data. Table 3 describes all inputs to this function with default values. Only inputs with no default values must be provided.

2.1.3 Bayesian Analysis

The optional Bayesian inference component of the analysis in Appendix A.3 is responsible for sampling forward and error model parameters, and benchmark source errors-in-variables yields (if relevant) from their joint posterior distribution using benchmark data. This Bayesian component must be run if multiple imputation of forward and error model parameters is desired in second stage Bayesian inference for new event device parameters.

- Line 6: Indicate if first stage Bayesian analysis is to be conducted. If second stage multiple imputation is desired, `iBayes` must be `TRUE`.
- Line 10: Read in code performing Bayesian analysis on forward and error model parameters, and benchmark source errors-in-variables yields (if relevant), using benchmark data.
- Line 14: Indicate if a log-prior density for the signature within phenomenology forward model parameters is supplied by the user (WPA, §5.5, p. 15). If `iBetaPrior` is `FALSE`, a “flat prior” (uniform on the domain) on these parameters is assumed.
- Lines 16-43: If relevant, specify details of user-provided log-prior distributions for signature within phenomenology forward model parameters. For each relevant phenomenology, the list object `lp_beta` is used for common coefficients, while the list object `lp_betat` is used for emplacement-dependent coefficients (as demonstrated below in this application).
 - Line 18: Specify location(s) of log-prior function(s). Must be provided if `iBetaPrior` is `TRUE` (Line 14). In this example, two log-prior functions are provided for the seismic (‘s’) and optical (‘o’) phenomenologies, located at

```
../Code/lp_beta_s.r
```

Table 3: Inputs to `calc_mle_cal` function.

Input	Default	Brief Description
<code>p_cal</code>	<code>none</code>	environment storing all objects needed in log-likelihood calculations
<code>gdir</code>	<code>none</code>	directory location of global subroutines
<code>adir</code>	<code>none</code>	directory location of application subroutines
<code>f</code>	<code>none</code>	names of forward model functions for each signature by phenomenology
<code>nst</code>	<code>10</code>	number of starting values for log-likelihood maximization
<code>ncor</code>	<code>1</code>	number of cores for log-likelihood maximization
<code>ci_lev</code>	<code>0.95</code>	confidence interval levels for global forward model parameter inference
<code>igrad</code>	<code>TRUE</code>	forward model Jacobian provided
<code>bfgs</code>	<code>TRUE</code>	log-likelihood maximization uses BFGS methods
<code>igrck</code>	<code>TRUE</code>	conduct log-likelihood function gradient verification
<code>g</code>	<code>NULL</code>	names of forward model Jacobian functions for each signature by phenomenology
<code>iresp</code>	<code>NULL</code>	flags for modified calculation by signature in a common forward model for each relevant phenomenology
<code>fp_fm</code>	<code>NULL</code>	fixed inputs required by forward models
<code>fopt_in</code>	<code>NULL</code>	location of input R data file(s) providing an initial starting value for log-likelihood maximization (if multiple files, starting value created by concatenating over phenomenologies)
<code>Xst</code>	<code>NULL</code>	matrix of starting values for log-likelihood maximization if not generated by this function
<code>cst</code>	<code>NULL</code>	vector of starting values for global forward model parameters in log-likelihood maximization
<code>fopt_out</code>	<code>NULL</code>	location to write output R data file with results of log-likelihood maximization
<code>phen</code>	<code>NULL</code>	phenomenology number and type (if needed for postprocessing)
<code>pl</code>	<code>"multicore"</code>	strategy for running parallel jobs using the <code>future</code> package

```
../Code/lp_beta_o.r
```

- Lines 20-21: If `igrad` is `TRUE` (Line 20 of Appendix A.2), specify location(s) of the log-prior gradient function(s). In this example, two log-prior gradient functions are provided, located at

```
../Code/glp_beta_s.r
../Code/glp_beta_o.r
```

- Line 26: For each relevant phenomenology, initialize a null list `lp_betat` of length equal to the number of emplacement conditions containing distinct forward model parameters.

- Line 28: For each relevant phenomenology and emplacement condition, provide the name(s) of the log-prior function(s) for each signature. In this example, the *seismic* phenomenology utilizes a log-prior function `lp_s` for each signature within each emplacement condition.
- Line 30: If `igrad` is `TRUE` (Line 20 of Appendix A.2), then for each relevant phenomenology and emplacement condition, provide the name(s) of the log-prior gradient function(s) for each signature. In this example, the *seismic* phenomenology utilizes a log-prior gradient function `lq_s` for each signature within each emplacement condition.
- Line 36: The list object `lp.beta` is used to specify log-prior distributions for common coefficients. In this example, the *optical* phenomenology utilizes a log-prior function `lp_o` for each signature.
- Line 38: If `igrad` is `TRUE` (Line 20 of Appendix A.2), then in this example, the *optical* phenomenology utilizes a log-prior gradient function `lq_o` for each signature.
- Line 46: If `calp` is `TRUE` (Line 60 of Appendix A.1), indicate if a log-prior density for the global forward model parameters is supplied by the user. If `iCalPrior` is `FALSE`, a “flat prior” (uniform on the domain) on these parameters is assumed.
- Lines 48-65: If relevant, specify details of user-provided log-prior distributions for global forward model parameters.
 - Line 50: Specify location of log-prior function. If `NULL`, utilize the default log-prior function contained in the file `lp_c.r` placed in the application code directory; in this example,


```
MultiPEM_Toolbox/Applications/Code/IYDT-gsrp
```
 - Line 52: If `igrad` is `TRUE` (Line 20 of Appendix A.2), specify location of the log-prior gradient function. If `NULL`, utilize the default log-prior gradient function contained in the file `glp_c.r` placed in the application code directory; in this example,


```
MultiPEM_Toolbox/Applications/Code/IYDT-gsrp
```
 - Line 56: Provide the name of the log-prior function.
 - Line 57: If `igrad` is `TRUE` (Line 20 of Appendix A.2), provide the name of the log-prior gradient function.
 - Lines 60-61: Specify all fixed quantities required for calculation of the log-prior density.
- Line 69: Specify a *fixed* value for the scale parameter `A` of half-Cauchy prior distribution(s) for the *source* and *path* variance component parameters if relevant (WPA, §5.5, p. 15; §5.6, p. 17). Prior distributions for a non-empty collection of variance

component parameters are taken to be mutually independent. Comment out if this parameter is to be sampled from its posterior distribution.

- Line 73: Specify a *fixed* value for the shape parameter η of the Lewandowski-Kurowicka-Joe (LKJ) prior distribution for the observational error model correlation parameters (WPA, §5.5, p. 15; §5.6, p. 17).
- Line 81: If `eiv` is `TRUE` (Line 126 of Appendix A.1), specify a *fixed* value for the parameter that controls the number of modes in the *flexible generalized skew-normal* (FGSN) prior distribution for the errors-in-variables yields of the benchmark events (WPA, §5.5, p. 15; §5.6, p. 26).
- Line 87: Select the Markov chain Monte Carlo (MCMC) algorithm to use for posterior sampling, from one of three options: `RAM`, `FME`, and `NUTS`. `RAM` is the robust adaptive Metropolis algorithm of Vihola⁶ implemented in the R package `adaptMCMC`. `FME` is the delayed rejection adaptive Metropolis algorithm of Haario, Laine, and Mira⁷ implemented in the R package `FME`. `NUTS` is the No-U-Turn Sampler of Hoffman and Gelman⁸. The `NUTS` option requires the analytical gradient of the log-posterior density, which in turn requires `igrad` to be `TRUE` (Line 20 of Appendix A.2).
- Line 90: Specify the per core sample size of the burn-in period for MCMC sampling (pre-equilibrium stage of Markov chain). These samples are discarded prior to any inference using the posterior samples.
- Line 93: Specify the sample size of the MCMC production run. These samples are kept for posterior inference.
- Line 96: Specify the rate at which MCMC production samples are thinned for estimation of the Deviance Information Criterion⁹ (DIC) and the Predictive Information Criterion¹⁰ (PIC). In this example, the `nthin` value of 20 indicates that every 20-th production sample is kept for DIC and PIC estimation.
- Line 99: Specify the number of cores to use for parallel optimization (across distinct starting values) of the benchmark data log-posterior function. By default, this optimization uses the same number of distinct starting values as optimization of the benchmark data log-likelihood function (Line 56 of Appendix A.2).
- Line 102: Specify the number of cores used to run parallel MCMC chains. The burn-in period for each chain is determined by `nburn` (Line 90), while the `nmcmc` (Line 93) production runs are split between the `ncores_mc` processors and combined at the conclusion of the runs.

⁶Vihola, M. (2012). Robust adaptive Metropolis algorithm with coerced acceptance rate. *Stat Comput* 22:997-1008.

⁷Haario, H., Laine, M., and Mira, A. (2006). DRAM: Efficient adaptive MCMC. *Stat Comput* 16:339-354.

⁸Hoffman, M. D. and Gelman, A. (2014). The No-U-Turn Sampler: Adaptively setting path lengths in Hamiltonian Monte Carlo. *J Mach Learn Res* 15:1593-1623.

⁹Spiegelhalter, D.J., Best, N.G., Carlin, B.P., & van der Linde, A. (2002). Bayesian measures of model complexity and fit (with discussion), *J R Stat Soc Ser B* 64:583-639.

¹⁰Ando, T. (2011). Predictive Bayesian model selection, *Am J Math Manag Sci* 31:13-38.

- Line 105: Indicate if gradient verification is to be conducted on the log-prior function. If `TRUE` and `igrad` is `TRUE` (Line 20 of Appendix A.2), analytical and numerical gradients at the maximum *a posteriori* parameter value, and other randomly sampled parameter values, are compared for consistency.
- Line 108: Indicate if gradient verification is to be conducted on the log-posterior function. If `TRUE` and `igrad` is `TRUE` (Line 20 of Appendix A.2), analytical and numerical gradients at the maximum *a posteriori* parameter value, and other randomly sampled parameter values, are compared for consistency.
- Lines 111-120: Calls the Bayesian analysis function `calc_bayes_cal` for the benchmark data. Table 4 describes all inputs to this function with default values. Only inputs with no default values must be provided.

2.1.4 Output

The output file `runMPEM.out` from the first stage analysis contains a summary of (if relevant) global forward model parameters, (if relevant) errors-in-variables yield estimates for the relevant benchmark sources, signature within phenomenology forward and error model parameter estimates, and (if relevant) posterior samples derived from the benchmark data. The desired output is supplied by the user function `print_sumstats.r`, placed in the application code directory; in this example,

`MultiPEM_Toolbox/Applications/Code/IYDT-gsrp`

The output presented in Appendix A.4 contains the most pertinent information extracted from the full file.

- Lines 1-13: Output from the preprocessing function `prepro_cal`. These warning messages explain which variance component models are allowed (if any) for each signature of each phenomenology based on the structure of the benchmark data. In this example, no source or path random effects are allowed for *optical* or *surface effects* phenomenologies (Lines 6-13). There are no warning messages for *seismic* or *acoustic* signatures, indicating source and path random effects are allowed.
- Lines 15-247: Output from the maximum likelihood estimation function `calc_mle_cal`:
 - Line 24: Convergence code from the R optimization function `optim`. In this example, ‘0’ indicates successful completion.
 - Line 25: Number of optimization restarts in which the relative absolute maximum log-likelihood difference is $\leq 10^{-8}$. The algorithm exits after 2 such restarts, which is attained in this example.
 - Lines 30-33: Maximum likelihood estimates of errors-in-variables yields for the relevant benchmark sources. Source names (Lines 30 and 32) are given above yield estimates (Lines 31 and 33). Errors-in-variables yields are only estimated if `eiv` is `TRUE` (Line 126 of Appendix A.1).

Table 4: Inputs to `calc_bayes_cal` function.

Input	Default	Brief Description
<code>p_cal</code>	<code>none</code>	environment storing all objects needed in log-posterior calculations
<code>gdir</code>	<code>none</code>	directory location of global subroutines
<code>adir</code>	<code>none</code>	directory location of application subroutines
<code>nst</code>	<code>10</code>	number of starting values for log-posterior maximization
<code>nburn</code>	<code>10000</code>	number of per core MCMC burn-in samples
<code>nmcmc</code>	<code>20000</code>	number of MCMC production samples
<code>nthin</code>	<code>20</code>	posterior sample thinning rate
<code>ncor_map</code>	<code>1</code>	number of cores for log-posterior maximization
<code>ncor_mc</code>	<code>1</code>	number of cores for generating parallel MCMC chains
<code>igrad</code>	<code>TRUE</code>	forward model Jacobian provided
<code>igrck_pr</code>	<code>TRUE</code>	conduct log-prior function gradient verification
<code>igrck_po</code>	<code>TRUE</code>	conduct log-posterior function gradient verification
<code>bfgs</code>	<code>TRUE</code>	log-posterior maximization uses BFGS methods
<code>ibpr</code>	<code>FALSE</code>	prior density function(s) provided for signature within phenomenology forward model coefficients
<code>icpr</code>	<code>FALSE</code>	prior density function(s) provided for global forward model coefficients
<code>fpr.b</code>	<code>NULL</code>	location of functions computing log-prior density for signature within phenomenology forward model coefficients
<code>fgpr.b</code>	<code>NULL</code>	location of functions computing gradients of log-prior density for signature within phenomenology forward model coefficients
<code>fpr.c</code>	<code>NULL</code>	location of functions computing log-prior density for global forward model coefficients
<code>fgpr.c</code>	<code>NULL</code>	location of functions computing gradients of log-prior density for global forward model coefficients
<code>Xnom</code>	<code>NULL</code>	matrix of starting values for hyperparameters in log-posterior maximization if not generated by this function
<code>imcmc</code>	<code>"FME"</code>	MCMC algorithm (current options: <code>"RAM"</code> , <code>"FME"</code> , <code>"NUTS"</code>)
<code>pl</code>	<code>"multicore"</code>	strategy for running parallel jobs using the <code>future</code> package

- Lines 37-59: Maximum likelihood estimates of *common* forward model parameters for each signature of each phenomenology (where present).
- Lines 63-109: Maximum likelihood estimates of *emplacement-dependent* forward model parameters for each signature of each phenomenology (where present).
- Lines 113-127: Maximum likelihood estimates of *source* random effect (error model) variance component parameters for each signature of each phenomenology (where present).
- Lines 131-145: Maximum likelihood estimates of *path* random effect (error model) variance component parameters for each signature of each phenomenology (where present).

present).

- Lines 149-195: Maximum likelihood estimates of observational error variances for each signature, and correlations between signatures, for each phenomenology (WPA, §A.5).
- Line 197: Akaike Information Criterion¹¹ (AIC) value based on benchmark data. Used for selecting among competing forward or error model specifications (WPA, §5.5, p. 16; §5.6, Tables 5 and 6, pp. 19, 21).
- Line 199: Bayesian Information Criterion¹² (BIC) value based on benchmark data. Used for selecting among competing forward or error model specifications (WPA, §5.5, p. 16; §5.6, Tables 5 and 6, pp. 19, 21).
- Lines 204-247: Example of log-likelihood gradient verification at a single sampled parameter vector. Additional checks were deleted for brevity.
 - * Lines 205-224: Analytic gradient calculation
 - * Lines 226-245: Numerical gradient calculation using the R package `numDeriv`
 - * Line 247: Largest negative (first entry) and positive (second entry) differences between the analytic and numerical gradients
- Lines 260-1376: Output from the Bayesian analysis function `calc_bayes_cal`:
 - Line 263: Convergence code from the R optimization function `optim`. In this example, ‘0’ indicates successful completion.
 - Line 264: Number of optimization restarts in which the relative absolute maximum log-posterior difference is $\leq 10^{-8}$. The algorithm exits after 2 such restarts, which is attained in this example.
 - Lines 269-272: Maximum *a posteriori* estimates of errors-in-variables yields for the relevant benchmark sources. Source names (Lines 269 and 271) are given above yield estimates (Lines 270 and 272). Errors-in-variables yields are only estimated if `eiv` is `TRUE` (Line 126 of Appendix A.1).
 - Lines 276-298: Maximum *a posteriori* estimates of *common* forward model parameters for each signature of each phenomenology (where present).
 - Lines 302-348: Maximum *a posteriori* estimates of *emplacement-dependent* forward model parameters for each signature of each phenomenology (where present).
 - Lines 352-366: Maximum *a posteriori* estimates of *source* random effect (error model) variance component parameters for each signature of each phenomenology (where present).

¹¹Akaike, H. (1973). Information Theory and an Extension of the Maximum Likelihood Principle. In: Petrov, B.N. & Csaki, F., Eds., International Symposium on Information Theory, 267-281.

¹²Schwarz, G. (1978). Estimating the dimension of a model, *Ann Stat* 6:461-464.

- Lines 370-384: Maximum *a posteriori* estimates of *path* random effect (error model) variance component parameters for each signature of each phenomenology (where present).
- Lines 388-434: Maximum *a posteriori* estimates of observational error variances for each signature, and correlations between signatures, for each phenomenology (WPA, §A.5).
- Lines 438-440: Maximum *a posteriori* estimates of FGSN prior distribution parameters (WPA, §5.6, p. 26; $\mathbf{Alpha} = \mu$, $\mathbf{Omega} = v$ (two coefficients)).
- Lines 444-489: Example of log-prior gradient verification at a single sampled parameter vector. Additional checks were deleted for brevity.
 - * Lines 445-465: Analytic gradient calculation
 - * Lines 467-487: Numerical gradient calculation using the R package `numDeriv`
 - * Line 489: Largest negative (first entry) and positive (second entry) differences between the analytic and numerical gradients
- Lines 493-538: Example of log-posterior gradient verification at a single sampled parameter vector. Additional checks were deleted for brevity.
 - * Lines 494-514: Analytic gradient calculation
 - * Lines 516-536: Numerical gradient calculation using the R package `numDeriv`
 - * Line 538: Largest negative (first entry) and positive (second entry) differences between the analytic and numerical gradients
- Lines 542-571: Acceptance rates on each core of the Robust Adaptive Metropolis (RAM) posterior sampling method implemented in R package `adaptMCMC`. The target acceptance rate is 0.234.
- Lines 577-629: Means and user specified quantiles of samples from the marginal posterior distributions of errors-in-variables yields for the relevant benchmark sources. The ordering of benchmark sources is provided with the maximum *a posteriori* estimates (Lines 269 and 271). Errors-in-variables yields are only estimated if `eiv` is `TRUE` (Line 126 of Appendix A.1).
- Lines 633-722: Means and user specified quantiles of samples from the marginal posterior distributions of *common* forward model parameters for each signature of each phenomenology (where present).
- Lines 726-937: Means and user specified quantiles of samples from the marginal posterior distributions of *emplacement-dependent* forward model parameters for each signature of each phenomenology (where present).
- Lines 941-995: Means and user specified quantiles of samples from the marginal posterior distributions of *source* random effect (error model) variance component parameters for each signature of each phenomenology (where present).

- Lines 999-1053: Means and user specified quantiles of samples from the marginal posterior distributions of *path* random effect (error model) variance component parameters for each signature of each phenomenology (where present).
- Lines 1057-1331: Means and user specified quantiles of samples from the marginal posterior distributions of observational error variances for each signature, and correlations between signatures, for each phenomenology (WPA, §A.5).
- Lines 1335-1372: Means and user specified quantiles of samples from the marginal posterior distributions of FGSN prior distribution parameters (WPA, §5.6, p. 26; $\text{Alpha} = \mu$, $\text{Omega} = v$ (two coefficients)).
- Line 1374: DIC value based on benchmark data. Used for selecting among competing forward or error model specifications (WPA, §5.5, p. 16; §5.6, Tables 5 and 6, pp. 19, 21).
- Line 1376: PIC value based on benchmark data. Used for selecting among competing forward or error model specifications (WPA, §5.5, p. 16; §5.6, Tables 5 and 6, pp. 19, 21).

The `p_cal` environment resulting from this run contains several elements of potential interest for additional post-processing:

- `p_cal$mle_cal`: Maximum likelihood estimate of (if relevant) unbounded global forward model parameters (i.e., on scale used by the optimizer), benchmark source errors-in-variables yields (if relevant), and signature within phenomenology forward and error model parameters based on benchmark data
- `p_cal$mle_calp`: If `calp` is `TRUE` (Line 60 of Appendix A.1), maximum likelihood estimate of unbounded global forward model parameters based on benchmark data
- `p_cal$Sigma_mle_cal$II_calp`: If `calp` is `TRUE` (Line 60 of Appendix A.1), estimated asymptotic covariance matrix of `p_cal$mle_calp`, adjusted for estimation of signature within phenomenology forward model parameters, and (if relevant) benchmark source errors-in-variables yields
- `p_cal$map_cal`: If `iBayes` is `TRUE` (Line 6 of Appendix A.3), maximum *a posteriori* estimate of (if relevant) unbounded global forward model parameters (i.e., on scale used by the optimizer), benchmark source errors-in-variables yields (if relevant), and signature within phenomenology forward and error model parameters based on benchmark data
- `p_cal$map_calp`: If `iBayes` is `TRUE` (Line 6 of Appendix A.3), and `calp` is `TRUE` (Line 60 of Appendix A.1), maximum *a posteriori* estimate of unbounded global forward model parameters based on benchmark data
- `p_cal$mpi`: If `iBayes` is `TRUE` (Line 6 of Appendix A.3), posterior samples of unbounded global forward model parameters (if relevant), benchmark source errors-in-variables yields (if relevant), and signature within phenomenology forward and error model parameters based on benchmark data

- `p_cal$mpi_calp`: If `iBayes` is `TRUE` (Line 6 of Appendix A.3), and `calp` is `TRUE` (Line 60 of Appendix A.1), posterior samples of unbounded global forward model parameters based on benchmark data

2.2 Second Stage

The second stage analysis is defined in the `runMPEM_0.r` file, provided in Appendix A with line numbers referred to in the ensuing discussion. Appendix A.5 provides the preprocessing component of the second stage, Appendix A.6 provides the code employed to maximize the likelihood function of the new event data with respect to the new event device parameters, while Appendix A.7 provides the code employed to optionally sample the posterior distribution of these parameters.

The second stage analysis is run in batch mode as follows,

```
% R CMD BATCH runMPEM_0.r runMPEM_0.out &
```

This job requires the `.RData` file from the completion of the first stage run to be copied into the second stage run directory. The main features of the output file `runMPEM_0.out` are provided in Appendix A.8.

2.2.1 Preprocessing

The preprocessing component of the second stage analysis in Appendix A.5 is primarily responsible for describing features of the new event data and device parameters of inferential interest.

- Line 26+: Load all R packages utilized by multiple supporting subroutines, most notably log-likelihood and log-prior calculations and their associated gradients.
- Line 37: Specify directory location (relative to run directory) of all global (application independent) subroutines.
- Line 40: Read in code performing second stage preprocessing of new event data.
- Line 43: Specify directory location (relative to run directory) of all application-specific subroutines.
- Line 46: Specify root directory (relative to run directory) containing all application-specific new event data files.
- Lines 49-52: A scalar or vector specifying the names of new event data files for each phenomenology, utilizing an ordering of the phenomenologies (for MultiPEM analysis) that is consistent with first stage preprocessing and maintained throughout the input deck (as indicated here in Lines 55-58). Data files are text files (CSV formatted) containing all measured signatures (in the first column(s)) and input covariates (in succeeding column(s)) including all those required in forward and error model calculations, but excepting the new event device parameters that are unknown and subject to second stage inference. Directories specifying the exact locations of these files relative to the root data directory (Line 46) may also be included in the filenames.

- Line 61: Specify the names of the new event device parameters of inferential interest as a vector of strings. This information is utilized in postprocessing.
- Line 65: Number of first stage within signature forward and error model parameter posterior samples utilized in the multiple imputation algorithm for generating second stage new event device parameter posterior samples. If `nimpute` is set to 1 (default), the first stage maximum likelihood estimate of the within signature forward and error model parameters is used in place of the imputation samples.
- Line 68: Specify if bounded optimization of new event device parameters is to be conducted. The default is to optimize all new event device parameters on an unbounded input space, transforming them to their input domain (specified subsequently in this preprocessing file) as necessary for forward model calculations. If `opt_B` is `TRUE`, the new event device parameters are optimized directly on their input domain.
- Line 71: Indicate if the new event device parameters are subjected to a user-provided bijective transformation supplied to assist likelihood maximization or posterior sampling. If `itransform` is `TRUE`, the code implementing this transformation is concatenated into a single file named `transform.r` and placed in the application code directory; in this example,

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The functions that must be provided in `transform.r` include the following:

- `tau`: Function $\tau(\cdot)$ applied to transformed variables $\tilde{\boldsymbol{\theta}}_0$ with the new event device parameters $\boldsymbol{\theta}_0$ as its image,

$$\boldsymbol{\theta}_0 = \tau(\tilde{\boldsymbol{\theta}}_0)$$

- `j_tau`: Jacobian matrix of $\tau(\cdot)$,

$$\mathbf{J}_{\tau}(\tilde{\boldsymbol{\theta}}_0) = \begin{bmatrix} \frac{\partial \tau_1(\tilde{\boldsymbol{\theta}}_0)}{\partial \tilde{\theta}_{0,1}} & \dots & \frac{\partial \tau_1(\tilde{\boldsymbol{\theta}}_0)}{\partial \tilde{\theta}_{0,q}} \\ \vdots & \ddots & \vdots \\ \frac{\partial \tau_q(\tilde{\boldsymbol{\theta}}_0)}{\partial \tilde{\theta}_{0,1}} & \dots & \frac{\partial \tau_q(\tilde{\boldsymbol{\theta}}_0)}{\partial \tilde{\theta}_{0,q}} \end{bmatrix}$$

where q is the dimension of $\boldsymbol{\theta}_0$.

- `log_absdet_j_tau`: Logarithm of the absolute value of the determinant of the Jacobian matrix computed from `j_tau`,

$$\log \text{abs}(\det(\mathbf{J}_{\tau}(\tilde{\boldsymbol{\theta}}_0)))$$

- `dlog_absdet_j_tau`: Gradient of the log absolute Jacobian determinant with respect to $\tilde{\boldsymbol{\theta}}_0$
- `inv_tau`: Inverse function of $\tau(\cdot)$

In this example, the new event device parameters of inferential interest are log-yield w and height-of-burst h , that is $\theta_0 = (w, h)$. The relevant forward models are functions of a scaled height-of-burst, $\tilde{h} = h \exp(-w/3)$, suggesting the possible utility of likelihood maximization or posterior sampling in terms of $\tilde{\theta}_0 = (\tilde{w}, \tilde{h})$ for $\tilde{w} = w$.

- Lines 74-81: If `itransform` is TRUE (Line 71), and if `tPars` is TRUE (Line 75), initialize `tPars` to a null list (Line 78). Subsequent lines provide the value(s) for all fixed inputs required to compute the function `tau` (see previous item). In this example, a `yield_scaling` value is required (Line 79).
- Lines 85-88: Specify lower and upper bounds for the new event device parameters if needed. By default, lower bounds are set to $-\infty$ (Line 85) and upper bounds to $+\infty$ (Line 87). In this example, the second parameter (height-of-burst) is restricted to the range (0,160) (Lines 86 and 88). Note that likelihood maximization and posterior sampling are conducted on an unbounded parameter space. If lower or upper bounds are specified for any parameter, they are applied just prior to objective function calculations using the `transform` function of the `transform.r` file located in the global code directory,

`MultiPEM_Toolbox/Code`

- Lines 91-96: If `tsub` is TRUE (Line 91), the forward model for at least one phenomenology depends only on a subset of the full vector θ_0 of new event device parameters. The `tsub` object is initialized to a null list with elements for each phenomenology in the proper order (Line 94). The `theta_names` vector (Line 61) describes the order of elements in θ_0 . For relevant phenomenologies, parameter subsets are specified as integer vectors identifying the extracted elements of θ_0 . The forward models of all other phenomenologies depend on the full θ_0 . In this example, the *surface effects* (*crater*) phenomenology only depends on log-yield (Line 95), while the other phenomenologies depend on both log-yield and height-of-burst.
- Lines 99-101: Calls the preprocessing function `prepro_0` for the new event data. Table 5 describes all inputs to this function with default values. Only inputs with no default values must be provided.
- Lines 102-108: If `opt_B` is TRUE (Line 68), the preprocessor function `prepro_0` returns a list (designated here as `tmp`) with objects `p_cal` and `t_cal`, which are then assigned as follows

```
% p_cal = tmp$p_cal
% t_cal = tmp$t_cal
```

and both are utilized for maximum likelihood estimation and Bayesian analysis. Otherwise, `p_cal` is the only object returned,

```
% p_cal = tmp$p_cal
```

and utilized in subsequent analyses.

Table 5: Inputs to `prepro_0` function.

Input	Default	Brief Description
<code>p_cal</code>	none	environment storing all objects needed in log-likelihood and log-posterior calculations
<code>gdir</code>	none	directory location of global subroutines
<code>adir</code>	none	directory location of application subroutines
<code>rdir</code>	none	root directory location of data files
<code>ndir</code>	none	directory locations (if relevant) and names of new event data files under <code>rdir</code>
<code>tnames</code>	none	names of new event parameters
<code>nimp</code>	1	number of first stage imputation samples used in second stage new event parameter posterior sampling
<code>bopt</code>	FALSE	new event parameter bounds supplied to log-likelihood maximization
<code>itr</code>	FALSE	bijective transform of new event parameters provided
<code>fp_tr</code>	NULL	fixed inputs to new event parameter transform
<code>tlb</code>	NULL	lower bounds for new event parameters
<code>tub</code>	NULL	upper bounds for new event parameters
<code>tsub</code>	NULL	list containing index sets identifying new event parameter subsets by phenomenology if relevant

2.2.2 Maximum Likelihood Estimation

The maximum likelihood estimation component of the second stage analysis in Appendix A.6 is responsible for integrating calibrated within signature forward and error model parameter values from the first stage with new event data to estimate new event device parameters of interest with uncertainty quantification (WPA, §A.2).

- Line 6: Read in code performing second stage maximum likelihood estimation and uncertainty quantification of new event device parameters, using calibrated within signature forward and error model parameters from the first stage.
- Line 9: User specified seed to ensure repeatability of maximum likelihood estimation.
- Lines 13-17: Provide names of forward models for each signature by phenomenology (WPA, §5.1-5.4). The `fm0` object is initialized as a null list with elements for each phenomenology in the proper order (Line 13). Subsequent lines specify the function names as vectors of strings having length equal to the number of signatures for each phenomenology (Lines 14-17). The code for all forward models from each phenomenology is concatenated into a single file named `forward_0.r` and placed in the application code directory; in this example,

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Note that these forward models compute the same signatures as those used in the first stage. However, they accept only the new event device parameters of inferential interest (designated θ_0 previously) as their main argument. Forward model parameter values

are passed in as fixed quantities. In this example, the *seismic* forward model $f_{sr}^0(\cdot)$ as a function of the new event device parameters $\theta_0 = (w, h)$ – for fixed parameters β_{sr} – is given by Equation (1). The function `f0_s` returns a vector of forward model calculations evaluated for the supplied value of θ_0 (fixed β_{sr} passed in as `params$beta`), each element corresponding to each row of a matrix of covariates (in this case, a column vector of ranges δ).

- Line 20: Indicate if forward model Jacobian matrices are provided for efficient log-likelihood maximization.
- Lines 22-30: If `igrad` is `TRUE` (Line 20), names of forward model Jacobian functions must be provided for each signature by phenomenology. The `gfm0` object is initialized as a null list with elements for each phenomenology in the proper order (Line 25). Subsequent lines specify the Jacobian function names as vectors of strings having length equal to the number of signatures for each phenomenology (Lines 26-29). The code for all forward model Jacobian functions from each phenomenology is concatenated into a single file named `jacobian_0.r` and placed in the application code directory; in this example,

`MultiPEM_Toolbox/Applications/Code/IYDT-gsrp`

Note that these Jacobian functions accept only the new event device parameters of inferential interest (designated θ_0 previously) as their main argument. Forward model parameter values are passed in as fixed quantities. In this example, the gradient vector of the *seismic* forward model (see description above) is computed from the partial derivatives of $f_{sr}^0(\cdot)$ for each new event device parameter in θ_0 , for fixed parameters β_{sr} . The partial derivative for log-yield w is given in Equation (2). For HOB/DOB h ,

$$\frac{\partial f_{sr}^0}{\partial h} = \beta_{sr,3} \beta_{sr,4} \exp(-w/3) \times \text{logistic}(\beta_{sr,4} \tilde{h}_s + \beta_{sr,5}) \times \text{logistic}(-\beta_{sr,4} \tilde{h}_s - \beta_{sr,5})$$

The function `g0_s` returns a Jacobian matrix (`jtheta`) of forward model gradients for the new event device parameters, evaluated at the supplied value of θ_0 (fixed β_{sr} passed in as `params$beta`), with rows corresponding to the rows of a matrix of covariates (in this case, a column vector of ranges δ).

- Line 33: Specify the number of starting new event device parameter vectors for the log-likelihood maximization routine.
- Line 36: Specify the number of cores to use for parallel optimization (across distinct starting values) of the new event data log-likelihood function.
- Line 39: Specify if the Broyden-Fletcher-Goldfarb-Shanno (BFGS) algorithm is to be used for maximization of the log-likelihood function. If `TRUE`, functions to compute forward model Jacobian matrices analytically must be provided, or numerical gradients will be utilized (generally increasing compute time). If `FALSE`, the gradient-free Nelder-Mead simplex algorithm will be utilized for optimization, which is generally much slower computationally than BFGS with analytical gradients.

- Line 43: If relevant, specifies the location (relative to the run directory) of parameter values or estimates to be used as the first starting value for log-likelihood maximization. These values are stored in a `.RData` object as a list, with an element corresponding to new event device parameters (`theta0`).
- Line 47: If desired, name of output `.RData` file to store optimization results from this run. The element of the list to be written are described in the previous item.
- Lines 50-55: If `tst` is `TRUE` (Line 50), specifies an initial starting value for the new event device parameters (Lines 53-54) for log-likelihood maximization. This value supersedes the value read in from `opt_files_in` (Line 43), if provided.
- Line 58: Specify the level of confidence intervals computed for the true values of each new event device parameter from the maximum likelihood estimate and the estimated Fisher information matrix (WPA, §A.2, Equation (23); §A.4, Equation (25)).
- Line 61: Indicate if gradient verification is to be conducted on the log-likelihood function. If `TRUE` and `igrad` is `TRUE` (Line 20), analytical and numerical gradients at the optimal parameter value, and other randomly sampled parameter values, are compared for consistency.
- Line 64: Specify the strategy for running parallel jobs using the `future` package in R. The available options are given by starting an R session and issuing the following commands,


```
% R
> require(future)
> help(plan)
```
- Lines 67-71: Calls the log-likelihood maximization function `calc_mle_0` for the new event data. Table 6 describes all inputs to this function with default values. Only inputs with no default values must be provided.

2.2.3 Bayesian Analysis

The optional Bayesian inference component of the second stage analysis in Appendix A.7 is responsible for integrating calibrated within signature forward and error model parameter values from the first stage with new event data to sample new event device parameters of interest from their posterior distribution. Imputation of first stage parameters results in more complete uncertainty quantification, but is computationally more intensive than employing the maximum likelihood estimate (default). Estimates of the new event device parameters with uncertainty quantification are computed from the posterior samples.

- Line 6: Indicate if Bayesian analysis is to be conducted.
- Line 10: Read in code performing second stage Bayesian analysis on new event device parameters, using calibrated within signature forward and error model parameters from the first stage.

Table 6: Inputs to `calc_mle_0` function.

Input	Default	Brief Description
<code>p_cal</code>	none	environment storing all objects needed in log-likelihood calculations
<code>gdir</code>	none	directory location of global subroutines
<code>adir</code>	none	directory location of application subroutines
<code>f0</code>	none	names of forward model functions for each signature by phenomenology
<code>nst</code>	10	number of starting values for log-likelihood maximization
<code>ncor</code>	1	number of cores for log-likelihood maximization
<code>ci_lev</code>	0.95	confidence interval levels for new event parameter inference
<code>igrad</code>	TRUE	forward model Jacobian provided
<code>bfgs</code>	TRUE	log-likelihood maximization uses BFGS methods
<code>igrck</code>	TRUE	conduct log-likelihood function gradient verification
<code>t_cal</code>	NULL	object required if bounds supplied to log-likelihood maximization
<code>g0</code>	NULL	names of forward model Jacobian functions for each signature by phenomenology
<code>fopt_in</code>	NULL	location of input R data file providing an initial starting value for log-likelihood maximization
<code>Xst</code>	NULL	matrix of starting values for log-likelihood maximization if not generated by this function
<code>tst</code>	NULL	vector of starting values for new event parameters in log-likelihood maximization
<code>fopt_out</code>	NULL	location to write output R data file with results of log-likelihood maximization
<code>pl</code>	"multicore"	strategy for running parallel jobs using the <code>future</code> package

- Line 13: Indicate if a log-prior density for the new event device parameters is supplied by the user (WPA, §5.5, p. 15; §5.6, p. 21). If `iTheta0Prior` is `FALSE`, a “flat prior” (uniform on the domain) on these parameters is assumed.
- Lines 15-35: If relevant, specify details of user-provided log-prior distributions for new event device parameters.
 - Line 17: Specify location of log-prior function. If `NULL`, utilize the default log-prior function contained in the file `lp_0.r` placed in the application code directory; in this example,

```
MultiPEM_Toolbox/Applications/Code/IYDT-gsrp
```
 - Line 19: If `igrad` is `TRUE` (Line 20 of Appendix A.6), specify location of the log-prior gradient function. If `NULL`, utilize the default log-prior gradient function contained in the file `glp_0.r` placed in the application code directory; in this example,

- Line 23: Provide the name of the log-prior function.
- Line 24: If `igrad` is `TRUE` (Line 20 of Appendix A.6), provide the name of the log-prior gradient function.
- Lines 26-31: Specify all fixed quantities required for calculation of the log-prior density.
- Line 39: Select the Markov chain Monte Carlo (MCMC) algorithm to use for posterior sampling, from one of four options: `RAM`, `FME`, `NUTS`, and `SMC`. `RAM` is the robust adaptive Metropolis algorithm of Vihola¹³ implemented in the R package `adaptMCMC`. `FME` is the delayed rejection adaptive Metropolis algorithm of Haario, Laine, and Mira¹⁴ implemented in the R package `FME`. `NUTS` is the No-U-Turn Sampler of Hoffman and Gelman¹⁵. The `NUTS` option requires the analytical gradient of the log-posterior density, which in turn requires `igrad` to be `TRUE` (Line 20 of Appendix A.6). `SMC` is a Sequential Monte Carlo (SMC) method adapted for sampling challenging posterior distributions (e.g. multi-modal) of low-dimensional parameter spaces¹⁶.
- Line 42: Specify the per core sample size of the burn-in period for MCMC sampling (pre-equilibrium stage of Markov chain). These samples are discarded prior to any inference using the posterior samples.
- Line 45: Specify the sample size of the MCMC production run. These samples are kept for posterior inference.
- Line 48: Specify the rate at which MCMC production samples are thinned when multiple imputation of first stage parameters is invoked for improved uncertainty quantification of second stage parameters.
- Line 51: Specify the number of cores to use for parallel optimization (across distinct starting values) of the new event data log-posterior function. By default, this optimization uses the same number of distinct starting values as optimization of the new event data log-likelihood function (Line 33 of Appendix A.6).
- Line 54: Specify the number of cores used to run multiple imputations simultaneously or parallel MCMC chains for a single imputation. The burn-in period for each chain is determined by `nburn` (Line 42), while `nmcmc` (Line 45) production runs are generated for each imputed first stage parameter value, or split between the `ncores_mc` processors and combined at the conclusion of the runs for a single imputation.
- Line 57: Indicate if gradient verification is to be conducted on the log-prior function. If

¹³Vihola, M. (2012). Robust adaptive Metropolis algorithm with coerced acceptance rate. *Stat Comput* 22:997-1008.

¹⁴Haario, H., Laine, M., and Mira, A. (2006). DRAM: Efficient adaptive MCMC. *Stat Comput* 16:339-354.

¹⁵Hoffman, M. D. and Gelman, A. (2014). The No-U-Turn Sampler: Adaptively setting path lengths in Hamiltonian Monte Carlo. *J Mach Learn Res* 15:1593-1623.

¹⁶Golchi, S. and Loepky, J.L. (2016). Monte Carlo based Designs for Constrained Domains. arXiv:1512.07328v2 [stat.ME], 8 Aug. 2016.

TRUE and `igrad` is TRUE (Line 20 of Appendix A.6), analytical and numerical gradients at the maximum *a posteriori* parameter value, and other randomly sampled parameter values, are compared for consistency.

- Line 60: Indicate if gradient verification is to be conducted on the log-posterior function. If TRUE and `igrad` is TRUE (Line 20 of Appendix A.6), analytical and numerical gradients at the maximum *a posteriori* parameter value, and other randomly sampled parameter values, are compared for consistency.
- Line 65: If `iMCMC` is "SMC" (Line 39), specify the number of cores to be used in the inner parallelization of the Sequential Monte Carlo (SMC) code for posterior sampling of the new event device parameters. SMC is advantageous if the posterior distribution of these parameters is multi-modal.
- Line 67: Lower bounds of new event device parameters (on infinite domain) for SMC sampling. For infinite values, lower bounds are determined from the maximum likelihood estimate (MLE) of these parameters and its uncertainty.
- Line 68: Upper bounds of new event device parameters (on infinite domain) for SMC sampling. For infinite values, upper bounds are determined from the MLE of these parameters and its uncertainty.
- Lines 71-79: Calls the Bayesian analysis function `calc_bayes_0` for the new event data. Table 7 describes all inputs to this function with default values. Only inputs with no default values must be provided.

2.2.4 Output

The output file `runMPEM_0.out` from the second stage analysis contains a summary of new event device parameter estimates and (if relevant) posterior samples derived from the new event data, based on fixed (or multiply imputed) within signature forward and error model parameter values from the first stage analysis derived from the benchmark data. The desired output is supplied by the user function `print_sumstats_0.r`, placed in the application code directory; in this example,

MultiPEM_Toolbox/Applications/Code/IYDT-gsrp

The output presented in Appendix A.8 contains the most pertinent information extracted from the full file.

- Lines 7-63: Output from the maximum likelihood estimation function `calc_mle_0`:
 - Line 9: Convergence code from the R optimization function `optim`. In this example, '0' indicates successful completion.
 - Line 10: Number of optimization restarts in which the relative absolute maximum log-likelihood difference is $\leq 10^{-8}$. The algorithm exits after 2 such restarts, which is attained in this example.
 - Line 18: Maximum likelihood estimates of the new event device parameters; in this example, log-yield (`W`) and height-of-burst (`HOB`).

Table 7: Inputs to `calc_bayes_0` function.

Input	Default	Brief Description
<code>p_cal</code>	none	environment storing all objects needed in log-posterior calculations
<code>gdir</code>	none	directory location of global subroutines
<code>adir</code>	none	directory location of application subroutines
<code>nst</code>	10	number of starting values for log-posterior maximization
<code>nburn</code>	10000	number of per core MCMC burn-in samples
<code>nmcmc</code>	20000	number of MCMC production samples
<code>nthin</code>	1	posterior sample thinning rate per imputation, for multiple imputation
<code>ncor_map</code>	1	number of cores for log-posterior maximization
<code>ncor_mc</code>	1	number of cores for multiple imputation, or for generating parallel MCMC chains if single imputation
<code>igrad</code>	TRUE	forward model Jacobian provided
<code>igrck_pr</code>	TRUE	conduct log-prior function gradient verification
<code>igrck_po</code>	TRUE	conduct log-posterior function gradient verification
<code>bfgs</code>	TRUE	log-posterior maximization uses BFGS methods
<code>itpr</code>	FALSE	prior density function provided for new event parameters
<code>fpr_t</code>	NULL	location of function computing log-prior density for new event parameters
<code>fgpr_t</code>	NULL	location of function computing gradients of log-prior density for new event parameters
<code>imcmc</code>	"FME"	MCMC algorithm (current options: "RAM", "FME", "NUTS", "SMC")
<code>pl</code>	"multicore"	strategy for running parallel jobs using the <code>future</code> package
<code>ncor_smc</code>	NULL	number of cores for inner parallelization of SMC code
<code>lb_smc</code>	NULL	lower bounds of new event parameters for SMC sampling
<code>ub_smc</code>	NULL	upper bounds of new event parameters for SMC sampling
<code>t_cal</code>	NULL	object required if bounds supplied to log-posterior maximization

- Line 23: Standard errors of the maximum likelihood estimates of the new event device parameters, assuming the forward model parameters and (if relevant) benchmark source errors-in-variables yields are known with certainty (set to their first stage values) (WPA, §A.2, p. 42, calculated from $(\mathcal{I}_{\theta_0, \theta_0}^0)^{-1}$).
- Lines 27-29: Correlation matrix of the maximum likelihood estimates of the new event device parameters, assuming the forward model parameters and (if relevant) benchmark source errors-in-variables yields are known with certainty (set to their first stage values) (WPA, §A.2, p. 42, calculated from $(\mathcal{I}_{\theta_0, \theta_0}^0)^{-1}$).
- Line 34: Standard errors of the maximum likelihood estimates of the new event device parameters, adjusted for estimation of the forward model parameters and

- (if relevant) benchmark source errors-in-variables yields in the first stage (WPA, §A.2, Equation (23); §A.4, Equation (25)).
- Lines 38-40: Correlation matrix of the maximum likelihood estimates of the new event device parameters, adjusted for estimation of the forward model parameters and (if relevant) benchmark source errors-in-variables yields in the first stage (WPA, §A.2, Equation (23); §A.4, Equation (25)).
 - Lines 45-46: 95% confidence intervals for the unknown true values of the new event device parameters, based on standard errors assuming the forward model parameters and (if relevant) benchmark source errors-in-variables yields are known with certainty (set to their first stage values) (WPA, §A.2, p. 42, calculated from $(\mathcal{I}_{\theta_0, \theta_0}^0)^{-1}$).
 - Lines 51-52: 95% confidence intervals for the unknown true values of the new event device parameters, based on standard errors adjusted for estimation of the forward model parameters and (if relevant) benchmark source errors-in-variables yields in the first stage (WPA, §A.2, Equation (23); §A.4, Equation (25)).
 - Lines 58-63: Example of log-likelihood gradient verification at a single sampled parameter vector. Additional checks were deleted for brevity.
 - * Line 59: Analytic gradient calculation
 - * Line 61: Numerical gradient calculation using the R package `numDeriv`
 - * Line 63: Largest negative (first entry) and positive (second entry) differences between the analytic and numerical gradients
 - Lines 75-137: Output from the Bayesian analysis function `calc_bayes_0`:
 - Line 77: Convergence code from the R optimization function `optim`. In this example, ‘0’ indicates successful completion.
 - Line 78: Number of optimization restarts in which the relative absolute maximum log-posterior difference is $\leq 10^{-8}$. The algorithm exits after 2 such restarts, which is attained in this example.
 - Line 86: Maximum *a posteriori* estimates of the new event device parameters.
 - Lines 91-96: Example of log-prior gradient verification at a single sampled parameter vector. Additional checks were deleted for brevity.
 - * Line 92: Analytic gradient calculation
 - * Line 94: Numerical gradient calculation using the R package `numDeriv`
 - * Line 96: Largest negative (first entry) and positive (second entry) differences between the analytic and numerical gradients
 - Lines 100-105: Example of log-posterior gradient verification at a single sampled parameter vector. Additional checks were deleted for brevity.

- * Line 101: Analytic gradient calculation
- * Line 103: Numerical gradient calculation using the R package `numDeriv`
- * Line 105: Largest negative (first entry) and positive (second entry) differences between the analytic and numerical gradients
- Lines 109-113: Acceptance rates of the Delayed Rejection Adaptive Metropolis (DRAM) posterior sampling method implemented in R package `FME` for each imputation of first stage parameters. Note that one delayed rejection step is allowed in the default implementation.
- Line 119: Means of samples from the new event device parameter marginal posterior distributions.
- Line 121: Standard deviations of samples from the new event device parameter marginal posterior distributions.
- Lines 123-131: User specified quantiles of samples from the new event device parameter marginal posterior distributions.
- Lines 135-137: Correlation matrix of samples from the new event device parameter joint posterior distribution.

The `p_cal` environment resulting from this run contains several elements of potential interest for additional post-processing:

- `p_cal$mle`: Maximum likelihood estimate of unbounded new event device parameters (i.e., on scale used by the optimizer)
- `p_cal$Sigma_mle_0$II_nev_it`: Estimated asymptotic covariance matrix of `p_cal$mle`, adjusted for first stage estimation of quantities stated below
- `p_cal$Sigma_mle_0$II_nev_0_it`: Estimated asymptotic covariance matrix of `p_cal$mle`, assuming first stage estimates of quantities stated below are known with certainty
- `p_cal$tmle_0`: Maximum likelihood estimate of transformed new event device parameters (i.e., on correct scale)
- `p_cal$Sigma_mle_0$II_nev`: Estimated asymptotic covariance matrix of `p_cal$tmle_0`, adjusted for first stage estimation of quantities stated below
- `p_cal$Sigma_mle_0$II_nev_0`: Estimated asymptotic covariance matrix of `p_cal$tmle_0`, assuming first stage estimates of quantities stated below are known with certainty
- `p_cal$Sigma_mle_0$II_calp`: If `calp` is `TRUE` (Line 60 of Appendix A.1), estimated asymptotic covariance matrix of `p_cal$mle_calp`, adjusted for estimation of new event device parameters and first stage estimation of quantities stated below
- `p_cal$map`: If `iBayes` is `TRUE` (Line 6 of Appendix A.7), maximum *a posteriori* estimate of unbounded new event device parameters (i.e., on scale used by the optimizer)

- `p_cal$tmapi_0`: If `iBayes` is `TRUE` (Line 6 of Appendix A.7), maximum *a posteriori* estimate of transformed new event device parameters (i.e., on correct scale)
- `p_cal$mpi`: For multiple imputation (`nimpute` > 1; Line 65 of Appendix A.5)), first stage posterior samples of (if relevant) unbounded global forward model parameters, (if relevant) benchmark source errors-in-variables yields, and signature within phenomenology forward and error model parameters based on benchmark data, used as second stage imputation values of these parameters if `iBayes` is `TRUE` (Line 6 of Appendix A.7)
- `p_cal$tmapi_0`: If `iBayes` is `TRUE` (Line 6 of Appendix A.7), posterior samples of transformed new event device parameters (i.e., on correct scale)

The maximum likelihood-based quantities use maximum likelihood estimates for the forward and error model parameters and (if relevant) the benchmark source errors-in-variables yields from the first stage analysis.

3 Complete Assessment

Complete assessments will be illustrated by examining the run files associated with a multi-phenomenology analysis in which signals from four phenomenologies are combined to infer the log-yield and height-of-burst (HOB) of a near-surface nuclear explosion (WPA, §5).

```
% cd ./Runfiles/IYDT-gsrp/4-Phen/I-EIV-SUGAR-hob
```

Complete assessments involve combining benchmark and (if relevant) new event data to simultaneously estimate global (if relevant) and signature within phenomenology forward model parameters (e.g. regression coefficients), error model parameters (e.g. source bias, path bias, observational error covariance), errors-in-variables yield values of benchmark sources (if relevant), and (if relevant) new event device parameters (e.g. yield, HOB/DOB, geolocation, event time) with uncertainty quantification. Complete assessments are more computationally intensive than rapid assessments, as they require that all parameters are inferred simultaneously for each new event of interest.

The analysis is defined in the `runMPEM.r` file, provided in Appendix B with line numbers referred to in the ensuing discussion. Appendix B.1 provides the preprocessing component, Appendix B.2 provides the code employed to maximize the likelihood function of the data with respect to all of the forward model, error model, (if relevant) errors-in-variables yield, and (if relevant) new event device parameters, while Appendix B.3 provides the code employed to optionally sample the posterior distribution of these parameters.

The complete analysis is run in batch mode as follows,

```
% R CMD BATCH runMPEM.r runMPEM.out &
```

The main features of the output file `runMPEM.out` are provided in Appendix B.4.

3.0.1 Preprocessing

The preprocessing component of the analysis in Appendix B.1 is primarily responsible for describing features of the benchmark and (if relevant) new event data and all parameters of inferential interest.

- Line 25+: Load all R packages utilized by multiple supporting subroutines, most notably log-likelihood and log-prior calculations and their associated gradients.
- Line 36: Specify directory location (relative to run directory) of all global (application independent) subroutines.
- Line 39: Read in code performing preprocessing of benchmark and (if relevant) new event data.
- Line 42: Specify directory location (relative to run directory) of all application-specific subroutines.
- Line 45: Specify root directory (relative to run directory) containing all application-specific benchmark and (if relevant) new event data files.

- Lines 48-51: A scalar or vector specifying the names of benchmark data files for each phenomenology, utilizing an ordering of the phenomenologies (for MultiPEM analysis) that is maintained throughout the input deck (as indicated here in Lines 54-57). Data files are text files (CSV formatted) containing all measured signatures (in the first column(s)) and input covariates (in succeeding column(s)) including all those required in forward and error model calculations. Directories specifying the exact locations of these files relative to the root data directory (Line 45) may also be included in the filenames.
- Line 60: Indicate if forward model parameters global to multiple signatures within phenomenology or across phenomenologies will be modeled. If **TRUE**, nominal values for these parameters may optionally be placed in the benchmark data file(s). If a subset of sources are to be assigned default values for some or all of these parameters, the value **NA** should be assigned to these parameters in the benchmark data file(s) for these sources, and the default values provided in the relevant forward model(s).
- Lines 62-65: If **calp** is **TRUE** (Line 60), provide a string vector of names for each of the global forward model parameters (Line 64).
- Line 68: A scalar or vector specifying the number of observed signatures for each phenomenology; in this example, 2 for each phenomenology.
- Lines 72-79: Specify the number of *common* forward model parameters within each phenomenology (WPA, §4.1, first paragraph). For a given forward model, common parameters maintain the same constant value within signature for every log-likelihood calculation. The **pbeta** object is initialized as a null list with elements for each phenomenology in the proper order (Line 72), initialized to zero vectors of length equal to the number of observed signatures (Line 73). If relevant, subsequent lines specify the number of common forward model parameters for each signature within each phenomenology. For example, the *acoustic* forward model for each signature contains 2 common forward model parameters (Line 75).
- Lines 82-85: Specify if the forward model(s) for any phenomenology depend on event emplacement conditions (Line 82), followed by (if relevant) a vector indicating the number of distinct emplacement conditions considered for each phenomenology in the proper order (Line 84). This specification allows distinct forward model parameters to be associated with different emplacement conditions (as specified subsequently). If **Th** is **TRUE** (Line 82), a factor named **Type** must be present in the benchmark and (if relevant) new event data files for each relevant phenomenology, indicating the emplacement condition pertaining to each entry. In this example, the *seismic* and *acoustic* forward model parameters may vary for 3 distinct emplacements (“soft”, “hard”, and “wet” rock types), while the *optical* and *surface effects* forward models are independent of emplacement condition.
- Lines 89-104: Specify the number of *emplacement* dependent forward model parameters within each phenomenology if relevant (WPA, §4.1, first paragraph). For a given forward model, emplacement parameters remain constant within signature for log-likelihood calculations with a given emplacement condition, but may be modified within

signature for each distinct emplacement. The `pbetat` object is initialized as a null list with elements for each phenomenology in the proper order (Line 90), initialized as null lists with elements for each emplacement condition (Line 92) if multiple emplacements are present. Subsequent lines specify the number of forward model parameters for each signature within each emplacement condition for each phenomenology. For example, the *seismic* forward model for each signature within each emplacement contains 5 forward model parameters (Line 97) allowed to vary across emplacements. `pbetat` must be specified if multiple emplacements are present for any phenomenology (at least one element of `Th` is greater than 1).

- Lines 108-123: Specify the location of *common* forward model parameters within the full parameter vector, for phenomenologies possessing both common and emplacement dependent parameters. The `ibetar` object is initialized as a null list with elements for each phenomenology in the proper order (Line 109), initialized as null lists with elements for each signature within each emplacement condition (Line 114) if multiple emplacements are present. Subsequent lines specify the position of common parameters in the full forward model parameter vector, for phenomenologies possessing both common and emplacement dependent forward model parameters. For example, the *acoustic* forward model parameter vector takes common parameter values in its first two positions for each signature within each emplacement condition (Line 120).
- Line 126: Indicate if errors-in-variables yield values for benchmark events will be modeled (WPA, §2, Equation (3); §A.4). If `TRUE`, this allows uncertain yields for benchmark events (often assumed known with certainty) to vary within user-specified guidelines.
- Lines 129-145: If relevant, specify details of errors-in-variables yield models for benchmark events.
 - Line 132: Specify phenomenologies for application of errors-in-variables yield models to benchmark events
 - Lines 136-137: Provide the sources subject to errors-in-variables yield models for each phenomenology. The `seiv` object is initialized as a null list with elements for each phenomenology in the proper order (Line 136), with vectors indicating the relevant sources for each relevant phenomenology (Line 137). The "ALL" designation indicates that every source in the benchmark data set for the indicated phenomenology will be modeled with an errors-in-variables yield. `seiv` must be specified if `ieiv` is provided.
 - Line 140: The standard deviation of the errors-in-variables Gaussian distribution for each benchmark event log-yield. For each event, the mean of this distribution is taken to be its provided (design or measured) log-yield. In this example, a "total" error (3 standard deviations) of 30% in each provided yield is allowed. Note that this error is relative because yields are treated on a logarithmic scale. `eiv_w_sd` must be specified if `ieiv` is provided.
- Lines 149-158: Specify if *source* random effects (WPA, §2, Equation (2); §3; §A.5) should be included in the error model (Line 149). If so, the `pvc_1` object is initialized

as a null list with elements for each phenomenology in the proper order (Line 152), initialized to zero vectors of length equal to the number of observed signatures (Line 153). Subsequent lines specify the number of source random effects for each signature within each phenomenology. For example, the *seismic* error model for each signature contains a single source bias term (Line 155). If `pvc_1` is `TRUE` (Line 149), a factor named `Source` may be provided in the benchmark and (if relevant) new event data files for each relevant phenomenology, identifying the source pertaining to each entry. This factor must be present if there is more than one data entry for any source. In order to include source random effects in the error model for an observed signature, the benchmark data must contain more than one source, with at least one source containing more than one observation. A warning message will be printed to the log file if one of these conditions is violated.

- Lines 161-177: Specify if *path* random effects (WPA, §2, Equation (2); §3; §A.5), also referred to as *station* random effects, should be included in the error model (Line 161). If so, the `pvc_2` object is initialized as a null list with elements for each phenomenology in the proper order (Line 164), initialized to zero vectors of length equal to the number of observed signatures (Line 165). Subsequent lines specify the number of path random effects for each signature within each phenomenology. For example, the *seismic* error model for each signature contains a single path bias term (Line 167). The type of path random effect desired is specified by the `p_type` object, initialized as a null list with elements for each phenomenology in the proper order (Line 172). In this application, both the seismic and acoustic error models contain crossed path random effects (Lines 174 and 176). If `pvc_2` is `TRUE` (Line 161), a factor named `Path` must be provided in the benchmark (and new event) data file for each relevant phenomenology, identifying the source-to-sensor path pertaining to each entry. In order to include path random effects in the error model for an observed signature, the benchmark data must contain more than one path, with at least one path containing more than one observation. Additionally, specification of crossed path effects (`p_type` is `"Crossed"`) requires the signature to be observed from at least one common path for two or more sources, while specification of nested (within source) path effects requires more than one path for at least one source, with more than one observation for at least one of those paths. A warning message will be printed to the log file if one of these conditions is violated.
- Line 181: Indicate if the user is providing code to compute coefficient matrices for *source* or *path* random effects (WPA, §4.1). If `FALSE`, the function `calc_zmat.r` located in the global code directory,

`MultiPEM_Toolbox/Code`

computes default coefficient matrices for the benchmark and (if relevant) new event data. If `TRUE`, then a user-provided function of the same name must be placed in the application code directory; in this example,

`MultiPEM_Toolbox/Applications/Code/IYDT-gsrp`

Table 8 shows notional data for two *seismic* benchmark sources.

Table 8: Data for seismic benchmark sources HRI-1 and HRI-2.

Y1	Y2	Source	Path	Type	lRange	W	HOB
-15.091	-9.252	HRI-1	Path_1	1	6.932	6.291	5
-15.089	-9.180	HRI-1	Path_1	1	6.932	6.291	5
-15.836	-10.218	HRI-1	Path_2	1	7.570	6.291	5
-15.892	-10.180	HRI-1	Path_2	1	7.570	6.291	5
-16.176	-10.557	HRI-1	Path_2	1	7.800	6.291	5
-16.907	-11.366	HRI-1	Path_2	1	8.371	6.291	5
-16.931	-11.338	HRI-1	Path_2	1	8.371	6.291	5
-14.835	-9.199	HRI-2	Path_1	1	6.930	6.291	3
-14.860	-9.184	HRI-2	Path_1	1	6.930	6.291	3
-15.674	-10.089	HRI-2	Path_1	1	7.568	6.291	3
-15.754	-10.197	HRI-2	Path_1	1	7.568	6.291	3
-16.002	-10.530	HRI-2	Path_2	1	7.802	6.291	3
-16.060	-10.605	HRI-2	Path_2	1	7.802	6.291	3
-16.534	-11.115	HRI-2	Path_2	1	8.239	6.291	3
-16.741	-11.230	HRI-2	Path_3	1	8.373	6.291	3
-16.737	-11.288	HRI-2	Path_3	1	8.373	6.291	3
-17.208	-11.656	HRI-2	Path_3	1	8.738	6.291	3

If source and nested (within source) path random effects are included in the error model, the source and path bias vectors (WPA, §4.1, p. 7) associated with these sources (with HRI-1 and HRI-2 designated as 1 and 2) are given by

$$\begin{aligned}
\mathbf{E}_{S,11r} &= \begin{pmatrix} \mathbf{Z}_{11r,1} \\ \mathbf{Z}_{11r,2} \end{pmatrix} b_{1r,1}^{(S)} & \mathbf{E}_{P,11r} &= \begin{pmatrix} \mathbf{E}_{P,111r} \\ \mathbf{E}_{P,112r} \end{pmatrix} = \begin{bmatrix} \mathbf{Z}_{111r} & \mathbf{0}_2 \\ \mathbf{0}_5 & \mathbf{Z}_{112r} \end{bmatrix} \begin{pmatrix} b_{1r,11}^{(P)} \\ b_{1r,12}^{(P)} \end{pmatrix} \\
\mathbf{E}_{S,12r} &= \begin{pmatrix} \mathbf{Z}_{12r,1} \\ \mathbf{Z}_{12r,2} \\ \mathbf{Z}_{12r,3} \end{pmatrix} b_{1r,2}^{(S)} & \mathbf{E}_{P,12r} &= \begin{pmatrix} \mathbf{E}_{P,121r} \\ \mathbf{E}_{P,122r} \\ \mathbf{E}_{P,123r} \end{pmatrix} = \begin{bmatrix} \mathbf{Z}_{121r} & \mathbf{0}_4 & \mathbf{0}_4 \\ \mathbf{0}_3 & \mathbf{Z}_{122r} & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{Z}_{123r} \end{bmatrix} \begin{pmatrix} b_{1r,21}^{(P)} \\ b_{1r,22}^{(P)} \\ b_{1r,23}^{(P)} \end{pmatrix},
\end{aligned}$$

where the default coefficient matrices are given by

$$\begin{aligned}
\mathbf{Z}_{11r,1} &= \mathbf{1}_2 & \mathbf{Z}_{111r} &= \mathbf{1}_2 \\
\mathbf{Z}_{11r,2} &= \mathbf{1}_5 & \mathbf{Z}_{112r} &= \mathbf{1}_5 \\
\mathbf{Z}_{12r,1} &= \mathbf{1}_4 & \mathbf{Z}_{121r} &= \mathbf{1}_4 \\
\mathbf{Z}_{12r,2} &= \mathbf{1}_3 & \mathbf{Z}_{122r} &= \mathbf{1}_3 \\
\mathbf{Z}_{12r,3} &= \mathbf{1}_3 & \mathbf{Z}_{123r} &= \mathbf{1}_3
\end{aligned}$$

for $\mathbf{1}_q$ and $\mathbf{0}_q$ the q -vectors of ones and zeros, respectively. The source and path random effects $\{b_{1r,1}^{(S)}, b_{1r,2}^{(S)}\}$ and $\{b_{1r,11}^{(P)}, b_{1r,12}^{(P)}, b_{1r,21}^{(P)}, b_{1r,22}^{(P)}, b_{1r,23}^{(P)}\}$ are mutually independent realizations of their respective random effects distributions (WPA, §4.1, Equation (5), p. 7). For each signature, this structure indicates that there is a single source bias effect applied to every observation within each source, while observations from each path are

adjusted by distinct (and independently distributed) path bias effects (signatures are collected from two and three paths respectively for HRI-1 and HRI-2).

If instead a signature is observed from one or more common source-to-sensor paths across two or more sources, referred to as “crossed paths” (**ptype** is “Crossed” (Line 172)) – assuming the source-to-sensor paths observed for the sources HRI-1 and HRI-2 are not present for any other source – the path bias vectors ($\mathbf{E}_{P,11r}, \mathbf{E}_{P,12r}$) corresponding to the two sources above are replaced by the single path bias vector

$$\mathbf{E}_{P,1\{1,2\}r} = \begin{bmatrix} \mathbf{Z}_{111r} & \mathbf{0}_2 & \mathbf{0}_2 \\ \mathbf{0}_5 & \mathbf{Z}_{112r} & \mathbf{0}_5 \\ \mathbf{Z}_{121r} & \mathbf{0}_4 & \mathbf{0}_4 \\ \mathbf{0}_3 & \mathbf{Z}_{122r} & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{Z}_{123r} \end{bmatrix} \begin{pmatrix} b_{1r,\{1,2\}1}^{(P)} \\ b_{1r,\{1,2\}2}^{(P)} \\ b_{1r,\{1,2\}3}^{(P)} \end{pmatrix}.$$

The entry $\{1, 2\}$ for the source index indicates that sources 1 (HRI-1) and 2 (HRI-2) must be considered jointly as a group, due to covariance between their observed signatures induced by the common source-to-sensor propagation paths **Path_1** and **Path_2**. The path random effects $\{b_{1r,\{1,2\}1}^{(P)}, b_{1r,\{1,2\}2}^{(P)}, b_{1r,\{1,2\}3}^{(P)}\}$ are mutually independent realizations of the path random effect distribution (WPA, §4.1, Equation (5), p. 7).

- Line 185: Specify if bounded optimization of any unknown parameters is to be conducted. This option is currently only supported for new event device parameters. The default is to optimize all new event device parameters on an unbounded input space, transforming them to their input domain (specified subsequently in this preprocessing file) as necessary for forward model calculations. If **opt_B** is **TRUE**, the new event device parameters are optimized directly on their input domain.
- Line 188: Indicate if new event device parameters are to be estimated with uncertainty quantification simultaneously from the benchmark and new event data. If **nev** is **FALSE**, only forward and error model parameters, and benchmark source errors-in-variables yields (if relevant), are inferred from the benchmark data.
- Lines 191-236: If relevant, specify details of new event device parameters and location(s) of new event data.
 - Lines 193-196: A scalar or vector specifying the names of new event data files for each phenomenology, utilizing an ordering of the phenomenologies (for MultiPEM analysis) that is consistent with the benchmark data files and maintained throughout the input deck (as indicated here in Lines 54-57). Data files are text files (CSV formatted) containing all measured signatures (in the first column(s)) and input covariates (in succeeding column(s)) including all those required in forward and error model calculations, but excepting the new event device parameters that are unknown and subject to inference. Directories specifying the exact locations of these files relative to the root data directory (Line 45) may also be included in the filenames. Must be provided if **nev** is **TRUE** (Line 188).
 - Line 199: Specify the names of the new event device parameters of inferential interest as a vector of strings. This information is utilized in postprocessing, and

must be provided if `nev` is `TRUE` (Line 188).

- Line 202: Indicate if the new event device parameters are subjected to a user-provided bijective transformation supplied to assist likelihood maximization or posterior sampling. If `itransform` is `TRUE`, the code implementing this transformation is concatenated into a single file named `transform.r` and placed in the application code directory; in this example,

`MultiPEM_Toolbox/Applications/Code/IYDT-gsrp`

The functions that must be provided in `transform.r` include the following:

- * `tau`: Function $\tau(\cdot)$ applied to transformed variables $\tilde{\theta}_0$ with the new event device parameters θ_0 as its image,

$$\theta_0 = \tau(\tilde{\theta}_0)$$

- * `j_tau`: Jacobian matrix of $\tau(\cdot)$,

$$\mathbf{J}_\tau(\tilde{\theta}_0) = \begin{bmatrix} \frac{\partial \tau_1(\tilde{\theta}_0)}{\partial \tilde{\theta}_{0,1}} & \dots & \frac{\partial \tau_1(\tilde{\theta}_0)}{\partial \tilde{\theta}_{0,q}} \\ \vdots & \ddots & \vdots \\ \frac{\partial \tau_q(\tilde{\theta}_0)}{\partial \tilde{\theta}_{0,1}} & \dots & \frac{\partial \tau_q(\tilde{\theta}_0)}{\partial \tilde{\theta}_{0,q}} \end{bmatrix}$$

where q is the dimension of θ_0 .

- * `log_absdet_j_tau`: Logarithm of the absolute value of the determinant of the Jacobian matrix computed from `j_tau`,

$$\log \text{abs}(\det(\mathbf{J}_\tau(\tilde{\theta}_0)))$$

- * `dlog_absdet_j_tau`: Gradient of the log absolute Jacobian determinant with respect to $\tilde{\theta}_0$

- * `inv_tau`: Inverse function of $\tau(\cdot)$

In this example, the new event device parameters of inferential interest are log-yield w and height-of-burst h , that is $\theta_0 = (w, h)$. The relevant forward models are functions of a scaled height-of-burst, $\tilde{h} = h \exp(-w/3)$, suggesting the possible utility of likelihood maximization or posterior sampling in terms of $\tilde{\theta}_0 = (\tilde{w}, \tilde{h})$ for $\tilde{w} = w$.

- Lines 205-212: If `itransform` is `TRUE` (Line 202), and if `tPars` is `TRUE` (Line 206), initialize `tPars` to a null list (Line 209). Subsequent lines provide the value(s) for all fixed inputs required to compute the function `tau` (see previous item). In this example, a `yield_scaling` value is required (Line 210).

- Lines 216-219: Specify lower and upper bounds for the new event device parameters if needed. By default, lower bounds are set to $-\infty$ (Line 216) and upper bounds to $+\infty$ (Line 218). In this example, the second parameter (height-of-burst) is restricted to the range (0, 160) (Lines 217 and 219). Note that likelihood maximization and posterior sampling are conducted on an unbounded parameter space. If lower or upper bounds are specified for any parameter, they are applied just prior to objective function calculations using the `transform` function of the `transform.r` file located in the global code directory,

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- Lines 222-236: If `tsub` is `TRUE` (Line 222), the forward model for at least one phenomenology depends only on a subset of the full vector θ_0 of new event device parameters. The `tsub` object is initialized to a null list with elements for each phenomenology in the proper order (Line 225). The `theta_names` vector (Line 199) describes the order of elements in θ_0 . For relevant phenomenologies, parameter subsets are specified as integer vectors identifying the extracted elements of θ_0 . The forward models of all other phenomenologies depend on the full θ_0 . In this example, the *surface effects* (*crater*) phenomenology only depends on log-yield (Line 226), while the other phenomenologies depend on both log-yield and height-of-burst.
- Lines 239-244: Calls the preprocessing function `prepro` for the benchmark and (if relevant) new event data. Table 9 describes all inputs to this function with default values. Only inputs with no default values must be provided.
- Lines 245-251: If `opt_B` is `TRUE` (Line 185), the preprocessor function `prepro` returns a list (designated here as `tmp`) with objects `p_cal` and `t_cal`, which are then assigned as follows

```
% p_cal = tmp$p_cal
% t_cal = tmp$t_cal
```

and both are utilized for maximum likelihood estimation and Bayesian analysis. Otherwise, `p_cal` is the only object returned,

```
% p_cal = tmp$p_cal
```

and utilized in subsequent analyses.

3.0.2 Maximum Likelihood Estimation

The maximum likelihood estimation component of the complete analysis in Appendix B.2 is responsible for utilizing benchmark and (if relevant) new event data to simultaneously estimate the parameters of the forward and error models, possibly the yield of each benchmark source for phenomenologies adopting the errors-in-variables yield model, and (if relevant) the new event device parameters (WPA, §A.1-A.2). If `nev` is `TRUE` (Line 188 of Appendix B.1), quantification of uncertainty in the new event device parameter estimates is provided, adjusting for asymptotically dependent quantities.

Table 9: Inputs to **prepro** function.

Input	Default	Brief Description
gdir	none	directory location of global subroutines
adir	none	directory location of application subroutines
rdir	none	root directory location of data files
cdir	none	directory locations (if relevant) and names of benchmark data files under rdir
Rh	none	vector with number of signatures for each phenomenology
pbeta	none	list containing empirical model common parameter counts by phenomenology
bopt	FALSE	parameter bounds supplied to log-likelihood maximization (currently implemented only for new event parameters)
nev	FALSE	analysis of new event
itr	FALSE	bijective transform of new event parameters provided
izmat	FALSE	user-provided code for computing variance component coefficient matrices
ieiv	NULL	numerical identifier of phenomenologies utilizing errors-in-variables yields in analysis of benchmark data
seiv	NULL	list containing identifiers of benchmark sources assigned errors-in-variables yields by phenomenology (ALL – every source)
ewsd	NULL	standard deviation of errors-in-variables Gaussian likelihood
Th	NULL	number of emplacement conditions for each phenomenology
pbetat	NULL	list containing empirical model emplacement-dependent parameter counts by phenomenology
ibetar	NULL	list containing locations of empirical model common parameters in full parameter vector by phenomenology
pvc_1	NULL	list containing source variance component parameter counts by phenomenology
pvc_2	NULL	list containing path variance component parameter counts by phenomenology
pctype	NULL	list indicating treatment of path variance component parameter by phenomenology (Crossed – common paths present across sources)
tnames	NULL	names of new event parameters
cnames	NULL	names of global forward model parameters
fp_tr	NULL	fixed inputs to new event parameter transform
tlb	NULL	lower bounds for new event parameters
tub	NULL	upper bounds for new event parameters
ndir	NULL	directory locations (if relevant) and names of new event data files under rdir
tsub	NULL	list containing index sets identifying new event parameter subsets by phenomenology if relevant

- Line 6: Read in code performing simultaneous maximum likelihood estimation of forward and error model parameters, benchmark source errors-in-variables yields (if relevant), and new event device parameters (if relevant), based on benchmark and (if relevant) new event data.
- Line 9: User specified seed to ensure repeatability of maximum likelihood estimation.
- Lines 13-17: Provide names of forward models for each signature by phenomenology (WPA, §5.1-5.4). The `fm` object is initialized as a null list with elements for each phenomenology in the proper order (Line 13). Subsequent lines specify the function names as vectors of strings having length equal to the number of signatures for each phenomenology (Lines 14-17). The code for all forward models from each phenomenology is concatenated into a single file named `forward.r` and placed in the application code directory; in this example,

`MultiPEM_Toolbox/Applications/Code/IYDT-gsrp`

Note that these forward models accept a vector of calibration and device parameters as their main argument. In this example, the *seismic* forward model $f_{sr}(\cdot)$ as a function of the parameters β_{sr} and device parameters log-yield (w) and HOB/DOB (h) is given as follows (WPA, §5.2, pp. 11-12),

$$\begin{aligned}\log(\tilde{d}_{sr}(\beta_{sr}, (w, h))) &= \beta_{sr,1} + \beta_{sr,2} \log(\tilde{\delta}_s) + \beta_{sr,3} \text{logistic}(\beta_{sr,4} \tilde{h}_s + \beta_{sr,5}) \\ f_{sr}(\beta_{sr}, (w, h)) &= \log(d_{sr}(\beta_{sr}, (w, h)))\end{aligned}\tag{3}$$

for

$$\text{logistic}(x) = \frac{1}{1 + \exp(-x)}.$$

The scaled signatures and covariates of this forward model are given by

$$\begin{aligned}\tilde{d}_{s1} &= d_{s1} \exp(-w/3) & \tilde{d}_{s2} &= d_{s2} \\ \tilde{\delta}_s &= \delta \exp(-w/3) & \tilde{h}_s &= h \exp(-w/3),\end{aligned}$$

where d_{s1} and d_{s2} are P-wave displacement and maximum velocity, and the covariate is range δ . The function `f_s` returns a vector of forward model calculations evaluated for the supplied value of $(\beta_{sr}, (w, h))$, each element corresponding to each row of a matrix of covariates (in this case, a column vector of ranges δ).

- Line 20: Indicate if forward model Jacobian matrices are provided for efficient log-likelihood maximization.
- Lines 22-30: If `igrad` is `TRUE` (Line 20), names of forward model Jacobian functions must be provided for each signature by phenomenology. The `gfm` object is initialized as a null list with elements for each phenomenology in the proper order (Line 25). Subsequent lines specify the Jacobian function names as vectors of strings having length equal to the number of signatures for each phenomenology (Lines 26-29). The code for all forward model Jacobian functions from each phenomenology is concatenated into a single file named `jacobian.r` and placed in the application code directory; in this example,

Note that these Jacobian functions accept a vector of calibration and device parameters as their main argument. In this example, the gradient vector of the *seismic* forward model of Equation (3) is computed from the partial derivatives of $f_{sr}(\cdot)$ for each parameter as follows,

$$\begin{aligned}\frac{\partial f_{sr}}{\partial \beta_{sr,1}} &= 1 \\ \frac{\partial f_{sr}}{\partial \beta_{sr,2}} &= \log(\tilde{\delta}_s) \\ \frac{\partial f_{sr}}{\partial \beta_{sr,3}} &= \text{logistic}(\beta_{sr,4}\tilde{h}_s + \beta_{sr,5}) \\ \frac{\partial f_{sr}}{\partial \beta_{sr,4}} &= \beta_{sr,3}\tilde{h}_s \times \text{logistic}(\beta_{sr,4}\tilde{h}_s + \beta_{sr,5}) \times \text{logistic}(-\beta_{sr,4}\tilde{h}_s - \beta_{sr,5}) \\ \frac{\partial f_{sr}}{\partial \beta_{sr,5}} &= \beta_{sr,3} \times \text{logistic}(\beta_{sr,4}\tilde{h}_s + \beta_{sr,5}) \times \text{logistic}(-\beta_{sr,4}\tilde{h}_s - \beta_{sr,5}),\end{aligned}$$

and each device parameter as follows,

$$\begin{aligned}\frac{\partial f_{sr}}{\partial w} &= -\frac{1}{3} \left(\beta_{sr,2} + \beta_{sr,3}\beta_{sr,4}\tilde{h}_s \times \text{logistic}(\beta_{sr,4}\tilde{h}_s + \beta_{sr,5}) \times \text{logistic}(-\beta_{sr,4}\tilde{h}_s - \beta_{sr,5}) \right) \\ &\quad + \frac{1}{3} \delta_1(r) \\ \frac{\partial f_{sr}}{\partial h} &= \beta_{sr,3}\beta_{sr,4} \exp(-w/3) \times \text{logistic}(\beta_{sr,4}\tilde{h}_s + \beta_{sr,5}) \times \text{logistic}(-\beta_{sr,4}\tilde{h}_s - \beta_{sr,5})\end{aligned}$$

for $\delta_A(x)$ the indicator function of set A . The function `g_s` returns a Jacobian matrix (`jbeta`) of forward model gradients for the parameters, evaluated at the supplied value of $(\beta_{sr}, (w, h))$, with rows corresponding to the rows of a matrix of covariates (in this case, a column vector of ranges δ). If `calp` is `TRUE` (Line 60 of Appendix B.1), partial derivatives of $f_{sr}(\cdot)$ with respect to the global forward model parameters must also be calculated analogously to `jbeta` and returned by `g_s` as the object `jcalp`. If `eiv` is `TRUE` (Line 126 of Appendix B.1), or `nev` is `TRUE` (Line 188 of Appendix B.1), `g_s` will also return a Jacobian matrix (`jtheta`) of forward model gradients for log-yield w or the device parameters (in this case, including log-yield) as appropriate, evaluated at the supplied value of $(\beta_{sr}, (w, h))$, each element corresponding to each row of the same covariate matrix used in the calculation of `jbeta`.

- Line 35: Indicate if the same forward model function is used to compute multiple signatures, and signature-specific code within this function is required.
- Lines 37-41: If `iResponse` is `TRUE` (Line 35), initialize `iResponse` to a null list with elements for each phenomenology in the proper order (Line 38). For each relevant phenomenology, subsequent lines provide vectors of length equal to the number of

signatures, each element of which is a tag identifying code specific to the corresponding signature. This mechanism is utilized for the *seismic* (Line 39) and *acoustic* (Line 40) phenomenologies.

- Line 44: Indicate if fixed inputs are to be provided to the forward models for at least one phenomenology.
- Lines 46-53: If **fPars** is **TRUE** (Line 44), initialize **fPars** to a null list with elements for each phenomenology in the proper order (Line 47). For each relevant phenomenology, subsequent lines provide the value(s) of all fixed inputs. For example, the *acoustic* forward model requires fixed values for **yield_scaling** (Line 49), **pressure_scaling** (Line 50), and **temp_scaling** (Line 51).
- Line 56: Specify the number of starting parameter vectors for the log-likelihood maximization routine.
- Line 59: Specify the number of cores to use for parallel optimization (across distinct starting values) of the benchmark and (if relevant) new event data log-likelihood function.
- Line 62: Specify if the Broyden-Fletcher-Goldfarb-Shanno (BFGS) algorithm is to be used for maximization of the log-likelihood function. If **TRUE**, functions to compute forward model Jacobian matrices analytically must be provided, or numerical gradients will be utilized (generally increasing compute time). If **FALSE**, the gradient-free Nelder-Mead simplex algorithm will be utilized for optimization, which is generally much slower computationally than BFGS with analytical gradients.
- Lines 66-69: If relevant, specifies the location (relative to the run directory) of parameter values or estimates to be used as the first starting value for log-likelihood maximization. These values are stored in a **.RData** object as a list, with elements corresponding to forward model (e.g. global parameters **calp**, within signature common parameters **beta**, within signature emplacement-dependent parameters **tbeta**) and error model (e.g. source variance components **vc_1**, path variance components **vc_2**, observation error parameters **eps**) quantities of interest. If relevant, benchmark source errors-in-variables yield estimates (**w_eiv**) are also provided. For multi-phenomenology analyses, values or estimates from individual phenomenologies may be input in the proper order, and they will be concatenated appropriately.
- Line 73: If desired, name of output **.RData** file to store optimization results from this run. The elements of the list to be written are described in the previous item.
- Lines 75-83: If **calp** is **TRUE** (Line 60 of Appendix B.1), and if **cst** is **TRUE** (Line 77), specifies an initial starting value for the global forward model parameters (Lines 80-81) for log-likelihood maximization. This value supersedes the value read in from the first file provided in the string vector **opt_files_in** (Line 66), if the **calp** list element is provided.
- Lines 85-93: If **nev** is **TRUE** (Line 188 of Appendix B.1), and if **tst** is **TRUE** (Line 87), specifies an initial starting value for the new event device parameters (Lines 90-91)

for log-likelihood maximization. This value supersedes the value read in from the first file provided in the string vector `opt_files_in` (Line 66), if the `theta0` list element is provided.

- Line 95-98: Specify the level of confidence intervals for (a) the true values of each global forward model parameter from the maximum likelihood estimate and the estimated Fisher information matrix, and/or (b) the true values of each new event device parameter from the maximum likelihood estimate and the estimated Fisher information matrix (WPA, §A.2, Equation (23); §A.4, Equation (25)).
- Line 102: Indicate if phenomenology specific code is required in the postprocessing function.
- Lines 104-106: If `Phen` is `TRUE` (Line 102), specifies a matrix in which the first column provides the numerical phenomenology indicator (see Lines 54-57 of the preprocessing code in Appendix B.1), and the second column provides the phenomenology name in string format. In this example, specific code is required to process results for the *seismic* phenomenology (Line 105).
- Line 109: Indicate if gradient verification is to be conducted on the log-likelihood function. If `TRUE` and `igrad` is `TRUE` (Line 20), analytical and numerical gradients at the optimal parameter value, and other randomly sampled parameter values, are compared for consistency.
- Line 112: Specify the strategy for running parallel jobs using the `future` package in R. The available options are given by starting an R session and issuing the following commands,


```
% R
> require(future)
> help(plan)
```
- Lines 115-119: Calls the log-likelihood maximization function `calc_mle` for the benchmark and (if relevant) new event data. Table 10 describes all inputs to this function with default values. Only inputs with no default values must be provided.

3.0.3 Bayesian Analysis

The optional Bayesian inference component of the analysis in Appendix B.3 is responsible for sampling forward and error model parameters, benchmark source errors-in-variables yields (if relevant), and new event device parameters (if relevant) from their joint posterior distribution, using benchmark and (if relevant) new event data simultaneously. If `nev` is `TRUE` (Line 188 of Appendix B.1), estimates of new event device parameters with uncertainty quantification are computed from these samples.

- Line 6: Indicate if Bayesian analysis is to be conducted.
- Line 10: Read in code performing Bayesian analysis on forward and error model parameters, benchmark source errors-in-variables yields (if relevant), and new event device parameters (if relevant), using benchmark and (if relevant) new event data.

Table 10: Inputs to `calc.mle` function.

Input	Default	Brief Description
<code>p_cal</code>	none	environment storing all objects needed in log-likelihood calculations
<code>gdir</code>	none	directory location of global subroutines
<code>adir</code>	none	directory location of application subroutines
<code>f</code>	none	names of forward model functions for each signature by phenomenology
<code>nst</code>	10	number of starting values for log-likelihood maximization
<code>ncor</code>	1	number of cores for log-likelihood maximization
<code>ci_lev</code>	0.95	confidence interval levels for global forward model and new event parameter inference
<code>igrad</code>	TRUE	forward model Jacobian provided
<code>bfgs</code>	TRUE	log-likelihood maximization uses BFGS methods
<code>igrck</code>	TRUE	conduct log-likelihood function gradient verification
<code>t_cal</code>	NULL	object required if bounds supplied to log-likelihood maximization
<code>g</code>	NULL	names of forward model Jacobian functions for each signature by phenomenology
<code>iresp</code>	NULL	flags for modified calculation by signature in a common forward model for each relevant phenomenology
<code>fp_fm</code>	NULL	fixed inputs required by forward models
<code>fopt_in</code>	NULL	location of input R data file(s) providing an initial starting value for log-likelihood maximization (if multiple files, starting value created by concatenating over phenomenologies)
<code>Xst</code>	NULL	matrix of starting values for log-likelihood maximization if not generated by this function
<code>tst</code>	NULL	vector of starting values for new event parameters in log-likelihood maximization
<code>cst</code>	NULL	vector of starting values for global forward model parameters in log-likelihood maximization
<code>fopt_out</code>	NULL	location to write output R data file with results of log-likelihood maximization
<code>phen</code>	NULL	phenomenology number and type (if needed for postprocessing)
<code>pl</code>	"multicore"	strategy for running parallel jobs using the <code>future</code> package

- Line 14: Indicate if a log-prior density for the signature within phenomenology forward model parameters is supplied by the user (WPA, §5.5, p. 15). If `iBetaPrior` is `FALSE`, a “flat prior” (uniform on the domain) on these parameters is assumed.
- Lines 16-43: If relevant, specify details of user-provided log-prior distributions for signature within phenomenology forward model parameters. For each relevant phenomenology, the list object `lp_beta` is used for common coefficients, while the list

object `lp_betat` is used for emplacement-dependent coefficients (as demonstrated below in this application).

- Line 18: Specify location(s) of log-prior function(s). Must be provided if `iBetaPrior` is `TRUE` (Line 14). In this example, two log-prior functions are provided for the seismic (`'s'`) and optical (`'o'`) phenomenologies, located at

```
../Code/lp_beta_s.r
../Code/lp_beta_o.r
```
- Lines 20-21: If `igrad` is `TRUE` (Line 20 of Appendix B.2), specify location(s) of the log-prior gradient function(s). In this example, two log-prior gradient functions are provided, located at

```
../Code/glp_beta_s.r
../Code/glp_beta_o.r
```
- Line 26: For each relevant phenomenology, initialize a null list `lp_betat` of length equal to the number of emplacement conditions containing distinct forward model parameters.
- Line 28: For each relevant phenomenology and emplacement condition, provide the name(s) of the log-prior function(s) for each signature. In this example, the *seismic* phenomenology utilizes a log-prior function `lp_s` for each signature within each emplacement condition.
- Line 30: If `igrad` is `TRUE` (Line 20 of Appendix B.2), then for each relevant phenomenology and emplacement condition, provide the name(s) of the log-prior gradient function(s) for each signature. In this example, the *seismic* phenomenology utilizes a log-prior gradient function `lq_s` for each signature within each emplacement condition.
- Line 36: The list object `lp_beta` is used to specify log-prior distributions for common coefficients. In this example, the *optical* phenomenology utilizes a log-prior function `lp_o` for each signature.
- Line 38: If `igrad` is `TRUE` (Line 20 of Appendix B.2), then in this example, the *optical* phenomenology utilizes a log-prior gradient function `lq_o` for each signature.
- Line 46: If `calp` is `TRUE` (Line 60 of Appendix B.1), indicate if a log-prior density for the global forward model parameters is supplied by the user. If `iCalPrior` is `FALSE`, a “flat prior” (uniform on the domain) on these parameters is assumed.
- Lines 48-65: If relevant, specify details of user-provided log-prior distributions for global forward model parameters.
 - Line 50: Specify location of log-prior function. If `NULL`, utilize the default log-prior function contained in the file `lp_c.r` placed in the application code directory; in this example,

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- Line 52: If `igrad` is `TRUE` (Line 20 of Appendix B.2), specify location of the log-prior gradient function. If `NULL`, utilize the default log-prior gradient function contained in the file `glp_c.r` placed in the application code directory; in this example,

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- Line 56: Provide the name of the log-prior function.
- Line 57: If `igrad` is `TRUE` (Line 20 of Appendix B.2), provide the name of the log-prior gradient function.
- Lines 60-61: Specify all fixed quantities required for calculation of the log-prior density.
- Line 68: If `nev` is `TRUE` (Line 188 of Appendix B.1), indicate if a log-prior density for the new event device parameters is supplied by the user (WPA, §5.5, p. 15; §5.6, p. 21). If `iThetaOPrior` is `FALSE`, a “flat prior” (uniform on the domain) on these parameters is assumed.
- Lines 70-90: If relevant, specify details of user-provided log-prior distributions for new event device parameters.
 - Line 72: Specify location of log-prior function. If `NULL`, utilize the default log-prior function contained in the file `lp_0.r` placed in the application code directory; in this example,

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- Line 74: If `igrad` is `TRUE` (Line 20 of Appendix B.2), specify location of the log-prior gradient function. If `NULL`, utilize the default log-prior gradient function contained in the file `glp_0.r` placed in the application code directory; in this example,

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- Line 78: Provide the name of the log-prior function.
- Line 79: If `igrad` is `TRUE` (Line 20 of Appendix B.2), provide the name of the log-prior gradient function.
- Lines 81-86: Specify all fixed quantities required for calculation of the log-prior density.
- Line 94: Specify a *fixed* value for the scale parameter `A` of half-Cauchy prior distribution(s) for the *source* and *path* variance component parameters if relevant (WPA, §5.5, p. 15; §5.6, p. 17). Prior distributions for a non-empty collection of variance component parameters are taken to be mutually independent. Comment out if this parameter is to be sampled from its posterior distribution.

- Line 98: Specify a *fixed* value for the shape parameter η of the Lewandowski-Kurowicka-Joe (LKJ) prior distribution for the observational error model correlation parameters (WPA, §5.5, p. 15; §5.6, p. 17).
- Line 106: If `eiv` is `TRUE` (Line 126 of Appendix B.1), specify a *fixed* value for the parameter that controls the number of modes in the *flexible generalized skew-normal* (FGSN) prior distribution for the errors-in-variables yields of the benchmark events (WPA, §5.5, p. 15; §5.6, p. 26).
- Line 112: Select the Markov chain Monte Carlo (MCMC) algorithm to use for posterior sampling, from one of three options: `RAM`, `FME`, and `NUTS`. `RAM` is the robust adaptive Metropolis algorithm of Vihola¹⁷ implemented in the R package `adaptMCMC`. `FME` is the delayed rejection adaptive Metropolis algorithm of Haario, Laine, and Mira¹⁸ implemented in the R package `FME`. `NUTS` is the No-U-Turn Sampler of Hoffman and Gelman¹⁹. The `NUTS` option requires the analytical gradient of the log-posterior density, which in turn requires `igrad` to be `TRUE` (Line 20 of Appendix B.2).
- Line 115: Specify the per core sample size of the burn-in period for MCMC sampling (pre-equilibrium stage of Markov chain). These samples are discarded prior to any inference using the posterior samples.
- Line 118: Specify the sample size of the MCMC production run. These samples are kept for posterior inference.
- Line 121: Specify the rate at which MCMC production samples are thinned for estimation of the Deviance Information Criterion²⁰ (DIC) and the Predictive Information Criterion²¹ (PIC). In this example, the `nthin` value of 20 indicates that every 20-th production sample is kept for DIC and PIC estimation.
- Line 124: Specify the number of cores to use for parallel optimization (across distinct starting values) of the benchmark and (if relevant) new event data log-posterior function. By default, this optimization uses the same number of distinct starting values as optimization of the benchmark and (if relevant) new event data log-likelihood function (Line 56 of Appendix B.2).
- Line 127: Specify the number of cores used to run parallel MCMC chains. The burn-in period for each chain is determined by `nburn` (Line 115), while the `nmcmc` (Line 118) production runs are split between the `ncores_mc` processors and combined at the conclusion of the runs.
- Line 130: Indicate if gradient verification is to be conducted on the log-prior function. If `TRUE` and `igrad` is `TRUE` (Line 20 of Appendix B.2), analytical and numerical gradients

¹⁷Vihola, M. (2012). Robust adaptive Metropolis algorithm with coerced acceptance rate. *Stat Comput* 22:997-1008.

¹⁸Haario, H., Laine, M., and Mira, A. (2006). DRAM: Efficient adaptive MCMC. *Stat Comput* 16:339-354.

¹⁹Hoffman, M. D. and Gelman, A. (2014). The No-U-Turn Sampler: Adaptively setting path lengths in Hamiltonian Monte Carlo. *J Mach Learn Res* 15:1593-1623.

²⁰Spiegelhalter, D.J., Best, N.G., Carlin, B.P., & van der Linde, A. (2002). Bayesian measures of model complexity and fit (with discussion), *J R Stat Soc Ser B* 64:583-639.

²¹Ando, T. (2011). Predictive Bayesian model selection, *Am J Math Manag Sci* 31:13-38.

at the maximum *a posteriori* parameter value, and other randomly sampled parameter values, are compared for consistency.

- Line 133: Indicate if gradient verification is to be conducted on the log-posterior function. If `TRUE` and `igrad` is `TRUE` (Line 20 of Appendix B.2), analytical and numerical gradients at the maximum *a posteriori* parameter value, and other randomly sampled parameter values, are compared for consistency.
- Lines 136-146: Calls the Bayesian analysis function `calc_bayes` for the new event data. Table 11 describes all inputs to this function with default values. Only inputs with no default values must be provided.

3.0.4 Output

The output file `runMPEM.out` from the complete analysis contains a summary of (if relevant) global forward model parameters, (if relevant) errors-in-variables yield estimates for the relevant benchmark sources, signature within phenomenology forward and error model parameter estimates, as well as (if relevant) new event device parameter estimates derived from the benchmark and (if relevant) new event data simultaneously. The desired output is supplied by the user function `print_sumstats.r`, placed in the application code directory; in this example,

`MultiPEM_Toolbox/Applications/Code/IYDT-gsrp`

The output presented in Appendix B.4 contains the most pertinent information extracted from the full file.

- Lines 8-15: Output from the preprocessing function `prepro`. These warning messages explain which variance component models are allowed (if any) for each signature of each phenomenology based on the structure of the benchmark data. In this example, no source or path random effects are allowed for *optical* or *surface effects* phenomenologies (Lines 8-15). There are no warning messages for *seismic* or *acoustic* signatures, indicating source and path random effects are allowed.
- Lines 23-290: Output from the maximum likelihood estimation function `calc_mle`:
 - Line 25: Convergence code from the R optimization function `optim`. In this example, ‘0’ indicates successful completion.
 - Line 26: Number of optimization restarts in which the relative absolute maximum log-likelihood difference is $\leq 10^{-8}$. The algorithm exits after 2 such restarts, which is attained in this example.
 - Line 34: Maximum likelihood estimates of the new event device parameters; in this example, log-yield (`W`) and height-of-burst (`HOB`).
 - Line 39: Standard errors of the maximum likelihood estimates of the new event device parameters, assuming the forward model parameters and (if relevant) benchmark source errors-in-variables yields are known with certainty (WPA, §A.2, p. 42, calculated from $(\mathcal{I}_{\theta_0, \theta_0}^0)^{-1}$).

Table 11: Inputs to `calc_bayes` function.

Input	Default	Brief Description
<code>p_cal</code>	<code>none</code>	environment storing all objects needed in log-posterior calculations
<code>gdir</code>	<code>none</code>	directory location of global subroutines
<code>adir</code>	<code>none</code>	directory location of application subroutines
<code>nst</code>	<code>10</code>	number of starting values for log-posterior maximization
<code>nburn</code>	<code>10000</code>	number of per core MCMC burn-in samples
<code>nmcmc</code>	<code>20000</code>	number of MCMC production samples
<code>nthin</code>	<code>20</code>	posterior sample thinning rate
<code>ncor_map</code>	<code>1</code>	number of cores for log-posterior maximization
<code>ncor_mc</code>	<code>1</code>	number of cores for generating parallel MCMC chains
<code>igrad</code>	<code>TRUE</code>	forward model Jacobian provided
<code>igrck_pr</code>	<code>TRUE</code>	conduct log-prior function gradient verification
<code>igrck_po</code>	<code>TRUE</code>	conduct log-posterior function gradient verification
<code>bfgs</code>	<code>TRUE</code>	log-posterior maximization uses BFGS methods
<code>ibpr</code>	<code>FALSE</code>	prior density function(s) provided for forward model coefficients
<code>icpr</code>	<code>FALSE</code>	prior density function(s) provided for global forward model coefficients
<code>itpr</code>	<code>FALSE</code>	prior density function provided for new event parameters
<code>fpr_b</code>	<code>NULL</code>	location of functions computing log-prior density for forward model coefficients
<code>fgpr_b</code>	<code>NULL</code>	location of functions computing gradients of log-prior density for forward model coefficients
<code>fpr_c</code>	<code>NULL</code>	location of functions computing log-prior density for global forward model coefficients
<code>fgpr_c</code>	<code>NULL</code>	location of functions computing gradients of log-prior density for global forward model coefficients
<code>fpr_t</code>	<code>NULL</code>	location of function computing log-prior density for new event parameters
<code>fgpr_t</code>	<code>NULL</code>	location of function computing gradients of log-prior density for new event parameters
<code>Xnom</code>	<code>NULL</code>	matrix of starting values for hyperparameters in log-posterior maximization if not generated by this function
<code>imcmc</code>	<code>"FME"</code>	MCMC algorithm (current options: <code>"RAM"</code> , <code>"FME"</code> , <code>"NUTS"</code>)
<code>pl</code>	<code>"multicore"</code>	strategy for running parallel jobs using the <code>future</code> package
<code>t_cal</code>	<code>NULL</code>	object required if bounds supplied to log-posterior maximization

- Lines 43-45: Correlation matrix of the maximum likelihood estimates of the new event device parameters, assuming the forward model parameters and (if relevant) benchmark source errors-in-variables yields are known with certainty (WPA, §A.2,

- p. 42, calculated from $(\mathcal{I}_{\theta_0, \theta_0}^0)^{-1}$).
- Line 50: Standard errors of the maximum likelihood estimates of the new event device parameters, adjusted for estimation of the forward model parameters and (if relevant) benchmark source errors-in-variables yields (WPA, §A.2, Equation (23); §A.4, Equation (25)).
 - Lines 54-56: Correlation matrix of the maximum likelihood estimates of the new event device parameters, adjusted for estimation of the forward model parameters and (if relevant) benchmark source errors-in-variables yields (WPA, §A.2, Equation (23); §A.4, Equation (25)).
 - Lines 61-62: 95% confidence intervals for the unknown true values of the new event device parameters, based on standard errors assuming the forward model parameters and (if relevant) benchmark source errors-in-variables yields are known with certainty (WPA, §A.2, p. 42, calculated from $(\mathcal{I}_{\theta_0, \theta_0}^0)^{-1}$).
 - Lines 67-68: 95% confidence intervals for the unknown true values of the new event device parameters, based on standard errors adjusted for estimation of the forward model parameters and (if relevant) benchmark source errors-in-variables yields (WPA, §A.2, Equation (23); §A.4, Equation (25)).
 - Lines 73-76: Maximum likelihood estimates of errors-in-variables yields for the relevant benchmark sources. Source names (Lines 73 and 75) are given above yield estimates (Lines 74 and 76). Errors-in-variables yields are only estimated if `eiv` is `TRUE` (Line 126 of Appendix B.1).
 - Lines 80-102: Maximum likelihood estimates of *common* forward model parameters for each signature of each phenomenology (where present).
 - Lines 106-152: Maximum likelihood estimates of *emplacement-dependent* forward model parameters for each signature of each phenomenology (where present).
 - Lines 156-170: Maximum likelihood estimates of *source* random effect (error model) variance component parameters for each signature of each phenomenology (where present).
 - Lines 174-188: Maximum likelihood estimates of *path* random effect (error model) variance component parameters for each signature of each phenomenology (where present).
 - Lines 192-238: Maximum likelihood estimates of observational error variances for each signature, and correlations between signatures, for each phenomenology (WPA, §A.5).
 - Line 240: Akaike Information Criterion²² (AIC) value based on benchmark and (if relevant) new event data. Used for selecting among competing forward or error model specifications (WPA, §5.5, p. 16; §5.6, Tables 5 and 6, pp. 19, 21).

²²Akaike, H. (1973). Information Theory and an Extension of the Maximum Likelihood Principle. In: Petrov, B.N. & Csaki, F., Eds., International Symposium on Information Theory, 267-281.

- Line 242: Bayesian Information Criterion²³ (BIC) value based on benchmark and (if relevant) new event data. Used for selecting among competing forward or error model specifications (WPA, §5.5, p. 16; §5.6, Tables 5 and 6, pp. 19, 21).
- Lines 247-290: Example of log-likelihood gradient verification at a single sampled parameter vector. Additional checks were deleted for brevity.
 - * Lines 248-267: Analytic gradient calculation
 - * Lines 269-288: Numerical gradient calculation using the R package `numDeriv`
 - * Line 290: Largest negative (first entry) and positive (second entry) differences between the analytic and numerical gradients
- Lines 292-1478: Output from the Bayesian analysis function `calc_bayes`:
 - Line 309: Convergence code from the R optimization function `optim`. In this example, ‘0’ indicates successful completion.
 - Line 310: Number of optimization restarts in which the relative absolute maximum log-posterior difference is $\leq 10^{-8}$. The algorithm exits after 2 such restarts, which is attained in this example.
 - Line 318: Maximum *a posteriori* estimates of the new event device parameters.
 - Lines 323-326: Maximum *a posteriori* estimates of errors-in-variables yields for the relevant benchmark sources. Source names (Lines 323 and 325) are given above yield estimates (Lines 324 and 326). Errors-in-variables yields are only estimated if `eiv` is `TRUE` (Line 126 of Appendix B.1).
 - Lines 330-352: Maximum *a posteriori* estimates of *common* forward model parameters for each signature of each phenomenology (where present).
 - Lines 356-402: Maximum *a posteriori* estimates of *emplacement-dependent* forward model parameters for each signature of each phenomenology (where present).
 - Lines 406-420: Maximum *a posteriori* estimates of *source* random effect (error model) variance component parameters for each signature of each phenomenology (where present).
 - Lines 424-438: Maximum *a posteriori* estimates of *path* random effect (error model) variance component parameters for each signature of each phenomenology (where present).
 - Lines 442-488: Maximum *a posteriori* estimates of observational error variances for each signature, and correlations between signatures, for each phenomenology (WPA, §A.5).
 - Lines 492-494: Maximum *a posteriori* estimates of FGSN prior distribution parameters (WPA, §5.6, p. 26; $\text{Alpha} = \mu$, $\text{Omega} = v$ (two coefficients)).

²³Schwarz, G. (1978). Estimating the dimension of a model, *Ann Stat* 6:461-464.

- Lines 498-543: Example of log-prior gradient verification at a single sampled parameter vector. Additional checks were deleted for brevity.
 - * Lines 499-519: Analytic gradient calculation
 - * Lines 521-541: Numerical gradient calculation using the R package `numDeriv`
 - * Line 543: Largest negative (first entry) and positive (second entry) differences between the analytic and numerical gradients
- Lines 547-592: Example of log-posterior gradient verification at a single sampled parameter vector. Additional checks were deleted for brevity.
 - * Lines 548-568: Analytic gradient calculation
 - * Lines 570-590: Numerical gradient calculation using the R package `numDeriv`
 - * Line 592: Largest negative (first entry) and positive (second entry) differences between the analytic and numerical gradients
- Lines 596-625: Acceptance rates on each core of the Robust Adaptive Metropolis (RAM) posterior sampling method implemented in R package `adaptMCMC`. The target acceptance rate is 0.234.
- Line 631: Means of samples from the new event device parameter marginal posterior distributions.
- Line 633: Standard deviations of samples from the new event device parameter marginal posterior distributions.
- Lines 635-643: User specified quantiles of samples from the new event device parameter marginal posterior distributions.
- Lines 647-649: Correlation matrix of samples from the new event device parameter joint posterior distribution.
- Lines 653-705: Means and user specified quantiles of samples from the marginal posterior distributions of errors-in-variables yields for the relevant benchmark sources. The ordering of benchmark sources is provided with the maximum *a posteriori* estimates (Lines 323 and 325). Errors-in-variables yields are only estimated if `eiv` is `TRUE` (Line 126 of Appendix B.1).
- Lines 709-798: Means and user specified quantiles of samples from the marginal posterior distributions of *common* forward model parameters for each signature of each phenomenology (where present).
- Lines 802-1039: Means and user specified quantiles of samples from the marginal posterior distributions of *emplacement-dependent* forward model parameters for each signature of each phenomenology (where present).
- Lines 1043-1097: Means and user specified quantiles of samples from the marginal posterior distributions of *source* random effect (error model) variance component parameters for each signature of each phenomenology (where present).

- Lines 1101-1155: Means and user specified quantiles of samples from the marginal posterior distributions of *path* random effect (error model) variance component parameters for each signature of each phenomenology (where present).
- Lines 1159-1433: Means and user specified quantiles of samples from the marginal posterior distributions of observational error variances for each signature, and correlations between signatures, for each phenomenology (WPA, §A.5).
- Lines 1437-1474: Means and user specified quantiles of samples from the marginal posterior distributions of FGSN prior distribution parameters (WPA, §5.6, p. 26; $\text{Alpha} = \mu$, $\text{Omega} = v$ (two coefficients)).
- Line 1476: DIC value based on benchmark and (if relevant) new event data. Used for selecting among competing forward or error model specifications (WPA, §5.5, p. 16; §5.6, Tables 5 and 6, pp. 19, 21).
- Line 1478: PIC value based on benchmark and (if relevant) new event data. Used for selecting among competing forward or error model specifications (WPA, §5.5, p. 16; §5.6, Tables 5 and 6, pp. 19, 21).

The `p_cal` environment resulting from this run contains several elements of potential interest for additional post-processing:

- `p_cal$mle`: Maximum likelihood estimate of (if relevant) unbounded new event device parameters (i.e., on scale used by the optimizer), unbounded global forward model parameters (if relevant), benchmark source errors-in-variables yields (if relevant), and signature within phenomenology forward and error model parameters
- `p_cal$Sigma_mle_0$II_nev_it`: If relevant, estimated asymptotic covariance matrix of new event device parameter elements of `p_cal$mle`
- `p_cal$tmle_0`: If relevant, maximum likelihood estimate of transformed new event device parameters (i.e., on correct scale)
- `p_cal$Sigma_mle_0$II_nev`: If relevant, estimated asymptotic covariance matrix of `p_cal$tmle_0`
- `p_cal$mle_calp`: If `calp` is `TRUE` (Line 60 of Appendix B.1), maximum likelihood estimate of unbounded global forward model parameters
- `p_cal$Sigma_mle_cal$II_calp`: If `nev` is `FALSE` (Line 188 of Appendix B.1) and `calp` is `TRUE` (Line 60 of Appendix B.1), estimated asymptotic covariance matrix of `p_cal$mle_calp`, adjusted for estimation of signature within phenomenology forward model parameters, and (if relevant) benchmark source errors-in-variables yields
- `p_cal$Sigma_mle_0$II_calp`: If `nev` is `TRUE` (Line 188 of Appendix B.1) and `calp` is `TRUE` (Line 60 of Appendix B.1), estimated asymptotic covariance matrix of `p_cal$mle_calp`, adjusted for estimation of new event device parameters, signature within phenomenology forward model parameters, and (if relevant) benchmark source errors-in-variables yields

- `p_cal$map`: If `iBayes` is `TRUE` (Line 6 of Appendix B.3), maximum *a posteriori* estimate of (if relevant) unbounded new event device parameters (i.e., on scale used by the optimizer), unbounded global forward model parameters (if relevant), benchmark source errors-in-variables yields (if relevant), and signature within phenomenology forward and error model parameters
- `p_cal$map_0`: If `iBayes` is `TRUE` (Line 6 of Appendix B.3), maximum *a posteriori* estimate of transformed new event device parameters (i.e., on correct scale)
- `p_cal$map_calp`: If `iBayes` is `TRUE` (Line 6 of Appendix B.3) and `calp` is `TRUE` (Line 60 of Appendix B.1), maximum *a posteriori* estimate of unbounded global forward model parameters
- `p_cal$mpi_0`: If `iBayes` is `TRUE` (Line 6 of Appendix B.3), posterior samples of transformed new event device parameters (i.e., on correct scale)
- `p_cal$mpi_calp`: If `iBayes` is `TRUE` (Line 6 of Appendix B.3) and `calp` is `TRUE` (Line 60 of Appendix B.1), posterior samples of unbounded global forward model parameters

A Rapid Assessment Run Files

This appendix provides example run files and output files for rapid assessments.

A.1 Benchmark Data: Preprocessing

```
1 #####
2 #
3 # This file is the input deck for MultiPEM Toolbox estimation of
4 # forward and error model parameters based on calibration data.
5 #
6 # © 2023. Triad National Security, LLC. All rights reserved.
7 # This program was produced under U.S. Government contract
8 # 89233218CNA000001 for Los Alamos National Laboratory (LANL), which
9 # is operated by Triad National Security, LLC for the U.S. Department
10 # of Energy/National Nuclear Security Administration. All rights in
11 # the program are reserved by Triad National Security, LLC, and the
12 # U.S. Department of Energy/National Nuclear Security Administration.
13 # The Government is granted for itself and others acting on its behalf
14 # a nonexclusive, paid-up, irrevocable worldwide license in this
15 # material to reproduce, prepare derivative works, distribute copies
16 # to the public, perform publicly and display publicly, and to permit
17 # others to do so.
18 #
19 #####
20
21 #
22 # REQUIRED R PACKAGES
23 #
24
25 require(Matrix)
26
27 #
28 # END REQUIRED R PACKAGES
29 #
30
31 #
32 # PREPROCESSING
33 #
34
35 # Specify directory for general subroutines
36 gen_dir = "../.../Code"
37
38 # Source supporting R function
39 source(paste(gen_dir,"/prepro_cal.r",sep=""))
```

```

40
41 # Specify directory for application subroutines
42 app_dir = "../..Code"
43
44 # Specify root data directory
45 dat_dir = "../..Data"
46
47 # Specify calibration data directories
48 dat_cal = c("seismic_cal.csv",
49             "acoustic_cal.csv",
50             "optical_cal.csv",
51             "crater_cal.csv")
52
53 # Phenomenologies for this analysis
54 # 1 - seismic
55 # 2 - acoustic
56 # 3 - optical
57 # 4 - crater (surface effects)
58
59 # Indicate presence of calibration inference parameters
60 calp = FALSE
61
62 if( calp ){
63     # Names of calibration inference parameters
64     #cal_par_names =
65 } else { cal_par_names = NULL }
66
67 # Specify number of responses for each phenomenology
68 Rh = c(2,2,2,2)
69
70 # Empirical model parameter count: common
71 # list with elements corresponding to phenomenologies
72 pbeta = vector("list",length(Rh))
73 for( hh in 1:length(Rh) ){ pbeta[[hh]] = numeric(Rh[hh]) }
74 # phenomenology 2
75 pbeta[[2]] = c(2,2)
76 # phenomenology 3
77 pbeta[[3]] = c(4,4)
78 # phenomenology 4
79 pbeta[[4]] = c(2,2)
80
81 # Specify number of emplacement conditions for each phenomenology
82 Th = TRUE
83
84 if( Th ){ Th = c(3,3,0,0)

```

```

85 } else { Th = NULL }
86
87 # Empirical model parameter count: emplacement condition
88 # list with elements corresponding to phenomenologies
89 if( !is.null(Th) ){
90   pbetat = vector("list",length(Rh))
91   for( hh in 1:length(Rh) ){
92     if( Th[hh] > 1 ){ pbetat[[hh]] = vector("list",Th[hh]) }
93   }
94   # phenomenology 1
95   for( tt in 1:Th[1] ){
96     pbetat[[1]][[tt]] = numeric(Rh[1])
97     pbetat[[1]][[tt]] = c(5,5)
98   }
99   # phenomenology 2
100   for( tt in 1:Th[2] ){
101     pbetat[[2]][[tt]] = numeric(Rh[2])
102     pbetat[[2]][[tt]] = c(1,1)
103   }
104 } else { pbetat = NULL }
105
106 # Locations of common parameters in full parameter vector
107 # list with elements corresponding to phenomenologies
108 if( !is.null(Th) ){
109   ibetar = vector("list",length(Rh))
110   for( hh in 1:length(Rh) ){
111     if( Th[hh] > 1 ){
112       # lists with elements for each response within
113       # emplacement condition
114       ibetar[[hh]] = vector("list",Th[hh]*Rh[hh])
115     }
116   }
117   # phenomenology 2
118   for( tt in 1:Th[2] ){
119     for( rr in 1:Rh[2] ){
120       ibetar[[2]][[(tt-1)*Rh[2]+rr]] = 1:2
121     }
122   }
123 } else { ibetar = NULL }
124
125 # Indicate analysis with errors-in-variables (eiv)
126 eiv = TRUE
127
128 # Specifications for errors-in-variables
129 if( eiv ){

```

```

130 # Specify phenomenologies utilizing
131 # errors-in-variables yields
132 ieiv = 3:4
133
134 # Errors-in-variables source lists by
135 # phenomenology
136 seiv = vector("list",length(Rh))
137 for( hh in ieiv ){ seiv[[hh]] = "ALL" }
138
139 # Set standard deviation of eiv Gaussian likelihood
140 eiv_w_sd = 0.3/3
141 } else {
142   ieiv = NULL
143   seiv = NULL
144   eiv_w_sd = NULL
145 }
146
147 # Specify Error Model
148 # Source variance component parameter count
149 pvc_1 = TRUE
150
151 if( pvc_1 ){
152   pvc_1 = vector("list",length(Rh))
153   for( hh in 1:length(Rh) ){ pvc_1[[hh]] = numeric(Rh[hh]) }
154   # phenomenology 1
155   pvc_1[[1]] = c(1,1)
156   # phenomenology 2
157   pvc_1[[2]] = c(1,1)
158 } else { pvc_1 = NULL }
159
160 # Path variance component parameter count
161 pvc_2 = TRUE
162
163 if( pvc_2 ){
164   pvc_2 = vector("list",length(Rh))
165   for( hh in 1:length(Rh) ){ pvc_2[[hh]] = numeric(Rh[hh]) }
166   # phenomenology 1
167   pvc_2[[1]] = c(1,1)
168   # phenomenology 2
169   pvc_2[[2]] = c(1,1)
170
171 # path error models by phenomenology
172 ptype = vector("list",length(Rh))
173 # phenomenology 1
174 ptype[[1]] = "Crossed"

```



```

175     # phenomenology 2
176     ptype[[2]] = "Crossed"
177 } else { pvc_2 = NULL; ptype = NULL; }
178
179 # Set flag for user-provided code to calculate variance
180 # component coefficient matrices
181 calc_Z = FALSE
182
183 # Preprocessing for statistical analysis routines
184 p_cal = prepro_cal(gen_dir,app_dir,dat_dir,dat_cal,Rh,pbeta,
185                   izmat=calc_Z,ieiv=ieiv,seiv=seiv,ewsd=eiv_w_sd,
186                   Th=Th,pbetat=pbetat,ibetar=ibetar,pvc_1=pvc_1,
187                   pvc_2=pvc_2,ptype=ptype,cnames=cal_par_names)
188 save.image()
189
190 #
191 # END PREPROCESSING
192 #

```

A.2 Benchmark Data: Maximum Likelihood Estimation

```
1  #
2  # MAXIMUM LIKELIHOOD CALCULATION
3  #
4
5  # Source supporting R function
6  source(paste(gen_dir,"/calc_mle_cal.r",sep=""))
7
8  # Set seed for repeatability of analysis
9  set.seed(601)
10
11 # Names of forward models for each response
12 # by phenomenology
13 fm = vector("list",length(Rh))
14 fm[[1]] = c("f_s","f_s")
15 fm[[2]] = c("f_a","f_a")
16 fm[[3]] = c("f_o","f_o")
17 fm[[4]] = c("f_c","f_c")
18
19 # Indicate if forward model gradients provided
20 igrad = TRUE
21
22 if( igrad ){
23   # Names of forward model gradients for each response
24   # by phenomenology
25   gfm = vector("list",length(Rh))
26   gfm[[1]] = c("g_s","g_s")
27   gfm[[2]] = c("g_a","g_a")
28   gfm[[3]] = c("g_o","g_o")
29   gfm[[4]] = c("g_c","g_c")
30 } else { gfm = NULL }
31
32 # Specifications for forward model calculations
33 # a) flags for modified forward model calculation by
34 #   response for each relevant phenomenology
35 iResponse = TRUE
36
37 if( iResponse ){
38   iResponse = vector("list",length(Rh))
39   iResponse[[1]] = c(TRUE,FALSE)
40   iResponse[[2]] = c(TRUE,FALSE)
41 } else { iResponse = NULL }
42
43 # b) fixed quantities required by forward models
```

```

44 fPars = TRUE
45
46 if( fPars ){
47   fPars = vector("list",length(Rh))
48   fPars[[1]]$yield_scaling = 1/3
49   fPars[[2]]$yield_scaling = 1/3
50   fPars[[2]]$pressure_scaling = 1/3
51   fPars[[2]]$temp_scaling = 1/2
52   fPars[[3]]$yield_scaling = 1/3
53 } else { fPars = NULL }
54
55 # Specify number of starting values for optimization
56 nstart = 30
57
58 # number of cores to use for optimization
59 ncores_mle = 30
60
61 # Indicate use of BFGS optimization methods
62 bfgs = TRUE
63
64 # Location of R data files with starting values
65 # for input to MLE optimization
66 opt_files_in = c("../Opt/opt_1_0.RData",
67                  "../Opt/opt_2_0.RData",
68                  "../Opt/opt_3_eiv_0.RData",
69                  "../Opt/opt_4_eiv_0.RData")
70
71 # Location of R data file to write the results of
72 # MLE optimization
73 opt_files_out = "../opt.RData"
74
75 if( calp ){
76   # Initial start value for calibration inference parameters
77   cst = FALSE
78
79   if( cst ){
80     cst = numeric(p_cal$ncalp)
81     #cst[1] =
82   } else { cst = NULL }
83
84   # Confidence interval levels for calibration parameter inference
85   ci_lev = 0.95
86 } else {
87   cst = NULL
88   ci_lev = NULL

```

```

89  }
90
91  # Indicate phenomenology number and type (if needed
92  # for postprocessing)
93  Phen = TRUE
94
95  if( Phen ){
96    Phen = matrix(c(1,3,"Seismic","Optical"),nrow=2)
97  } else { Phen = NULL }
98
99  # Indicator of MLE gradient check
100  mle_grad_ck = TRUE
101
102  # Strategy for running parallel jobs (future package)
103  parallel_plan = "multicore"
104
105  # MLE calculations
106  p_cal = calc_mle_cal(p_cal,gen_dir,app_dir,fm,nst=nstart,
107                      ncor=ncores_mle,ci_lev=ci_lev,igrad=igrad,
108                      bfgs=bfgs,igrck=mle_grad_ck,g=gfm,iresp=iResponse,
109                      fp_fm=fPars,fopt_in=opt_files_in,Xst=NULL,
110                      cst=cst,fopt_out=opt_files_out,phen=Phen,
111                      pl=parallel_plan)
112  save.image()
113
114  #
115  # END MAXIMUM LIKELIHOOD CALCULATION
116  #

```

A.3 Benchmark Data: Bayesian Analysis

```
1 #
2 # BAYESIAN ANALYSIS
3 #
4
5 # Specify if Bayesian analysis is to be conducted
6 iBayes = TRUE
7
8 if( iBayes ){
9   # Source supporting R function
10  source(paste(gen_dir,"/calc_bayes_cal.r",sep=""))
11
12  # Indicator of prior distribution for forward model
13  # coefficients
14  iBetaPrior = TRUE
15
16  if( iBetaPrior ){
17    # location of code for computing log-prior densities and gradients
18    prior_files_beta = c("../Code/lp_beta_s.r","../Code/lp_beta_o.r")
19    if( igrad ){
20      gr_prior_files_beta = c("../Code/glp_beta_s.r",
21                              "../Code/glp_beta_o.r")
22    } else { gr_prior_files_beta = NULL }
23
24    # prior distribution for phenomenology 1
25    # forward model coefficients
26    p_cal$h[[1]]$lp_betat = vector("list",Th[1])
27    for( tt in 1:Th[1] ){
28      p_cal$h[[1]]$lp_betat[[tt]]$f = c("lp_s","lp_s")
29      if( igrad ){
30        p_cal$h[[1]]$lp_betat[[tt]]$g = c("lq_s","lq_s")
31      }
32    }
33
34    # prior distribution for phenomenology 3
35    # forward model coefficients
36    p_cal$h[[3]]$lp_beta$f = c("lp_o","lp_o")
37    if( igrad ){
38      p_cal$h[[3]]$lp_beta$g = c("lq_o","lq_o")
39    }
40  } else {
41    prior_files_beta = NULL
42    gr_prior_files_beta = NULL
43  }
```

```

44
45 # Indicator of prior distribution for calibration parameters
46 iCalPrior = FALSE
47
48 if( calp && iCalPrior ){
49     # location of code for computing log-prior densities and gradients
50     prior_files_calp = NULL
51     if( igrad ){
52         gr_prior_files_calp = NULL
53     } else { gr_prior_files_calp = NULL }
54
55     # prior distribution for calibration parameters (calp)
56     p_cal$lp_calp$f = "lp_c"
57     if( igrad ){ p_cal$lp_calp$g = "lq_c" }
58
59     # parameters for calibration parameter prior distribution
60     #p_cal$pi_c_mu =
61     #p_cal$pi_c_sd =
62 } else {
63     prior_files_calp = NULL
64     gr_prior_files_calp = NULL
65 }
66
67 # fixed scale parameters for variance component prior
68 # comment out if these parameters should vary
69 p_cal$A = 20
70
71 # eta parameter in Lewandowski-Kurowicka-Joe (LKJ) prior
72 # distribution for correlation parameters
73 p_cal$lp_corr$eta = 1
74
75 # FGSN parameters for errors-in-variables yields prior
76 # number of components
77 p_cal$K = 0
78 # total number of FGSN parameters
79 p_cal$p_fgsn = 0
80 if( eiv ){
81     p_cal$K = 2
82     p_cal$p_fgsn = p_cal$K + 2
83 }
84
85 # specify Markov chain Monte Carlo (MCMC) algorithm
86 # options: "RAM", "FME", or "NUTS"
87 iMCMC = "RAM"
88

```

```

89  # burn-in
90  nburn = 10000
91
92  # production
93  nmcmc = 20000
94
95  # posterior sample thinning rate
96  nthin = 20
97
98  # number of cores to use for optimization
99  ncores_map = 30
100
101  # number of cores to use for generating parallel MCMC chains
102  ncores_mc = 30
103
104  # Indicator of prior gradient check
105  prior_grad_ck = TRUE
106
107  # Indicator of posterior gradient check
108  post_grad_ck = TRUE
109
110  # Bayesian calculations
111  p_cal = calc_bayes_cal(p_cal,gen_dir,app_dir,nst=nstart,nburn=nburn,
112                        nmcmc=nmcmc,nthin=nthin,ncor_map=ncores_map,
113                        ncor_mc=ncores_mc,igrad=igrad,
114                        igrck_pr=prior_grad_ck,igrck_po=post_grad_ck,
115                        bfgs=bfgs,ibpr=iBetaPrior,icpr=iCalPrior,
116                        fpr_b=prior_files_beta,
117                        fgpr_b=gr_prior_files_beta,
118                        fpr_c=prior_files_calp,
119                        fgpr_c=gr_prior_files_calp,
120                        Xnom=NULL,imcmc=iMCMC,pl=parallel_plan)
121  save.image()
122 }
123
124 #
125 # END BAYESIAN ANALYSIS
126 #

```

A.4 Benchmark Data: Output File

```

1 > # Preprocessing for statistical analysis routines
2 > p_cal = prepro_cal(gen_dir,app_dir,dat_dir,dat_cal,Rh,pbeta,
3 +                   izmat=calc_Z,ieiv=ieiv,seiv=seiv,ewsd=eiv_w_sd,
4 +                   Th=Th,pbetat=pbetat,ibetar=ibetar,pvc_1=pvc_1,
5 +                   pvc_2=pvc_2,ptype=ptype,cnames=cal_par_names)
6 [1] "Warning: Insufficient number of observations per Source for Variance
7     Component models with Phenomenology 3 and Response 1."
8 [1] "Warning: Insufficient number of observations per Source for Variance
9     Component models with Phenomenology 3 and Response 2."
10 [1] "Warning: Insufficient number of observations per Source for Variance
11     Component models with Phenomenology 4 and Response 1."
12 [1] "Warning: Insufficient number of observations per Source for Variance
13     Component models with Phenomenology 4 and Response 2."
14
15 > # MLE calculations
16 > p_cal = calc_mle_cal(p_cal,gen_dir,app_dir,fm,nst=nstart,
17 +                   ncor=ncores_mle,ci_lev=ci_lev,igrad=igrad,
18 +                   bfgs=bfgs,igrck=mle_grad_ck,g=gfm,iresp=iResponse,
19 +                   fp_fm=fPars,fopt_in=opt_files_in,Xst=NULL,
20 +                   cst=cst,fopt_out=opt_files_out,phen=Phen,
21 +                   pl=parallel_plan)
22 [1] "MLE CONVERGENCE STATUS"
23
24 [1] 0
25 [1] 2
26 [1] "MAXIMUM LIKELIHOOD SUMMARY"
27
28 [1] "ERRORS-IN-VARIABLES YIELDS"
29
30      7      8      9     10     11     13     14     16     17     20     21     22     23
31 16.31 16.21 16.50 16.59 16.99 12.25 17.57 17.28 16.54 14.50 15.75 17.59 15.19
32     24     25     28     29     30     31     33     34     35     36     37     38     39
33 15.89 16.45 14.50 12.14 17.66 23.10 23.47 17.46 21.91 22.33 16.74 21.04 18.51
34
35 [1] "COMMON COEFFICIENTS"
36
37      [1] "Phenomenology: 2; Response: 1"
38
39      [1] 6.13 -1.13
40
41      [1] "Phenomenology: 2; Response: 2"
42
43      [1] -5.25 0.23

```


44
45 [1] "Phenomenology: 3; Response: 1"
46
47 [1] -10.52 0.37 0.63 0.42
48
49 [1] "Phenomenology: 3; Response: 2"
50
51 [1] -8.23 0.38 0.43 0.02
52
53 [1] "Phenomenology: 4; Response: 1"
54
55 [1] -3.11 0.42
56
57 [1] "Phenomenology: 4; Response: 2"
58
59 [1] -2.19 0.27
60
61 [1] "EMPLACEMENT CONDITION DEPENDENT COEFFICIENTS"
62
63 [1] "Phenomenology: 1; Emplacement: 1; Response: 1"
64
65 [1] -9.60 -1.42 -1.14 22.04 4.04
66
67 [1] "Phenomenology: 1; Emplacement: 1; Response: 2"
68
69 [1] 0.64 -1.85 -0.94 272.09 39.43
70
71 [1] "Phenomenology: 1; Emplacement: 2; Response: 1"
72
73 [1] -10.53 -1.25 -226.48 1.73 -4.81
74
75 [1] "Phenomenology: 1; Emplacement: 2; Response: 2"
76
77 [1] 0.75 -1.84 -871.41 2.74 -6.82
78
79 [1] "Phenomenology: 1; Emplacement: 3; Response: 1"
80
81 [1] -6.61 -1.64 -4.16 7.34 0.76
82
83 [1] "Phenomenology: 1; Emplacement: 3; Response: 2"
84
85 [1] 3.88 -1.87 -2.43 134.04 4.71
86
87 [1] "Phenomenology: 2; Emplacement: 1; Response: 1"
88

89 [1] 4.81

90

91 [1] "Phenomenology: 2; Emplacement: 1; Response: 2"

92

93 [1] -0.2

94

95 [1] "Phenomenology: 2; Emplacement: 2; Response: 1"

96

97 [1] 3.86

98

99 [1] "Phenomenology: 2; Emplacement: 2; Response: 2"

100

101 [1] -1.14

102

103 [1] "Phenomenology: 2; Emplacement: 3; Response: 1"

104

105 [1] 2.44

106

107 [1] "Phenomenology: 2; Emplacement: 3; Response: 2"

108

109 [1] -1.42

110

111 [1] "SOURCE VARIANCE COMPONENTS"

112

113 [1] "Phenomenology: 1; Response: 1"

114

115 [1] 0.0385

116

117 [1] "Phenomenology: 1; Response: 2"

118

119 [1] 0.1035

120

121 [1] "Phenomenology: 2; Response: 1"

122

123 [1] 0.1596

124

125 [1] "Phenomenology: 2; Response: 2"

126

127 [1] 0.0326

128

129 [1] "PATH VARIANCE COMPONENTS"

130

131 [1] "Phenomenology: 1; Response: 1"

132

133 [1] 0.1371

```

134
135         [1] "Phenomenology: 1; Response: 2"
136
137         [1] 0.1484
138
139         [1] "Phenomenology: 2; Response: 1"
140
141         [1] 0.0209
142
143         [1] "Phenomenology: 2; Response: 2"
144
145         [1] 0.0125
146
147 [1] "OBSERVATIONAL ERROR COVARIANCE PARAMETERS"
148
149 [1] "Phenomenology 1"
150
151 [1] "Variances"
152
153 [1] 0.0838 0.1855
154
155 [1] "Correlations"
156
157         [,1] [,2]
158 [1,]      1 0.41
159 [2,]      0 1.00
160
161 [1] "Phenomenology 2"
162
163 [1] "Variances"
164
165 [1] 0.0727 0.0168
166
167 [1] "Correlations"
168
169         [,1] [,2]
170 [1,]      1 -0.14
171 [2,]      0 1.00
172
173 [1] "Phenomenology 3"
174
175 [1] "Variances"
176
177 [1] 0.1866 0.1700
178

```

```

179 [1] "Correlations"
180
181      [,1] [,2]
182 [1,]    1 0.98
183 [2,]    0 1.00
184
185 [1] "Phenomenology 4"
186
187 [1] "Variances"
188
189 [1] 0.0149 0.0256
190
191 [1] "Correlations"
192
193      [,1] [,2]
194 [1,]    1 -0.13
195 [2,]    0 1.00
196
197 [1] "AIC = 1169.56"
198
199 [1] "BIC = 1520.85"
200
201 Loading required package: numDeriv
202 [1] "CHECK LOG-LIKELIHOOD GRADIENTS"
203
204 [1] "Analytic gradient"
205 [1] 4.188191e-04 -1.556874e-03 4.994343e-05 -1.027171e-03 -6.350244e-04
206 [6] 1.338318e-03 1.490413e-03 1.822911e-04 -8.781176e-05 -1.701393e-03
207 [11] -4.054896e-04 -7.994541e-04 1.325234e-04 -1.741897e-04 2.792636e-04
208 [16] -9.606525e-04 1.063311e-03 -3.157558e-04 8.514972e-04 3.354002e-04
209 [21] 3.638315e-04 9.931704e-04 -3.814680e-05 -1.555627e-03 -1.337597e-03
210 [26] -8.453346e-04 2.329911e-03 -1.515201e-03 2.635193e-03 -2.247225e-03
211 [31] 4.962788e-02 8.847048e-01 1.274354e-02 8.297122e-03 -5.205946e-02
212 [36] -9.278998e-01 -4.303701e-03 -4.130706e-04 8.968790e-04 2.401641e-03
213 [41] 1.118191e-03 -1.337758e-03 -3.501021e-05 1.277223e-03 2.204184e-03
214 [46] 2.931608e-05 -2.640035e-04 2.052454e-03 3.571648e-03 2.027448e-04
215 [51] 1.577952e-05 -1.079429e-04 -9.293545e-02 -4.560626e-01 3.368823e-02
216 [56] 4.059000e-02 2.137229e-01 -1.312622e-02 -6.693443e-02 1.761745e-03
217 [61] 5.619397e-03 2.169909e-02 -1.353354e-03 -1.499636e-03 8.976902e-04
218 [66] -1.358511e-04 2.812572e-04 -1.248915e-03 1.689187e-03 1.410754e-03
219 [71] -6.710970e-05 2.365466e-03 -4.556320e-04 5.121147e-05 -1.663705e-04
220 [76] 8.800844e-05 -1.087562e-04 -8.913278e-04 1.102226e-03 3.854918e-03
221 [81] 3.093168e-04 -6.789418e-04 -7.161085e-04 1.334897e-03 3.684368e-05
222 [86] 9.980138e-04 -1.477236e-03 1.778982e-03 -7.424502e-04 6.772199e-05
223 [91] 4.284607e-04 -6.758684e-03 1.061669e-03 5.782710e-04 3.310316e-04

```

```

224 [96] 1.364596e-03 -3.417969e-04 -1.286467e-03
225 [1] "Numerical gradient"
226 [1] 4.188192e-04 -1.556874e-03 4.994356e-05 -1.027171e-03 -6.350255e-04
227 [6] 1.338319e-03 1.490413e-03 1.822914e-04 -8.781189e-05 -1.701392e-03
228 [11] -4.054893e-04 -7.994541e-04 1.325236e-04 -1.741894e-04 2.792640e-04
229 [16] -9.606530e-04 1.063311e-03 -3.157562e-04 8.514979e-04 3.353999e-04
230 [21] 3.638308e-04 9.931698e-04 -3.814650e-05 -1.555627e-03 -1.337597e-03
231 [26] -8.453339e-04 2.329906e-03 -1.515215e-03 2.635188e-03 -2.247175e-03
232 [31] 4.962788e-02 8.847048e-01 1.274348e-02 8.297168e-03 -5.205947e-02
233 [36] -9.278998e-01 -4.303705e-03 -4.130724e-04 8.968807e-04 2.401647e-03
234 [41] 1.118189e-03 -1.337717e-03 -3.501020e-05 1.277219e-03 2.204184e-03
235 [46] 2.931500e-05 -2.639989e-04 2.052430e-03 3.571633e-03 2.025130e-04
236 [51] 1.577948e-05 -1.079426e-04 -9.293546e-02 -4.560625e-01 3.368823e-02
237 [56] 4.059000e-02 2.137229e-01 -1.312625e-02 -6.693443e-02 1.761745e-03
238 [61] 5.619398e-03 2.169909e-02 -1.353354e-03 -1.499640e-03 8.976936e-04
239 [66] -1.358528e-04 2.812669e-04 -1.248928e-03 1.689198e-03 1.410786e-03
240 [71] -6.710967e-05 2.365467e-03 -4.556348e-04 5.131107e-05 -1.663693e-04
241 [76] 8.800439e-05 -1.087570e-04 -8.913444e-04 1.102231e-03 3.854924e-03
242 [81] 3.093061e-04 -6.789373e-04 -7.161076e-04 1.334888e-03 3.683721e-05
243 [86] 9.980131e-04 -1.477232e-03 1.778972e-03 -7.427133e-04 6.771442e-05
244 [91] 4.284644e-04 -6.755292e-03 1.061612e-03 5.782736e-04 3.310533e-04
245 [96] 1.364593e-03 -3.417913e-04 -1.284712e-03
246 [1] "Difference"
247 [1] -3.392416e-06 2.631184e-07
248
249 + # Bayesian calculations
250 + p_cal = calc_bayes_cal(p_cal,gen_dir,app_dir,nst=nstart,nburn=nburn,
251 + nmcmc=nmcmc,nthin=nthin,ncor_map=ncores_map,
252 + ncor_mc=ncores_mc,igrad=igrad,
253 + igrck_pr=prior_grad_ck,igrck_po=post_grad_ck,
254 + bfgs=bfgs,ibpr=iBetaPrior,icpr=iCalPrior,
255 + fpr_b=prior_files_beta,
256 + fgpr_b=gr_prior_files_beta,
257 + fpr_c=prior_files_calp,
258 + fgpr_c=gr_prior_files_calp,
259 + Xnom=NULL,imcmc=iMCMC,pl=parallel_plan)
260 [1] "Perturbation added to Hessian diagonals: 1e-06"
261 [1] "MAP CONVERGENCE STATUS"
262
263 [1] 0
264 [1] 2
265 [1] "MAXIMUM A POSTERIORI SUMMARY"
266
267 [1] "ERRORS-IN-VARIABLES YIELDS"
268

```

	7	8	9	10	11	13	14	16	17	20	21	22	23
269	16.29	16.21	16.51	16.58	16.99	12.28	17.57	17.27	16.54	14.49	15.75	17.57	15.19
270	24	25	28	29	30	31	33	34	35	36	37	38	39
271	15.89	16.44	14.51	12.17	17.66	23.09	23.47	17.46	21.92	22.34	16.74	21.05	18.50
272													
273													
274	[1] "COMMON COEFFICIENTS"												
275													
276													
277													
278													
279													
280													
281													
282													
283													
284													
285													
286													
287													
288													
289													
290													
291													
292													
293													
294													
295													
296													
297													
298													
299													
300	[1] "EMPLACEMENT CONDITION DEPENDENT COEFFICIENTS"												
301													
302													
303													
304													
305													
306													
307													
308													
309													
310													
311													
312													
313													

```

314      [1] "Phenomenology: 1; Emplacement: 2; Response: 2"
315
316      [1]      0.64      -1.82 -611113.40      2.86      -13.42
317
318      [1] "Phenomenology: 1; Emplacement: 3; Response: 1"
319
320      [1] -6.59 -1.64 -4.19  7.28  0.75
321
322      [1] "Phenomenology: 1; Emplacement: 3; Response: 2"
323
324      [1]  3.91  -1.87  -2.48 359.57  13.34
325
326      [1] "Phenomenology: 2; Emplacement: 1; Response: 1"
327
328      [1] 4.81
329
330      [1] "Phenomenology: 2; Emplacement: 1; Response: 2"
331
332      [1] -0.2
333
334      [1] "Phenomenology: 2; Emplacement: 2; Response: 1"
335
336      [1] 3.86
337
338      [1] "Phenomenology: 2; Emplacement: 2; Response: 2"
339
340      [1] -1.14
341
342      [1] "Phenomenology: 2; Emplacement: 3; Response: 1"
343
344      [1] 2.44
345
346      [1] "Phenomenology: 2; Emplacement: 3; Response: 2"
347
348      [1] -1.43
349
350 [1] "SOURCE VARIANCE COMPONENTS"
351
352      [1] "Phenomenology: 1; Response: 1"
353
354      [1] 0.0419
355
356      [1] "Phenomenology: 1; Response: 2"
357
358      [1] 0.149

```

```

359
360         [1] "Phenomenology: 2; Response: 1"
361
362         [1] 0.1674
363
364         [1] "Phenomenology: 2; Response: 2"
365
366         [1] 0.034
367
368 [1] "PATH VARIANCE COMPONENTS"
369
370         [1] "Phenomenology: 1; Response: 1"
371
372         [1] 0.1382
373
374         [1] "Phenomenology: 1; Response: 2"
375
376         [1] 0.1495
377
378         [1] "Phenomenology: 2; Response: 1"
379
380         [1] 0.0217
381
382         [1] "Phenomenology: 2; Response: 2"
383
384         [1] 0.0128
385
386 [1] "OBSERVATIONAL ERROR COVARIANCE PARAMETERS"
387
388 [1] "Phenomenology 1"
389
390 [1] "Variances"
391
392 [1] 0.0833 0.1841
393
394 [1] "Correlations"
395
396         [,1] [,2]
397 [1,]      1 0.41
398 [2,]      0 1.00
399
400 [1] "Phenomenology 2"
401
402 [1] "Variances"
403

```



```

404 [1] 0.0724 0.0166
405
406 [1] "Correlations"
407
408      [,1] [,2]
409 [1,]    1 -0.13
410 [2,]    0  1.00
411
412 [1] "Phenomenology 3"
413
414 [1] "Variances"
415
416 [1] 0.1661 0.1681
417
418 [1] "Correlations"
419
420      [,1] [,2]
421 [1,]    1  0.97
422 [2,]    0  1.00
423
424 [1] "Phenomenology 4"
425
426 [1] "Variances"
427
428 [1] 0.0152 0.0217
429
430 [1] "Correlations"
431
432      [,1] [,2]
433 [1,]    1 -0.09
434 [2,]    0  1.00
435
436 [1] "FGSN PRIOR PARAMETERS"
437
438 [1] "Alpha = 17.34"
439 [1] "Lambda squared = 8"
440 [1] "Omega = -1.72" "Omega = 0.59"
441
442 [1] "CHECK LOG-PRIOR GRADIENTS"
443
444 [1] "Analytic gradient"
445 [1] -1.083895e-01 -8.001752e-02 -1.825562e-01 -2.082898e-01 -3.560802e-01
446 [6]  1.985120e+00 -5.649089e-01 -4.608885e-01 -1.937528e-01  3.608033e-01
447 [11]  5.837263e-02 -5.678288e-01  2.018810e-01  1.912606e-02 -1.586782e-01
448 [16]  3.568148e-01  2.235514e+00 -5.952330e-01 -4.140344e-01 -6.896776e-01

```

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449 [21] -5.296586e-01 4.463627e-01 2.809968e-01 -2.648248e-01 2.124770e-01
450 [26] -7.827891e-01 0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00
451 [31] 0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00
452 [36] 0.000000e+00 0.000000e+00 -6.660679e-02 0.000000e+00 0.000000e+00
453 [41] 0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00 1.000000e+00
454 [46] 0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00 1.998576e-03
455 [51] 0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00 1.145573e-03
456 [56] 0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00 1.491323e-03
457 [61] 0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00 6.933747e-01
458 [66] 0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00 1.000000e+00
459 [71] 0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00
460 [76] 0.000000e+00 0.000000e+00 0.000000e+00 4.998954e-01 4.996275e-01
461 [81] 4.995818e-01 4.999150e-01 4.996546e-01 4.996265e-01 4.999459e-01
462 [86] 4.999680e-01 2.220446e-16 -5.044408e-01 -2.841793e+00 -2.220446e-16
463 [91] -9.462299e-01 3.115060e+00 0.000000e+00 1.801975e+00 -7.070504e+00
464 [96] -4.440892e-16 -9.750215e-01 1.857409e+00 1.396281e-04 -1.049843e-03
465 [101] -2.513265e-04 -4.505594e-04
466 [1] "Numerical gradient"
467 [1] -1.083895e-01 -8.001752e-02 -1.825562e-01 -2.082898e-01 -3.560802e-01
468 [6] 1.985120e+00 -5.649089e-01 -4.608885e-01 -1.937528e-01 3.608033e-01
469 [11] 5.837263e-02 -5.678288e-01 2.018810e-01 1.912606e-02 -1.586782e-01
470 [16] 3.568148e-01 2.235514e+00 -5.952330e-01 -4.140344e-01 -6.896776e-01
471 [21] -5.296586e-01 4.463627e-01 2.809968e-01 -2.648248e-01 2.124770e-01
472 [26] -7.827891e-01 0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00
473 [31] 0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00
474 [36] 0.000000e+00 0.000000e+00 -6.660679e-02 0.000000e+00 0.000000e+00
475 [41] 0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00 1.000000e+00
476 [46] 0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00 1.998576e-03
477 [51] 0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00 1.145573e-03
478 [56] 0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00 1.491323e-03
479 [61] 0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00 6.933747e-01
480 [66] 0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00 1.000000e+00
481 [71] 0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00
482 [76] 0.000000e+00 0.000000e+00 0.000000e+00 4.998954e-01 4.996275e-01
483 [81] 4.995818e-01 4.999150e-01 4.996546e-01 4.996265e-01 4.999459e-01
484 [86] 4.999680e-01 0.000000e+00 -5.044408e-01 -2.841793e+00 0.000000e+00
485 [91] -9.462299e-01 3.115061e+00 0.000000e+00 1.801975e+00 -7.070504e+00
486 [96] 1.306509e-09 -9.750215e-01 1.857409e+00 1.396281e-04 -1.049841e-03
487 [101] -2.513257e-04 -4.505617e-04
488 [1] "Difference"
489 [1] -1.121221e-07 8.044184e-09
490
491 [1] "CHECK LOG-POSTERIOR GRADIENTS"
492
493 [1] "Analytic gradient"

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494 [1] -1.149266e-03 -1.838546e-03 1.300391e-03 -3.703966e-03 -5.097952e-03
495 [6] 1.272357e-04 -3.177396e-03 -2.127513e-03 -3.030570e-03 1.723859e-04
496 [11] -1.926781e-03 -3.432909e-03 -2.438480e-03 -8.139090e-03 -1.985777e-03
497 [16] -6.073749e-03 -2.637149e-02 -3.853780e-03 -1.884415e-04 5.581015e-04
498 [21] -2.625654e-04 -4.404396e-03 4.581790e-03 -8.997515e-04 -8.930853e-04
499 [26] -2.632595e-03 -5.107163e-04 -2.469743e-03 1.196296e-03 -3.640142e-04
500 [31] -3.614957e-02 -4.343029e-01 -1.360530e-02 -1.411979e-01 1.254912e-03
501 [36] -1.209147e-01 -2.216253e-03 -2.900101e-02 -1.215675e-03 -5.974240e-03
502 [41] 1.497459e-03 1.368800e-02 2.329651e-03 9.599978e-03 1.323139e-03
503 [46] 4.021822e-04 1.416579e-03 7.426873e-05 -2.378385e-03 1.964426e-03
504 [51] -4.297665e-02 -8.539621e-03 1.809286e-03 9.907109e-05 1.144601e-03
505 [56] -9.229589e-04 -4.246081e-04 -4.528716e-03 -1.678044e-02 1.500407e-03
506 [61] 3.599532e-03 3.044843e-03 1.454813e-03 3.863822e-03 -2.902271e-03
507 [66] -3.411561e-05 -2.428791e-04 -1.514788e-04 -1.771897e-03 5.794661e-04
508 [71] 1.114292e-05 -2.909921e-04 7.606831e-04 -4.591566e-04 -7.116044e-04
509 [76] 1.204441e-04 5.667956e-04 -6.795606e-04 -2.911962e-05 1.159801e-03
510 [81] -5.502487e-04 -5.177448e-04 -3.804081e-04 1.314520e-03 8.743294e-05
511 [86] 3.644565e-04 -3.162584e-03 -2.810660e-04 6.117701e-04 1.101393e-03
512 [91] 2.034093e-03 -1.043395e-04 4.453171e-02 -1.434648e-02 -1.047941e-01
513 [96] 5.990844e-04 3.530632e-04 -2.891184e-03 1.396281e-04 -1.049843e-03
514 [101] -2.513265e-04 -4.505594e-04
515 [1] "Numerical gradient"
516 [1] -1.149266e-03 -1.838546e-03 1.300391e-03 -3.703965e-03 -5.097954e-03
517 [6] 1.272357e-04 -3.177395e-03 -2.127514e-03 -3.030570e-03 1.723858e-04
518 [11] -1.926780e-03 -3.432909e-03 -2.438479e-03 -8.139091e-03 -1.985778e-03
519 [16] -6.073749e-03 -2.637149e-02 -3.853780e-03 -1.884412e-04 5.581013e-04
520 [21] -2.625668e-04 -4.404395e-03 4.581789e-03 -8.997507e-04 -8.930859e-04
521 [26] -2.632594e-03 -5.107188e-04 -2.469731e-03 1.196301e-03 -3.641744e-04
522 [31] -3.614957e-02 -4.343030e-01 -1.360514e-02 -1.411981e-01 1.254910e-03
523 [36] -1.209147e-01 -2.216243e-03 -2.900104e-02 -1.215670e-03 -5.974262e-03
524 [41] 1.497449e-03 1.368798e-02 2.329657e-03 9.599997e-03 1.323294e-03
525 [46] 4.021837e-04 1.416597e-03 7.426902e-05 -2.378375e-03 1.964426e-03
526 [51] -4.295985e-02 -8.539617e-03 1.809285e-03 9.907354e-05 1.144601e-03
527 [56] -9.229615e-04 -4.246078e-04 -4.528720e-03 -1.678044e-02 1.500407e-03
528 [61] 3.599541e-03 3.044845e-03 1.454808e-03 3.863828e-03 -2.902276e-03
529 [66] -3.411619e-05 -2.428746e-04 -1.514816e-04 -1.771907e-03 5.794637e-04
530 [71] 1.114288e-05 -2.909918e-04 7.606828e-04 -4.592070e-04 -7.116050e-04
531 [76] 1.204412e-04 5.667965e-04 -6.795506e-04 -2.910864e-05 1.159832e-03
532 [81] -5.502844e-04 -5.177281e-04 -3.804223e-04 1.314504e-03 8.744951e-05
533 [86] 3.644470e-04 -3.162563e-03 -2.810115e-04 6.118833e-04 1.101437e-03
534 [91] 2.034099e-03 -1.017935e-04 4.453148e-02 -1.434645e-02 -1.047940e-01
535 [96] 5.990814e-04 3.530695e-04 -2.890208e-03 1.396279e-04 -1.049842e-03
536 [101] -2.513385e-04 -4.505767e-04
537 [1] "Difference"
538 [1] -1.679777e-05 2.350832e-07

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539
540 [1] "ACCEPTANCE RATES:"
541
542 [1] "Core 1: 0.215"
543 [1] "Core 2: 0.221"
544 [1] "Core 3: 0.231"
545 [1] "Core 4: 0.227"
546 [1] "Core 5: 0.226"
547 [1] "Core 6: 0.228"
548 [1] "Core 7: 0.242"
549 [1] "Core 8: 0.224"
550 [1] "Core 9: 0.225"
551 [1] "Core 10: 0.227"
552 [1] "Core 11: 0.241"
553 [1] "Core 12: 0.222"
554 [1] "Core 13: 0.204"
555 [1] "Core 14: 0.091"
556 [1] "Core 15: 0.228"
557 [1] "Core 16: 0.213"
558 [1] "Core 17: 0.226"
559 [1] "Core 18: 0.208"
560 [1] "Core 19: 0.228"
561 [1] "Core 20: 0.237"
562 [1] "Core 21: 0.228"
563 [1] "Core 22: 0.207"
564 [1] "Core 23: 0.236"
565 [1] "Core 24: 0.219"
566 [1] "Core 25: 0.219"
567 [1] "Core 26: 0.221"
568 [1] "Core 27: 0.216"
569 [1] "Core 28: 0.176"
570 [1] "Core 29: 0.226"
571 [1] "Core 30: 0.229"
572
573 [1] "POSTERIOR SUMMARY"
574
575 [1] "ERRORS-IN-VARIABLES YIELDS"
576
577 [1] "POSTERIOR MEAN: 16.28" "POSTERIOR MEAN: 16.19" "POSTERIOR MEAN: 16.49"
578 [4] "POSTERIOR MEAN: 16.58" "POSTERIOR MEAN: 17" "POSTERIOR MEAN: 12.27"
579 [7] "POSTERIOR MEAN: 17.57" "POSTERIOR MEAN: 17.28" "POSTERIOR MEAN: 16.56"
580 [10] "POSTERIOR MEAN: 14.51" "POSTERIOR MEAN: 15.73" "POSTERIOR MEAN: 17.58"
581 [13] "POSTERIOR MEAN: 15.21" "POSTERIOR MEAN: 15.89" "POSTERIOR MEAN: 16.45"
582 [16] "POSTERIOR MEAN: 14.5" "POSTERIOR MEAN: 12.19" "POSTERIOR MEAN: 17.64"
583 [19] "POSTERIOR MEAN: 23.09" "POSTERIOR MEAN: 23.46" "POSTERIOR MEAN: 17.46"

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584 [22] "POSTERIOR MEAN: 21.96" "POSTERIOR MEAN: 22.36" "POSTERIOR MEAN: 16.72"

585 [25] "POSTERIOR MEAN: 21.06" "POSTERIOR MEAN: 18.52"

586

587 [1] "LEVEL 2.5%: 16.08" "LEVEL 2.5%: 16.03" "LEVEL 2.5%: 16.3"

588 [4] "LEVEL 2.5%: 16.44" "LEVEL 2.5%: 16.86" "LEVEL 2.5%: 12.12"

589 [7] "LEVEL 2.5%: 17.35" "LEVEL 2.5%: 17.09" "LEVEL 2.5%: 16.4"

590 [10] "LEVEL 2.5%: 14.29" "LEVEL 2.5%: 15.57" "LEVEL 2.5%: 17.38"

591 [13] "LEVEL 2.5%: 15.02" "LEVEL 2.5%: 15.75" "LEVEL 2.5%: 16.3"

592 [16] "LEVEL 2.5%: 14.34" "LEVEL 2.5%: 12.07" "LEVEL 2.5%: 17.5"

593 [19] "LEVEL 2.5%: 22.93" "LEVEL 2.5%: 23.31" "LEVEL 2.5%: 17.32"

594 [22] "LEVEL 2.5%: 21.8" "LEVEL 2.5%: 22.16" "LEVEL 2.5%: 16.51"

595 [25] "LEVEL 2.5%: 20.83" "LEVEL 2.5%: 18.24"

596

597 [1] "LEVEL 5%: 16.1" "LEVEL 5%: 16.05" "LEVEL 5%: 16.33" "LEVEL 5%: 16.45"

598 [5] "LEVEL 5%: 16.89" "LEVEL 5%: 12.14" "LEVEL 5%: 17.39" "LEVEL 5%: 17.12"

599 [9] "LEVEL 5%: 16.41" "LEVEL 5%: 14.34" "LEVEL 5%: 15.6" "LEVEL 5%: 17.43"

600 [13] "LEVEL 5%: 15.05" "LEVEL 5%: 15.77" "LEVEL 5%: 16.32" "LEVEL 5%: 14.38"

601 [17] "LEVEL 5%: 12.09" "LEVEL 5%: 17.52" "LEVEL 5%: 22.96" "LEVEL 5%: 23.32"

602 [21] "LEVEL 5%: 17.33" "LEVEL 5%: 21.84" "LEVEL 5%: 22.21" "LEVEL 5%: 16.59"

603 [25] "LEVEL 5%: 20.9" "LEVEL 5%: 18.38"

604

605 [1] "LEVEL 50%: 16.29" "LEVEL 50%: 16.19" "LEVEL 50%: 16.48" "LEVEL 50%: 16.58"

606 [5] "LEVEL 50%: 17.01" "LEVEL 50%: 12.26" "LEVEL 50%: 17.56" "LEVEL 50%: 17.29"

607 [9] "LEVEL 50%: 16.55" "LEVEL 50%: 14.52" "LEVEL 50%: 15.74" "LEVEL 50%: 17.59"

608 [13] "LEVEL 50%: 15.21" "LEVEL 50%: 15.89" "LEVEL 50%: 16.44" "LEVEL 50%: 14.49"

609 [17] "LEVEL 50%: 12.18" "LEVEL 50%: 17.64" "LEVEL 50%: 23.09" "LEVEL 50%: 23.48"

610 [21] "LEVEL 50%: 17.46" "LEVEL 50%: 21.96" "LEVEL 50%: 22.35" "LEVEL 50%: 16.71"

611 [25] "LEVEL 50%: 21.05" "LEVEL 50%: 18.52"

612

613 [1] "LEVEL 95%: 16.44" "LEVEL 95%: 16.31" "LEVEL 95%: 16.63" "LEVEL 95%: 16.71"

614 [5] "LEVEL 95%: 17.11" "LEVEL 95%: 12.38" "LEVEL 95%: 17.74" "LEVEL 95%: 17.41"

615 [9] "LEVEL 95%: 16.69" "LEVEL 95%: 14.62" "LEVEL 95%: 15.88" "LEVEL 95%: 17.72"

616 [13] "LEVEL 95%: 15.36" "LEVEL 95%: 16.01" "LEVEL 95%: 16.59" "LEVEL 95%: 14.69"

617 [17] "LEVEL 95%: 12.32" "LEVEL 95%: 17.75" "LEVEL 95%: 23.19" "LEVEL 95%: 23.61"

618 [21] "LEVEL 95%: 17.6" "LEVEL 95%: 22.08" "LEVEL 95%: 22.52" "LEVEL 95%: 16.85"

619 [25] "LEVEL 95%: 21.21" "LEVEL 95%: 18.68"

620

621 [1] "LEVEL 97.5%: 16.46" "LEVEL 97.5%: 16.36" "LEVEL 97.5%: 16.65"

622 [4] "LEVEL 97.5%: 16.72" "LEVEL 97.5%: 17.13" "LEVEL 97.5%: 12.39"

623 [7] "LEVEL 97.5%: 17.76" "LEVEL 97.5%: 17.43" "LEVEL 97.5%: 16.7"

624 [10] "LEVEL 97.5%: 14.63" "LEVEL 97.5%: 15.9" "LEVEL 97.5%: 17.73"

625 [13] "LEVEL 97.5%: 15.38" "LEVEL 97.5%: 16.08" "LEVEL 97.5%: 16.61"

626 [16] "LEVEL 97.5%: 14.72" "LEVEL 97.5%: 12.35" "LEVEL 97.5%: 17.77"

627 [19] "LEVEL 97.5%: 23.21" "LEVEL 97.5%: 23.62" "LEVEL 97.5%: 17.61"

628 [22] "LEVEL 97.5%: 22.1" "LEVEL 97.5%: 22.55" "LEVEL 97.5%: 16.86"

```

629 [25] "LEVEL 97.5%: 21.25" "LEVEL 97.5%: 18.69"
630
631 [1] "COMMON COEFFICIENTS"
632
633     [1] "Phenomenology: 2; Response: 1"
634
635     [1] "POSTERIOR MEAN: 6.14" "POSTERIOR MEAN: -1.13"
636
637     [1] "LEVEL 2.5%: 5.81" "LEVEL 2.5%: -1.16"
638
639     [1] "LEVEL 5%: 5.97" "LEVEL 5%: -1.16"
640
641     [1] "LEVEL 50%: 6.14" "LEVEL 50%: -1.14"
642
643     [1] "LEVEL 95%: 6.31" "LEVEL 95%: -1.11"
644
645     [1] "LEVEL 97.5%: 6.34" "LEVEL 97.5%: -1.09"
646
647     [1] "Phenomenology: 2; Response: 2"
648
649     [1] "POSTERIOR MEAN: -5.26" "POSTERIOR MEAN: 0.23"
650
651     [1] "LEVEL 2.5%: -5.36" "LEVEL 2.5%: 0.21"
652
653     [1] "LEVEL 5%: -5.34" "LEVEL 5%: 0.22"
654
655     [1] "LEVEL 50%: -5.26" "LEVEL 50%: 0.23"
656
657     [1] "LEVEL 95%: -5.14" "LEVEL 95%: 0.25"
658
659     [1] "LEVEL 97.5%: -5.12" "LEVEL 97.5%: 0.25"
660
661     [1] "Phenomenology: 3; Response: 1"
662
663     [1] "POSTERIOR MEAN: -10.86" "POSTERIOR MEAN: 0.37" "POSTERIOR MEAN: 1.18"
664 [4] "POSTERIOR MEAN: 0.75"
665
666     [1] "LEVEL 2.5%: -11.63" "LEVEL 2.5%: 0.31" "LEVEL 2.5%: 0.55"
667 [4] "LEVEL 2.5%: 0.56"
668
669     [1] "LEVEL 5%: -11.52" "LEVEL 5%: 0.34" "LEVEL 5%: 0.62" "LEVEL 5%: 0.59"
670
671     [1] "LEVEL 50%: -10.9" "LEVEL 50%: 0.37" "LEVEL 50%: 1.08" "LEVEL 50%: 0.76"
672
673     [1] "LEVEL 95%: -10.13" "LEVEL 95%: 0.4" "LEVEL 95%: 1.98"

```

674 [4] "LEVEL 95%: 0.89"

675

676 [1] "LEVEL 97.5%: -10.01" "LEVEL 97.5%: 0.43" "LEVEL 97.5%: 2.11"

677 [4] "LEVEL 97.5%: 0.93"

678

679 [1] "Phenomenology: 3; Response: 2"

680

681 [1] "POSTERIOR MEAN: -9.76" "POSTERIOR MEAN: 0.41" "POSTERIOR MEAN: 3.09"

682 [4] "POSTERIOR MEAN: 3.86"

683

684 [1] "LEVEL 2.5%: -12.5" "LEVEL 2.5%: 0.34" "LEVEL 2.5%: 0.76"

685 [4] "LEVEL 2.5%: 1.84"

686

687 [1] "LEVEL 5%: -11.57" "LEVEL 5%: 0.36" "LEVEL 5%: 0.85" "LEVEL 5%: 1.99"

688

689 [1] "LEVEL 50%: -9.64" "LEVEL 50%: 0.41" "LEVEL 50%: 1.89" "LEVEL 50%: 3.09"

690

691 [1] "LEVEL 95%: -8.79" "LEVEL 95%: 0.44" "LEVEL 95%: 14.86" "LEVEL 95%: 9.13"

692

693 [1] "LEVEL 97.5%: -8.6" "LEVEL 97.5%: 0.46" "LEVEL 97.5%: 17.67"

694 [4] "LEVEL 97.5%: 9.82"

695

696 [1] "Phenomenology: 4; Response: 1"

697

698 [1] "POSTERIOR MEAN: -3.06" "POSTERIOR MEAN: 0.42"

699

700 [1] "LEVEL 2.5%: -3.82" "LEVEL 2.5%: 0.38"

701

702 [1] "LEVEL 5%: -3.76" "LEVEL 5%: 0.38"

703

704 [1] "LEVEL 50%: -3.09" "LEVEL 50%: 0.42"

705

706 [1] "LEVEL 95%: -2.28" "LEVEL 95%: 0.45"

707

708 [1] "LEVEL 97.5%: -2.14" "LEVEL 97.5%: 0.46"

709

710 [1] "Phenomenology: 4; Response: 2"

711

712 [1] "POSTERIOR MEAN: -2.23" "POSTERIOR MEAN: 0.27"

713

714 [1] "LEVEL 2.5%: -2.9" "LEVEL 2.5%: 0.23"

715

716 [1] "LEVEL 5%: -2.83" "LEVEL 5%: 0.24"

717

718 [1] "LEVEL 50%: -2.26" "LEVEL 50%: 0.27"

719 [1] "LEVEL 95%: -1.66" "LEVEL 95%: 0.3"

720

721 [1] "LEVEL 97.5%: -1.57" "LEVEL 97.5%: 0.3"

722

723 [1] "EMPLACEMENT CONDITION DEPENDENT COEFFICIENTS"

724

725 [1] "Phenomenology: 1; Emplacement: 1; Response: 1"

726

727 [1] "POSTERIOR MEAN: -9.63" "POSTERIOR MEAN: -1.41" "POSTERIOR MEAN: -1.12"

728

729 [4] "POSTERIOR MEAN: 24.45" "POSTERIOR MEAN: 4.67"

730

731 [1] "LEVEL 2.5%: -10.57" "LEVEL 2.5%: -1.55" "LEVEL 2.5%: -1.34"

732 [4] "LEVEL 2.5%: 9.68" "LEVEL 2.5%: 1.83"

733

734 [1] "LEVEL 5%: -10.43" "LEVEL 5%: -1.53" "LEVEL 5%: -1.32" "LEVEL 5%: 10.84"

735 [5] "LEVEL 5%: 2.19"

736

737 [1] "LEVEL 50%: -9.59" "LEVEL 50%: -1.42" "LEVEL 50%: -1.11" "LEVEL 50%: 24.94"

738 [5] "LEVEL 50%: 4.51"

739

740 [1] "LEVEL 95%: -8.95" "LEVEL 95%: -1.28" "LEVEL 95%: -0.96" "LEVEL 95%: 40.07"

741 [5] "LEVEL 95%: 7.96"

742

743 [1] "LEVEL 97.5%: -8.76" "LEVEL 97.5%: -1.25" "LEVEL 97.5%: -0.94"

744 [4] "LEVEL 97.5%: 45.25" "LEVEL 97.5%: 8.35"

745

746 [1] "Phenomenology: 1; Emplacement: 1; Response: 2"

747

748 [1] "POSTERIOR MEAN: 138.31" "POSTERIOR MEAN: -1.86"

749 [3] "POSTERIOR MEAN: -366282.97" "POSTERIOR MEAN: 0.01"

750 [5] "POSTERIOR MEAN: -7.86"

751

752 [1] "LEVEL 2.5%: 81.12" "LEVEL 2.5%: -2.03" "LEVEL 2.5%: -604586.39"

753 [4] "LEVEL 2.5%: 0" "LEVEL 2.5%: -8.87"

754

755 [1] "LEVEL 5%: 86.18" "LEVEL 5%: -1.97" "LEVEL 5%: -575072.97"

756 [4] "LEVEL 5%: 0" "LEVEL 5%: -8.37"

757

758 [1] "LEVEL 50%: 141.26" "LEVEL 50%: -1.86" "LEVEL 50%: -352656.95"

759 [4] "LEVEL 50%: 0.01" "LEVEL 50%: -7.85"

760

761 [1] "LEVEL 95%: 187.55" "LEVEL 95%: -1.73" "LEVEL 95%: -192926.29"

762 [4] "LEVEL 95%: 0.01" "LEVEL 95%: -7.56"

763

764 [1] "LEVEL 97.5%: 198.51" "LEVEL 97.5%: -1.72"
 765 [3] "LEVEL 97.5%: -172574.34" "LEVEL 97.5%: 0.01"
 766 [5] "LEVEL 97.5%: -6.77"
 767
 768 [1] "Phenomenology: 1; Emplacement: 2; Response: 1"
 769
 770 [1] "POSTERIOR MEAN: -10.46" "POSTERIOR MEAN: -1.25"
 771 [3] "POSTERIOR MEAN: -1197432.49" "POSTERIOR MEAN: 1.72"
 772 [5] "POSTERIOR MEAN: -13.31"
 773
 774 [1] "LEVEL 2.5%: -11.38" "LEVEL 2.5%: -1.37"
 775 [3] "LEVEL 2.5%: -2098849.25" "LEVEL 2.5%: 1.06"
 776 [5] "LEVEL 2.5%: -14.2"
 777
 778 [1] "LEVEL 5%: -11.22" "LEVEL 5%: -1.35" "LEVEL 5%: -1973177.24"
 779 [4] "LEVEL 5%: 1.11" "LEVEL 5%: -14.06"
 780
 781 [1] "LEVEL 50%: -10.44" "LEVEL 50%: -1.25" "LEVEL 50%: -1165068.74"
 782 [4] "LEVEL 50%: 1.63" "LEVEL 50%: -13.36"
 783
 784 [1] "LEVEL 95%: -9.66" "LEVEL 95%: -1.15" "LEVEL 95%: -549361.2"
 785 [4] "LEVEL 95%: 2.51" "LEVEL 95%: -12.45"
 786
 787 [1] "LEVEL 97.5%: -9.53" "LEVEL 97.5%: -1.14"
 788 [3] "LEVEL 97.5%: -501739.14" "LEVEL 97.5%: 2.61"
 789 [5] "LEVEL 97.5%: -12.31"
 790
 791 [1] "Phenomenology: 1; Emplacement: 2; Response: 2"
 792
 793 [1] "POSTERIOR MEAN: 0.45" "POSTERIOR MEAN: -1.8"
 794 [3] "POSTERIOR MEAN: -960525.53" "POSTERIOR MEAN: 3.16"
 795 [5] "POSTERIOR MEAN: -13.96"
 796
 797 [1] "LEVEL 2.5%: -0.9" "LEVEL 2.5%: -1.99"
 798 [3] "LEVEL 2.5%: -1921549.46" "LEVEL 2.5%: 1.12"
 799 [5] "LEVEL 2.5%: -15.43"
 800
 801 [1] "LEVEL 5%: -0.56" "LEVEL 5%: -1.97" "LEVEL 5%: -1807267.89"
 802 [4] "LEVEL 5%: 1.43" "LEVEL 5%: -15.27"
 803
 804 [1] "LEVEL 50%: 0.49" "LEVEL 50%: -1.8" "LEVEL 50%: -893450.78"
 805 [4] "LEVEL 50%: 3.06" "LEVEL 50%: -13.92"
 806
 807 [1] "LEVEL 95%: 1.45" "LEVEL 95%: -1.66" "LEVEL 95%: -310056.11"
 808 [4] "LEVEL 95%: 5.26" "LEVEL 95%: -12.87"

809
810 [1] "LEVEL 97.5%: 1.54" "LEVEL 97.5%: -1.58"
811 [3] "LEVEL 97.5%: -232888.11" "LEVEL 97.5%: 5.5"
812 [5] "LEVEL 97.5%: -12.71"
813
814 [1] "Phenomenology: 1; Emplacement: 3; Response: 1"
815
816 [1] "POSTERIOR MEAN: -6.65" "POSTERIOR MEAN: -1.63" "POSTERIOR MEAN: -4.24"
817 [4] "POSTERIOR MEAN: 7.14" "POSTERIOR MEAN: 0.68"
818
819 [1] "LEVEL 2.5%: -7.48" "LEVEL 2.5%: -1.8" "LEVEL 2.5%: -4.94"
820 [4] "LEVEL 2.5%: 5.2" "LEVEL 2.5%: 0.31"
821
822 [1] "LEVEL 5%: -7.41" "LEVEL 5%: -1.77" "LEVEL 5%: -4.85" "LEVEL 5%: 5.63"
823 [5] "LEVEL 5%: 0.35"
824
825 [1] "LEVEL 50%: -6.64" "LEVEL 50%: -1.63" "LEVEL 50%: -4.21" "LEVEL 50%: 6.92"
826 [5] "LEVEL 50%: 0.69"
827
828 [1] "LEVEL 95%: -5.9" "LEVEL 95%: -1.5" "LEVEL 95%: -3.73" "LEVEL 95%: 9.01"
829 [5] "LEVEL 95%: 1.05"
830
831 [1] "LEVEL 97.5%: -5.78" "LEVEL 97.5%: -1.48" "LEVEL 97.5%: -3.69"
832 [4] "LEVEL 97.5%: 9.48" "LEVEL 97.5%: 1.1"
833
834 [1] "Phenomenology: 1; Emplacement: 3; Response: 2"
835
836 [1] "POSTERIOR MEAN: 3.99" "POSTERIOR MEAN: -1.88"
837 [3] "POSTERIOR MEAN: -2.48" "POSTERIOR MEAN: 1044.58"
838 [5] "POSTERIOR MEAN: 39.55"
839
840 [1] "LEVEL 2.5%: 2.37" "LEVEL 2.5%: -2.1" "LEVEL 2.5%: -3.08"
841 [4] "LEVEL 2.5%: 80.13" "LEVEL 2.5%: 3.13"
842
843 [1] "LEVEL 5%: 2.69" "LEVEL 5%: -2.08" "LEVEL 5%: -3.01" "LEVEL 5%: 128.57"
844 [5] "LEVEL 5%: 4.83"
845
846 [1] "LEVEL 50%: 3.94" "LEVEL 50%: -1.88" "LEVEL 50%: -2.48" "LEVEL 50%: 900.8"
847 [5] "LEVEL 50%: 33.81"
848
849 [1] "LEVEL 95%: 5.21" "LEVEL 95%: -1.69" "LEVEL 95%: -2.06"
850 [4] "LEVEL 95%: 2277.09" "LEVEL 95%: 86.63"
851
852 [1] "LEVEL 97.5%: 5.3" "LEVEL 97.5%: -1.64" "LEVEL 97.5%: -1.92"
853 [4] "LEVEL 97.5%: 2400.69" "LEVEL 97.5%: 91.28"

854
855 [1] "Phenomenology: 2; Emplacement: 1; Response: 1"
856
857 [1] "POSTERIOR MEAN: 4.86"
858
859 [1] "LEVEL 2.5%: 4.32"
860
861 [1] "LEVEL 5%: 4.37"
862
863 [1] "LEVEL 50%: 4.88"
864
865 [1] "LEVEL 95%: 5.32"
866
867 [1] "LEVEL 97.5%: 5.4"
868
869 [1] "Phenomenology: 2; Emplacement: 1; Response: 2"
870
871 [1] "POSTERIOR MEAN: -0.19"
872
873 [1] "LEVEL 2.5%: -0.54"
874
875 [1] "LEVEL 5%: -0.49"
876
877 [1] "LEVEL 50%: -0.18"
878
879 [1] "LEVEL 95%: 0.1"
880
881 [1] "LEVEL 97.5%: 0.14"
882
883 [1] "Phenomenology: 2; Emplacement: 2; Response: 1"
884
885 [1] "POSTERIOR MEAN: 3.91"
886
887 [1] "LEVEL 2.5%: 3.49"
888
889 [1] "LEVEL 5%: 3.54"
890
891 [1] "LEVEL 50%: 3.92"
892
893 [1] "LEVEL 95%: 4.38"
894
895 [1] "LEVEL 97.5%: 4.48"
896
897 [1] "Phenomenology: 2; Emplacement: 2; Response: 2"
898

899 [1] "POSTERIOR MEAN: -1.16"
900
901 [1] "LEVEL 2.5%: -1.89"
902
903 [1] "LEVEL 5%: -1.81"
904
905 [1] "LEVEL 50%: -1.19"
906
907 [1] "LEVEL 95%: -0.46"
908
909 [1] "LEVEL 97.5%: -0.39"
910
911 [1] "Phenomenology: 2; Emplacement: 3; Response: 1"
912
913 [1] "POSTERIOR MEAN: 2.44"
914
915 [1] "LEVEL 2.5%: 1.85"
916
917 [1] "LEVEL 5%: 1.93"
918
919 [1] "LEVEL 50%: 2.43"
920
921 [1] "LEVEL 95%: 2.91"
922
923 [1] "LEVEL 97.5%: 2.95"
924
925 [1] "Phenomenology: 2; Emplacement: 3; Response: 2"
926
927 [1] "POSTERIOR MEAN: -1.47"
928
929 [1] "LEVEL 2.5%: -2.63"
930
931 [1] "LEVEL 5%: -2.41"
932
933 [1] "LEVEL 50%: -1.47"
934
935 [1] "LEVEL 95%: -0.47"
936
937 [1] "LEVEL 97.5%: -0.36"
938
939 [1] "SOURCE VARIANCE COMPONENTS"
940
941 [1] "Phenomenology: 1; Response: 1"
942
943 [1] "POSTERIOR MEAN: 0.0523"

944 [1] "LEVEL 2.5%: 0.0248"

945

946 [1] "LEVEL 5%: 0.0278"

947

948 [1] "LEVEL 50%: 0.0503"

949

950 [1] "LEVEL 95%: 0.0872"

951

952 [1] "LEVEL 97.5%: 0.105"

953

954 [1] "Phenomenology: 1; Response: 2"

955

956 [1] "POSTERIOR MEAN: 0.1807"

957

958 [1] "LEVEL 2.5%: 0.1149"

959

960 [1] "LEVEL 5%: 0.1209"

961

962 [1] "LEVEL 50%: 0.1762"

963

964 [1] "LEVEL 95%: 0.2633"

965

966 [1] "LEVEL 97.5%: 0.2801"

967

968 [1] "Phenomenology: 2; Response: 1"

969

970 [1] "POSTERIOR MEAN: 0.1953"

971

972 [1] "LEVEL 2.5%: 0.1167"

973

974 [1] "LEVEL 5%: 0.1216"

975

976 [1] "LEVEL 50%: 0.1866"

977

978 [1] "LEVEL 95%: 0.3069"

979

980 [1] "LEVEL 97.5%: 0.3187"

981

982 [1] "Phenomenology: 2; Response: 2"

983

984 [1] "POSTERIOR MEAN: 0.0371"

985

986 [1] "LEVEL 2.5%: 0.0217"

987

988

989 [1] "LEVEL 5%: 0.0227"

990

991 [1] "LEVEL 50%: 0.0339"

992

993 [1] "LEVEL 95%: 0.06"

994

995 [1] "LEVEL 97.5%: 0.0771"

996

997 [1] "PATH VARIANCE COMPONENTS"

998

999 [1] "Phenomenology: 1; Response: 1"

1000

1001 [1] "POSTERIOR MEAN: 0.1448"

1002

1003 [1] "LEVEL 2.5%: 0.104"

1004

1005 [1] "LEVEL 5%: 0.1181"

1006

1007 [1] "LEVEL 50%: 0.1431"

1008

1009 [1] "LEVEL 95%: 0.1822"

1010

1011 [1] "LEVEL 97.5%: 0.194"

1012

1013 [1] "Phenomenology: 1; Response: 2"

1014

1015 [1] "POSTERIOR MEAN: 0.1563"

1016

1017 [1] "LEVEL 2.5%: 0.1166"

1018

1019 [1] "LEVEL 5%: 0.1224"

1020

1021 [1] "LEVEL 50%: 0.1505"

1022

1023 [1] "LEVEL 95%: 0.2134"

1024

1025 [1] "LEVEL 97.5%: 0.2204"

1026

1027 [1] "Phenomenology: 2; Response: 1"

1028

1029 [1] "POSTERIOR MEAN: 0.0217"

1030

1031 [1] "LEVEL 2.5%: 0.0127"

1032

1033 [1] "LEVEL 5%: 0.0149"

```

1034
1035         [1] "LEVEL 50%: 0.0211"
1036
1037         [1] "LEVEL 95%: 0.0298"
1038
1039         [1] "LEVEL 97.5%: 0.0314"
1040
1041         [1] "Phenomenology: 2; Response: 2"
1042
1043         [1] "POSTERIOR MEAN: 0.0128"
1044
1045         [1] "LEVEL 2.5%: 0.0086"
1046
1047         [1] "LEVEL 5%: 0.01"
1048
1049         [1] "LEVEL 50%: 0.0125"
1050
1051         [1] "LEVEL 95%: 0.0165"
1052
1053         [1] "LEVEL 97.5%: 0.017"
1054
1055     [1] "OBSERVATIONAL ERROR COVARIANCE PARAMETERS"
1056
1057     [1] "Phenomenology 1"
1058
1059     [1] "POSTERIOR MEAN:"
1060
1061     [1] "Variances"
1062
1063     [1] 0.0836 0.1862
1064
1065     [1] "Correlations"
1066
1067         [,1] [,2]
1068     [1,]    1 0.4
1069     [2,]    0 1.0
1070
1071     [1] "Variances"
1072
1073         [1] "LEVEL 2.5%: 0.0713" "LEVEL 2.5%: 0.1591"
1074
1075     [1] "Correlations"
1076
1077         [1] "LEVEL 2.5%:"
1078             [,1] [,2]

```

```

1079 [1,]    1 0.32
1080 [2,]    0 1.00
1081
1082 [1] "Variances"
1083
1084      [1] "LEVEL 5%: 0.0729" "LEVEL 5%: 0.1646"
1085
1086 [1] "Correlations"
1087
1088      [1] "LEVEL 5%:"
1089      [,1] [,2]
1090 [1,]    1 0.33
1091 [2,]    0 1.00
1092
1093 [1] "Variances"
1094
1095      [1] "LEVEL 50%: 0.083" "LEVEL 50%: 0.1852"
1096
1097 [1] "Correlations"
1098
1099      [1] "LEVEL 50%:"
1100      [,1] [,2]
1101 [1,]    1 0.41
1102 [2,]    0 1.00
1103
1104 [1] "Variances"
1105
1106      [1] "LEVEL 95%: 0.0957" "LEVEL 95%: 0.2116"
1107
1108 [1] "Correlations"
1109
1110      [1] "LEVEL 95%:"
1111      [,1] [,2]
1112 [1,]    1 0.49
1113 [2,]    0 1.00
1114
1115 [1] "Variances"
1116
1117      [1] "LEVEL 97.5%: 0.0977" "LEVEL 97.5%: 0.2141"
1118
1119 [1] "Correlations"
1120
1121      [1] "LEVEL 97.5%:"
1122      [,1] [,2]
1123 [1,]    1 0.53

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1124 [2,]    0 1.00
1125
1126 [1] "Phenomenology 2"
1127
1128 [1] "POSTERIOR MEAN:"
1129
1130 [1] "Variances"
1131
1132 [1] 0.0725 0.0168
1133
1134 [1] "Correlations"
1135
1136      [,1] [,2]
1137 [1,]    1 -0.14
1138 [2,]    0  1.00
1139
1140 [1] "Variances"
1141
1142      [1] "LEVEL 2.5%: 0.0613" "LEVEL 2.5%: 0.0144"
1143
1144 [1] "Correlations"
1145
1146      [1] "LEVEL 2.5%:"
1147      [,1] [,2]
1148 [1,]    1 -0.27
1149 [2,]    0  1.00
1150
1151 [1] "Variances"
1152
1153      [1] "LEVEL 5%: 0.062" "LEVEL 5%: 0.0146"
1154
1155 [1] "Correlations"
1156
1157      [1] "LEVEL 5%:"
1158      [,1] [,2]
1159 [1,]    1 -0.23
1160 [2,]    0  1.00
1161
1162 [1] "Variances"
1163
1164      [1] "LEVEL 50%: 0.0728" "LEVEL 50%: 0.0169"
1165
1166 [1] "Correlations"
1167
1168      [1] "LEVEL 50%:"

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1169          [,1] [,2]
1170 [1,]      1 -0.14
1171 [2,]      0  1.00
1172
1173 [1] "Variances"
1174
1175          [1] "LEVEL 95%: 0.0811" "LEVEL 95%: 0.0194"
1176
1177 [1] "Correlations"
1178
1179          [1] "LEVEL 95%:"
1180          [,1] [,2]
1181 [1,]      1 -0.01
1182 [2,]      0  1.00
1183
1184 [1] "Variances"
1185
1186          [1] "LEVEL 97.5%: 0.0824" "LEVEL 97.5%: 0.0198"
1187
1188 [1] "Correlations"
1189
1190          [1] "LEVEL 97.5%:"
1191          [,1] [,2]
1192 [1,]      1      0
1193 [2,]      0      1
1194
1195 [1] "Phenomenology 3"
1196
1197 [1] "POSTERIOR MEAN:"
1198
1199 [1] "Variances"
1200
1201 [1] 0.172 0.174
1202
1203 [1] "Correlations"
1204
1205          [,1] [,2]
1206 [1,]      1 0.96
1207 [2,]      0 1.00
1208
1209 [1] "Variances"
1210
1211          [1] "LEVEL 2.5%: 0.1003" "LEVEL 2.5%: 0.0946"
1212
1213 [1] "Correlations"

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1214
1215         [1] "LEVEL 2.5%:"
1216             [,1] [,2]
1217 [1,]      1 0.93
1218 [2,]      0 1.00
1219
1220 [1] "Variances"
1221
1222         [1] "LEVEL 5%: 0.1049" "LEVEL 5%: 0.1003"
1223
1224 [1] "Correlations"
1225
1226         [1] "LEVEL 5%:"
1227             [,1] [,2]
1228 [1,]      1 0.94
1229 [2,]      0 1.00
1230
1231 [1] "Variances"
1232
1233         [1] "LEVEL 50%: 0.1731" "LEVEL 50%: 0.1746"
1234
1235 [1] "Correlations"
1236
1237         [1] "LEVEL 50%:"
1238             [,1] [,2]
1239 [1,]      1 0.96
1240 [2,]      0 1.00
1241
1242 [1] "Variances"
1243
1244         [1] "LEVEL 95%: 0.2348" "LEVEL 95%: 0.2333"
1245
1246 [1] "Correlations"
1247
1248         [1] "LEVEL 95%:"
1249             [,1] [,2]
1250 [1,]      1 0.98
1251 [2,]      0 1.00
1252
1253 [1] "Variances"
1254
1255         [1] "LEVEL 97.5%: 0.2428" "LEVEL 97.5%: 0.2413"
1256
1257 [1] "Correlations"
1258

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1259          [1] "LEVEL 97.5%:"
1260          [,1] [,2]
1261 [1,]      1 0.98
1262 [2,]      0 1.00
1263
1264 [1] "Phenomenology 4"
1265
1266 [1] "POSTERIOR MEAN:"
1267
1268 [1] "Variances"
1269
1270 [1] 0.0233 0.0323
1271
1272 [1] "Correlations"
1273
1274          [,1] [,2]
1275 [1,]      1 0.04
1276 [2,]      0 1.00
1277
1278 [1] "Variances"
1279
1280          [1] "LEVEL 2.5%: 0.0061" "LEVEL 2.5%: 0.0135"
1281
1282 [1] "Correlations"
1283
1284          [1] "LEVEL 2.5%:"
1285          [,1] [,2]
1286 [1,]      1 -0.4
1287 [2,]      0 1.0
1288
1289 [1] "Variances"
1290
1291          [1] "LEVEL 5%: 0.0086" "LEVEL 5%: 0.0144"
1292
1293 [1] "Correlations"
1294
1295          [1] "LEVEL 5%:"
1296          [,1] [,2]
1297 [1,]      1 -0.34
1298 [2,]      0 1.00
1299
1300 [1] "Variances"
1301
1302          [1] "LEVEL 50%: 0.0201" "LEVEL 50%: 0.0269"
1303

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1304 [1] "Correlations"
1305
1306         [1] "LEVEL 50%:"
1307             [,1] [,2]
1308 [1,]      1 -0.01
1309 [2,]      0  1.00
1310
1311 [1] "Variances"
1312
1313         [1] "LEVEL 95%: 0.0578" "LEVEL 95%: 0.0681"
1314
1315 [1] "Correlations"
1316
1317         [1] "LEVEL 95%:"
1318             [,1] [,2]
1319 [1,]      1  0.5
1320 [2,]      0  1.0
1321
1322 [1] "Variances"
1323
1324         [1] "LEVEL 97.5%: 0.0647" "LEVEL 97.5%: 0.0777"
1325
1326 [1] "Correlations"
1327
1328         [1] "LEVEL 97.5%:"
1329             [,1] [,2]
1330 [1,]      1 0.57
1331 [2,]      0 1.00
1332
1333 [1] "FGSN PRIOR PARAMETERS"
1334
1335 [1] "POSTERIOR MEAN:"
1336
1337 [1] "Alpha = 17.43"
1338 [1] "Lambda squared = 8.91"
1339 [1] "Omega = -1.91" "Omega = 0.7"
1340
1341 [1] "Alpha:"
1342 [1] "LEVEL 2.5%: 16.92"
1343
1344 [1] "LEVEL 5%: 16.99"
1345
1346 [1] "LEVEL 50%: 17.44"
1347
1348 [1] "LEVEL 95%: 17.89"

```

```

1349
1350 [1] "LEVEL 97.5%: 17.95"
1351
1352 [1] "Lambda squared:"
1353 [1] "LEVEL 2.5%: 5.81"
1354
1355 [1] "LEVEL 5%: 6.01"
1356
1357 [1] "LEVEL 50%: 8.45"
1358
1359 [1] "LEVEL 95%: 12.65"
1360
1361 [1] "LEVEL 97.5%: 16.11"
1362
1363 [1] "Omega:"
1364 [1] "LEVEL 2.5%: -3.03" "LEVEL 2.5%: 0.21"
1365
1366 [1] "LEVEL 5%: -2.83" "LEVEL 5%: 0.27"
1367
1368 [1] "LEVEL 50%: -1.96" "LEVEL 50%: 0.63"
1369
1370 [1] "LEVEL 95%: -0.77" "LEVEL 95%: 1.16"
1371
1372 [1] "LEVEL 97.5%: -0.57" "LEVEL 97.5%: 1.2"
1373
1374 [1] "DIC = 1120.11"
1375
1376 [1] "PIC = 1181.96"

```

A.5 New Event Data: Preprocessing

```
1 #####
2 #
3 # This file is the input deck for MultiPEM Toolbox rapid post-
4 # detonation analysis, based on using fixed values of the forward and
5 # error model parameters obtained from calibration data.
6 #
7 # © 2023. Triad National Security, LLC. All rights reserved.
8 # This program was produced under U.S. Government contract
9 # 89233218CNA000001 for Los Alamos National Laboratory (LANL), which
10 # is operated by Triad National Security, LLC for the U.S. Department
11 # of Energy/National Nuclear Security Administration. All rights in
12 # the program are reserved by Triad National Security, LLC, and the
13 # U.S. Department of Energy/National Nuclear Security Administration.
14 # The Government is granted for itself and others acting on its behalf
15 # a nonexclusive, paid-up, irrevocable worldwide license in this
16 # material to reproduce, prepare derivative works, distribute copies
17 # to the public, perform publicly and display publicly, and to permit
18 # others to do so.
19 #
20 #####
21
22 #
23 # REQUIRED R PACKAGES
24 #
25
26 require(Matrix)
27
28 #
29 # END REQUIRED R PACKAGES
30 #
31
32 #
33 # PREPROCESSING
34 #
35
36 # Specify directory for general subroutines
37 gen_dir = "../.../Code"
38
39 # Source supporting R function
40 source(paste(gen_dir,"/prepro_0.r",sep=""))
41
42 # Specify directory for application subroutines
43 app_dir = "../.../Code"
```

```

44
45 # Specify root data directory
46 dat_dir = "../..Data"
47
48 # Specify new event data directories
49 dat_new = c("seismic_new.csv",
50             "acoustic_new.csv",
51             "optical_new.csv",
52             "crater_new.csv")
53
54 # Phenomenologies for this analysis
55 # 1 - seismic
56 # 2 - acoustic
57 # 3 - optical
58 # 4 - crater (surface effects)
59
60 # Names of new event inference parameters
61 theta_names = c("W","HOB")
62
63 # Number of calibration parameter imputations utilized in
64 # Markov chain Monte Carlo (MCMC) for new event parameters
65 nimpute = 1000
66
67 # Set flag for bounded optimization
68 opt_B = FALSE
69
70 # Indicate nev parameter transform
71 itransform = FALSE
72
73 # Specify fixed parameters for nev parameter transform
74 if( itransform ){
75     tPars = TRUE
76
77     if( tPars ){
78         tPars = vector("list",0)
79         tPars$yield_scaling = 1/3
80     } else { tPars = NULL }
81 } else { tPars = NULL }
82
83 # Set up parameter constraints
84 # lower and upper bounds (use -Inf and Inf if unbounded)
85 lb_theta0 = rep(-Inf,length(theta_names))
86 lb_theta0[2] = 0
87 ub_theta0 = rep(Inf,length(theta_names))
88 ub_theta0[2] = 160

```



```

89
90 # Set up parameter subsets by phenomenology
91 tsub = TRUE
92
93 if( tsub ) {
94     tsub = vector("list",length(Rh))
95     tsub[[4]] = 1 # only log-yield for crater
96 } else { tsub = NULL }
97
98 # Preprocessing for statistical analysis routines
99 tmp = prepro_0(p_cal,gen_dir,app_dir,dat_dir,dat_new,theta_names,
100               nimp=nimpute,bopt=opt_B,itr=itransform,fp_tr=tPars,
101               tlb=lb_theta0,tub=ub_theta0,tsub=tsub)
102 if( opt_B ){
103     p_cal = tmp$p_cal
104     t_cal = tmp$t_cal
105 } else {
106     p_cal = tmp
107     t_cal = NULL
108 }
109 rm(tmp)
110 save.image()
111
112 #
113 # END PREPROCESSING
114 #

```

A.6 New Event Data: Maximum Likelihood Estimation

```
1  #
2  # MAXIMUM LIKELIHOOD CALCULATION
3  #
4
5  # Source supporting R function
6  source(paste(gen_dir,"/calc_mle_0.r",sep=""))
7
8  # Set seed for repeatability of analysis
9  set.seed(631)
10
11 # Names of forward models for each response
12 # by phenomenology
13 fm0 = vector("list",length(Rh))
14 fm0[[1]] = c("f0_s","f0_s")
15 fm0[[2]] = c("f0_a","f0_a")
16 fm0[[3]] = c("f0_o","f0_o")
17 fm0[[4]] = c("f0_c","f0_c")
18
19 # Indicate if forward model gradients provided
20 igrad = TRUE
21
22 if( igrad ){
23   # Names of forward model gradients for each response
24   # by phenomenology
25   gfm0 = vector("list",length(Rh))
26   gfm0[[1]] = c("g0_s","g0_s")
27   gfm0[[2]] = c("g0_a","g0_a")
28   gfm0[[3]] = c("g0_o","g0_o")
29   gfm0[[4]] = c("g0_c","g0_c")
30 } else { gfm0 = NULL }
31
32 # Specify number of starting values for optimization
33 nstart = 30
34
35 # number of cores to use for optimization
36 ncores_mle = 30
37
38 # Indicate use of BFGS optimization methods
39 bfgs = TRUE
40
41 # Location of R data files with starting values
42 # for input to MLE optimization
43 opt_files_in = NULL
```

```

44
45 # Location of R data file to write the results of
46 # MLE optimization
47 opt_files_out = "./opt_nev.RData"
48
49 # Initial start value for theta0
50 tst = FALSE
51
52 if( tst ){
53     tst = numeric(p_cal$ntheta0)
54     #tst[2] =
55 } else { tst = NULL }
56
57 # Confidence interval levels for new event parameter inference
58 ci_lev = 0.95
59
60 # Indicator of MLE gradient check
61 mle_grad_ck = TRUE
62
63 # Strategy for running parallel jobs (future package)
64 parallel_plan = "multicore"
65
66 # MLE calculations
67 p_cal = calc_mle_0(p_cal,gen_dir,app_dir,fm0,nst=nstart,ncor=ncores_mle,
68                   ci_lev=ci_lev,igrad=igrad,bfgs=bfgs,
69                   igrck=mle_grad_ck,t_cal=t_cal,g0=gfm0,
70                   fopt_in=opt_files_in,Xst=NULL,tst=tst,
71                   fopt_out=opt_files_out,pl=parallel_plan)
72 save.image()
73
74 #
75 # END MAXIMUM LIKELIHOOD CALCULATION
76 #

```

A.7 New Event Data: Bayesian Analysis

```
1  #
2  # BAYESIAN ANALYSIS
3  #
4
5  # Specify if Bayesian analysis is to be conducted
6  iBayes = TRUE
7
8  if( iBayes ){
9    # Source supporting R function
10    source(paste(gen_dir,"/calc_bayes_0.r",sep=""))
11
12    # Indicator of prior distribution for theta0
13    iTheta0Prior = FALSE
14
15    if( iTheta0Prior ){
16      # location of code for computing log-prior densities and gradients
17      prior_files_theta0 = NULL
18      if( igrad ){
19        gr_prior_files_theta0 = NULL
20      } else { gr_prior_files_theta0 = NULL }
21
22      # prior distribution for new event parameters (theta0)
23      p_cal$lp_theta0$f = "lp_0"
24      if( igrad ){ p_cal$lp_theta0$g = "lq_0" }
25
26      # parameters for log yield parameter prior (Gaussian)
27      #p_cal$pi_w_mu =
28      #p_cal$pi_w_sd =
29      # parameters for HOB/DOB parameter prior (Gaussian)
30      #p_cal$pi_h_mu =
31      #p_cal$pi_h_sd =
32    } else {
33      prior_files_theta0 = NULL
34      gr_prior_files_theta0 = NULL
35    }
36
37    # specify MCMC algorithm
38    # options: "RAM", "FME", "NUTS", or "SMC"
39    iMCMC = "FME"
40
41    # burn-in
42    nburn = 1000
43
```

```

44 # production
45 nmcmc = 4000
46
47 # posterior sample thinning rate (for multiple imputation)
48 nthin = 200
49
50 # number of cores to use for optimization
51 ncores_map = 30
52
53 # number of cores to use for generating parallel MCMC chains
54 ncores_mc = 30
55
56 # Indicator of prior gradient check
57 prior_grad_ck = TRUE
58
59 # Indicator of posterior gradient check
60 post_grad_ck = TRUE
61
62 # Options for Sequential Monte Carlo (SMC) sampling
63 # (iMCMC = "SMC")
64 # number of cores to use for parallelization within SMC algorithm
65 ncores_smc = NULL
66 # new event parameter ranges for initial SMC sample
67 lb_smc = rep(-Inf,length(theta_names))
68 ub_smc = rep(Inf,length(theta_names))
69
70 # Bayesian calculations
71 p_cal = calc_bayes_0(p_cal,gen_dir,app_dir,nst=nstart,nburn=nburn,
72                     nmcmc=nmcmc,nthin=nthin,ncor_map=ncores_map,
73                     ncor_mc=ncores_mc,igrad=igrad,
74                     igrck_pr=prior_grad_ck,igrck_po=post_grad_ck,
75                     bfgs=bfgs,itpr=iTheta0Prior,
76                     fpr_t=prior_files_theta0,
77                     fgpr_t=gr_prior_files_theta0,imcmc=iMCMC,
78                     pl=parallel_plan,ncor_smc=ncores_smc,
79                     lb_smc=lb_smc,ub_smc=ub_smc,t_cal=t_cal)
80 save.image()
81 }
82
83 #
84 # END BAYESIAN ANALYSIS
85 #

```

A.8 New Event Data: Output File

```
1 > # MLE calculations
2 > p_cal = calc_mle_0(p_cal,gen_dir,app_dir,fm0,nst=nstart,ncor=ncores_mle,
3 +                   ci_lev=ci_lev,igrad=igrad,bfgs=bfgs,
4 +                   igrck=mle_grad_ck,t_cal=t_cal,g0=gfm0,
5 +                   fopt_in=opt_files_in,Xst=NULL,tst=tst,
6 +                   fopt_out=opt_files_out,pl=parallel_plan)
7 [1] "MLE CONVERGENCE STATUS"
8
9 [1] 0
10 [1] 2
11 [1] "MAXIMUM LIKELIHOOD SUMMARY"
12
13 [1] "NEW EVENT INFERENCE PARAMETERS"
14
15 [1] "ESTIMATE: "
16
17      W   HOB
18 13.78  1.09
19
20 [1] "STANDARD DEVIATION FIXED MODEL PARAMETERS: "
21
22      W   HOB
23  0.19 0.44
24
25 [1] "CORRELATION MATRIX FIXED MODEL PARAMETERS: "
26
27      W   HOB
28 W   1.00 0.23
29 HOB 0.23 1.00
30
31 [1] "STANDARD DEVIATION: "
32
33      W   HOB
34  0.25 0.63
35
36 [1] "CORRELATION MATRIX: "
37
38      W   HOB
39 W   1.0 0.2
40 HOB 0.2 1.0
41
42 [1] "95%: CONFIDENCE INTERVAL FIXED MODEL PARAMETERS:"
43
```

```

44           W   HOB
45 lb_0 13.41 0.56
46 ub_0 14.14 2.95
47
48 [1] "95%: CONFIDENCE INTERVAL:"
49
50           W   HOB
51 lb 13.28 0.45
52 ub 14.27 5.50
53
54
55 Loading required package: numDeriv
56 [1] "CHECK LOG-LIKELIHOOD GRADIENTS"
57
58 [1] "Analytic gradient"
59 [1] 2.236547e-08 7.031679e-09
60 [1] "Numerical gradient"
61 [1] 2.241838e-08 7.056912e-09
62 [1] "Difference"
63 [1] -5.290787e-11 -2.523366e-11
64
65 + # Bayesian calculations
66 + p_cal = calc_bayes_0(p_cal,gen_dir,app_dir,nst=nstart,nburn=nburn,
67 +                     nmcmc=nmcmc,nthin=nthin,ncor_map=ncores_map,
68 +                     ncor_mc=ncores_mc,igrad=igrad,
69 +                     igrck_pr=prior_grad_ck,igrck_po=post_grad_ck,
70 +                     bfgs=bfgs,itpr=iTheta0Prior,
71 +                     fpr_t=prior_files_theta0,
72 +                     fgpr_t=gr_prior_files_theta0,imcmc=iMCMC,
73 +                     pl=parallel_plan,ncor_smc=ncores_smc,
74 +                     lb_smc=lb_smc,ub_smc=ub_smc,t_cal=t_cal)
75 [1] "MAP CONVERGENCE STATUS"
76
77 [1] 0
78 [1] 2
79 [1] "MAXIMUM A POSTERIORI SUMMARY"
80
81 [1] "NEW EVENT INFERENCE PARAMETERS"
82
83 [1] "ESTIMATE: "
84
85           W   HOB
86 13.75 37.22
87
88

```

```

89 [1] "CHECK LOG-PRIOR GRADIENTS"
90
91 [1] "Analytic gradient"
92 [1] 0.000000 0.673581
93 [1] "Numerical gradient"
94 [1] 0.000000 0.673581
95 [1] "Difference"
96 [1] -2.021971e-11 0.000000e+00
97
98 [1] "CHECK LOG-POSTERIOR GRADIENTS"
99
100 [1] "Analytic gradient"
101 [1] -2.852906e-08 -3.147993e-08
102 [1] "Numerical gradient"
103 [1] -2.849478e-08 -3.171866e-08
104 [1] "Difference"
105 [1] -3.427476e-11 2.387282e-10
106
107 [1] "ACCEPTANCE RATES:"
108
109 [1] "Imputation 1: 0.8456"
110 [1] "Imputation 2: 0.874"
111 ...
112 [1] "Imputation 999: 0.6364"
113 [1] "Imputation 1000: 0.7898"
114
115 [1] "POSTERIOR SUMMARY"
116
117 [1] "NEW EVENT INFERENCE PARAMETERS"
118
119 [1] "POSTERIOR MEAN: 13.77" "POSTERIOR MEAN: 38.75"
120
121 [1] "POSTERIOR SD: 0.28" "POSTERIOR SD: 21.46"
122
123 [1] "LEVEL 2.5%: 13.2" "LEVEL 2.5%: 3.59"
124
125 [1] "LEVEL 5%: 13.3" "LEVEL 5%: 6.38"
126
127 [1] "LEVEL 50%: 13.77" "LEVEL 50%: 38.33"
128
129 [1] "LEVEL 95%: 14.23" "LEVEL 95%: 71.27"
130
131 [1] "LEVEL 97.5%: 14.32" "LEVEL 97.5%: 80.8"
132
133 [1] "CORRELATION MATRIX:"

```


134				
135		W	HOB	
136	W	1.00	0.08	
137	HOB	0.08	1.00	

B Complete Assessment Run Files

This appendix provides an example run file and output file for complete assessments.

B.1 Preprocessing

```
1 #####
2 #
3 # This file is the input deck for MultiPEM Toolbox complete post-
4 # detonation analysis.
5 #
6 # © 2023. Triad National Security, LLC. All rights reserved.
7 # This program was produced under U.S. Government contract
8 # 89233218CNA000001 for Los Alamos National Laboratory (LANL), which
9 # is operated by Triad National Security, LLC for the U.S. Department
10 # of Energy/National Nuclear Security Administration. All rights in
11 # the program are reserved by Triad National Security, LLC, and the
12 # U.S. Department of Energy/National Nuclear Security Administration.
13 # The Government is granted for itself and others acting on its behalf
14 # a nonexclusive, paid-up, irrevocable worldwide license in this
15 # material to reproduce, prepare derivative works, distribute copies
16 # to the public, perform publicly and display publicly, and to permit
17 # others to do so.
18 #
19 #####
20
21 #
22 # REQUIRED R PACKAGES
23 #
24
25 require(Matrix)
26
27 #
28 # END REQUIRED R PACKAGES
29 #
30
31 #
32 # PREPROCESSING
33 #
34
35 # Specify directory for general subroutines
36 gen_dir = "../.../Code"
37
38 # Source supporting R function
39 source(paste(gen_dir, "/prepro.r", sep=""))
```

```

40
41 # Specify directory for application subroutines
42 app_dir = "../..Code"
43
44 # Specify root data directory
45 dat_dir = "../..Data"
46
47 # Specify calibration data directories
48 dat_cal = c("seismic_cal.csv",
49             "acoustic_cal.csv",
50             "optical_cal.csv",
51             "crater_cal.csv")
52
53 # Phenomenologies for this analysis
54 # 1 - seismic
55 # 2 - acoustic
56 # 3 - optical
57 # 4 - crater (surface effects)
58
59 # Indicate presence of calibration inference parameters
60 calp = FALSE
61
62 if( calp ){
63     # Names of calibration inference parameters
64     #cal_par_names =
65 } else { cal_par_names = NULL }
66
67 # Specify number of responses for each phenomenology
68 Rh = c(2,2,2,2)
69
70 # Empirical model parameter count: common
71 # list with elements corresponding to phenomenologies
72 pbeta = vector("list",length(Rh))
73 for( hh in 1:length(Rh) ){ pbeta[[hh]] = numeric(Rh[hh]) }
74 # phenomenology 2
75 pbeta[[2]] = c(2,2)
76 # phenomenology 3
77 pbeta[[3]] = c(4,4)
78 # phenomenology 4
79 pbeta[[4]] = c(2,2)
80
81 # Specify number of emplacement conditions for each phenomenology
82 Th = TRUE
83
84 if( Th ){ Th = c(3,3,0,0)

```

```

85 } else { Th = NULL }
86
87 # Empirical model parameter count: emplacement condition
88 # list with elements corresponding to phenomenologies
89 if( !is.null(Th) ){
90   pbetat = vector("list",length(Rh))
91   for( hh in 1:length(Rh) ){
92     if( Th[hh] > 1 ){ pbetat[[hh]] = vector("list",Th[hh]) }
93   }
94   # phenomenology 1
95   for( tt in 1:Th[1] ){
96     pbetat[[1]][[tt]] = numeric(Rh[1])
97     pbetat[[1]][[tt]] = c(5,5)
98   }
99   # phenomenology 2
100  for( tt in 1:Th[2] ){
101    pbetat[[2]][[tt]] = numeric(Rh[2])
102    pbetat[[2]][[tt]] = c(1,1)
103  }
104 } else { pbetat = NULL }
105
106 # Locations of common parameters in full parameter vector
107 # list with elements corresponding to phenomenologies
108 if( !is.null(Th) ){
109   ibetar = vector("list",length(Rh))
110   for( hh in 1:length(Rh) ){
111     if( Th[hh] > 1 ){
112       # lists with elements for each response within
113       # emplacement condition
114       ibetar[[hh]] = vector("list",Th[hh]*Rh[hh])
115     }
116   }
117   # phenomenology 2
118   for( tt in 1:Th[2] ){
119     for( rr in 1:Rh[2] ){
120       ibetar[[2]][[(tt-1)*Rh[2]+rr]] = 1:2
121     }
122   }
123 } else { ibetar = NULL }
124
125 # Indicate analysis with errors-in-variables (eiv)
126 eiv = TRUE
127
128 # Specifications for errors-in-variables
129 if( eiv ){

```

```

130 # Specify phenomenologies utilizing
131 # errors-in-variables yields
132 ieiv = 3:4
133
134 # Errors-in-variables source lists by
135 # phenomenology
136 seiv = vector("list",length(Rh))
137 for( hh in ieiv ){ seiv[[hh]] = "ALL" }
138
139 # Set standard deviation of eiv Gaussian likelihood
140 eiv_w_sd = 0.3/3
141 } else {
142   ieiv = NULL
143   seiv = NULL
144   eiv_w_sd = NULL
145 }
146
147 # Specify Error Model
148 # Source variance component parameter count
149 pvc_1 = TRUE
150
151 if( pvc_1 ){
152   pvc_1 = vector("list",length(Rh))
153   for( hh in 1:length(Rh) ){ pvc_1[[hh]] = numeric(Rh[hh]) }
154   # phenomenology 1
155   pvc_1[[1]] = c(1,1)
156   # phenomenology 2
157   pvc_1[[2]] = c(1,1)
158 } else { pvc_1 = NULL }
159
160 # Path variance component parameter count
161 pvc_2 = TRUE
162
163 if( pvc_2 ){
164   pvc_2 = vector("list",length(Rh))
165   for( hh in 1:length(Rh) ){ pvc_2[[hh]] = numeric(Rh[hh]) }
166   # phenomenology 1
167   pvc_2[[1]] = c(1,1)
168   # phenomenology 2
169   pvc_2[[2]] = c(1,1)
170
171   # path error models by phenomenology
172   ptype = vector("list",length(Rh))
173   # phenomenology 1
174   ptype[[1]] = "Crossed"

```

```

175   # phenomenology 2
176   ptype[[2]] = "Crossed"
177 } else { pvc_2 = NULL; ptype = NULL; }
178
179 # Set flag for user-provided code to calculate variance
180 # component coefficient matrices
181 calc_Z = FALSE
182
183 # Set flag for bounded optimization
184 # currently only supported for new event parameters
185 opt_B = FALSE
186
187 # Indicate analysis of new event (nev)
188 nev = TRUE
189
190 # Specifications for new event
191 if( nev ){
192   # Specify new event data directories
193   dat_new = c("seismic_new.csv",
194               "acoustic_new.csv",
195               "optical_new.csv",
196               "crater_new.csv")
197
198   # Names of new event inference parameters
199   theta_names = c("W","HOB")
200
201   # Indicate nev parameter transform
202   itransform = FALSE
203
204   # Specify fixed parameters for nev parameter transform
205   if( itransform ){
206     tPars = TRUE
207
208     if( tPars ){
209       tPars = vector("list",0)
210       tPars$yield_scaling = 1/3
211     } else { tPars = NULL }
212   } else { tPars = NULL }
213
214   # Set up parameter constraints
215   # lower and upper bounds (use -Inf and Inf if unbounded)
216   lb_theta0 = rep(-Inf,length(theta_names))
217   lb_theta0[2] = 0
218   ub_theta0 = rep(Inf,length(theta_names))
219   ub_theta0[2] = 160

```

```

220
221 # Set up parameter subsets by phenomenology
222 tsub = TRUE
223
224 if( tsub ){
225     tsub = vector("list",length(Rh))
226     tsub[[4]] = 1 # only log-yield for crater
227 } else { tsub = NULL }
228 } else {
229     dat_new = NULL
230     theta_names = NULL
231     itransform = FALSE
232     tPars = NULL
233     lb_theta0 = NULL
234     ub_theta0 = NULL
235     tsub = NULL
236 }
237
238 # Preprocessing for statistical analysis routines
239 tmp = prepro(gen_dir,app_dir,dat_dir,dat_cal,Rh,pbeta,bopt=opt_B,
240             nev=nev,itr=itransform,izmat=calc_Z,ieiv=ieiv,seiv=seiv,
241             ewsd=eiv_w_sd,Th=Th,pbetat=pbetat,ibetar=ibetar,
242             pvc_1=pvc_1,pvc_2=pvc_2,ptype=ptype,tnames=theta_names,
243             cnames=cal_par_names,fp_tr=tPars,tlb=lb_theta0,
244             tub=ub_theta0,ndir=dat_new,tsub=tsub)
245 if( opt_B ){
246     p_cal = tmp$p_cal
247     t_cal = tmp$t_cal
248 } else {
249     p_cal = tmp
250     t_cal = NULL
251 }
252 rm(tmp)
253 save.image()
254
255 #
256 # END PREPROCESSING
257 #

```

B.2 Maximum Likelihood Estimation

```
1  #
2  # MAXIMUM LIKELIHOOD CALCULATION
3  #
4
5  # Source supporting R function
6  source(paste(gen_dir,"/calc_mle.r",sep=""))
7
8  # Set seed for repeatability of analysis
9  set.seed(611)
10
11 # Names of forward models for each response
12 # by phenomenology
13 fm = vector("list",length(Rh))
14 fm[[1]] = c("f_s","f_s")
15 fm[[2]] = c("f_a","f_a")
16 fm[[3]] = c("f_o","f_o")
17 fm[[4]] = c("f_c","f_c")
18
19 # Indicate if forward model gradients provided
20 igrad = TRUE
21
22 if( igrad ){
23   # Names of forward model gradients for each response
24   # by phenomenology
25   gfm = vector("list",length(Rh))
26   gfm[[1]] = c("g_s","g_s")
27   gfm[[2]] = c("g_a","g_a")
28   gfm[[3]] = c("g_o","g_o")
29   gfm[[4]] = c("g_c","g_c")
30 } else { gfm = NULL }
31
32 # Specifications for forward model calculations
33 # a) flags for modified forward model calculation by
34 #    response for each relevant phenomenology
35 iResponse = TRUE
36
37 if( iResponse ){
38   iResponse = vector("list",length(Rh))
39   iResponse[[1]] = c(TRUE,FALSE)
40   iResponse[[2]] = c(TRUE,FALSE)
41 } else { iResponse = NULL }
42
43 # b) fixed quantities required by forward models
```



```

44 fPars = TRUE
45
46 if( fPars ){
47   fPars = vector("list",length(Rh))
48   fPars[[1]]$yield_scaling = 1/3
49   fPars[[2]]$yield_scaling = 1/3
50   fPars[[2]]$pressure_scaling = 1/3
51   fPars[[2]]$temp_scaling = 1/2
52   fPars[[3]]$yield_scaling = 1/3
53 } else { fPars = NULL }
54
55 # Specify number of starting values for optimization
56 nstart = 30
57
58 # number of cores to use for optimization
59 ncores_mle = 30
60
61 # Indicate use of BFGS optimization methods
62 bfgs = TRUE
63
64 # Location of R data files with starting values
65 # for input to MLE optimization
66 opt_files_in = c("../Opt/opt_1.RData",
67                  "../Opt/opt_2.RData",
68                  "../Opt/opt_3_eiv.RData",
69                  "../Opt/opt_4_eiv.RData")
70
71 # Location of R data file to write the results of
72 # MLE optimization
73 opt_files_out = "../opt.RData"
74
75 if( calp ){
76   # Initial start value for calibration inference parameters
77   cst = FALSE
78
79   if( cst ){
80     cst = numeric(p_cal$ncalp)
81     #cst[1] =
82   } else { cst = NULL }
83 } else { cst = NULL }
84
85 if( nev ){
86   # Initial start value for theta0
87   tst = FALSE
88

```

```

89   if( tst ){
90       tst = numeric(p_cal$ntheta0)
91       #tst[2] =
92   } else { tst = NULL }
93 } else { tst = NULL }
94
95 if( calp || nev ){
96     # Confidence interval levels
97     ci_lev = 0.95
98 } else { ci_lev = NULL }
99
100 # Indicate phenomenology number and type (if needed
101 # for postprocessing)
102 Phen = TRUE
103
104 if( Phen ){
105     Phen = matrix(c(1,"Seismic"),nrow=1)
106 } else { Phen = NULL }
107
108 # Indicator of MLE gradient check
109 mle_grad_ck = TRUE
110
111 # Strategy for running parallel jobs (future package)
112 parallel_plan = "multicore"
113
114 # MLE calculations
115 p_cal = calc_mle(p_cal,gen_dir,app_dir,fm,nst=nstart,ncor=ncores_mle,
116                 ci_lev=ci_lev,igrad=igrad,bfgs=bfgs,igrck=mle_grad_ck,
117                 t_cal=t_cal,g=gfm,iresp=iResponse,fp_fm=fPars,
118                 fopt_in=opt_files_in,Xst=NULL,tst=tst,cst=cst,
119                 fopt_out=opt_files_out,phen=Phen,pl=parallel_plan)
120 save.image()
121
122 #
123 # END MAXIMUM LIKELIHOOD CALCULATION
124 #

```

B.3 Bayesian Analysis

```
1  #
2  # BAYESIAN ANALYSIS
3  #
4
5  # Specify if Bayesian analysis is to be conducted
6  iBayes = TRUE
7
8  if( iBayes ){
9      # Source supporting R function
10     source(paste(gen_dir,"/calc_bayes.r",sep=""))
11
12     # Indicator of prior distribution for forward model
13     # coefficients
14     iBetaPrior = TRUE
15
16     if( iBetaPrior ){
17         # location of code for computing log-prior densities and gradients
18         prior_files_beta = c("../Code/lp_beta_s.r","../Code/lp_beta_o.r")
19         if( igrad ){
20             gr_prior_files_beta = c("../Code/glp_beta_s.r",
21                                     "../Code/glp_beta_o.r")
22         } else { gr_prior_files_beta = NULL }
23
24         # prior distribution for phenomenology 1
25         # forward model coefficients
26         p_cal$h[[1]]$lp_betat = vector("list",Th[1])
27         for( tt in 1:Th[1] ){
28             p_cal$h[[1]]$lp_betat[[tt]]$f = c("lp_s","lp_s")
29             if( igrad ){
30                 p_cal$h[[1]]$lp_betat[[tt]]$g = c("lq_s","lq_s")
31             }
32         }
33
34         # prior distribution for phenomenology 3
35         # forward model coefficients
36         p_cal$h[[3]]$lp_beta$f = c("lp_o","lp_o")
37         if( igrad ){
38             p_cal$h[[3]]$lp_beta$g = c("lq_o","lq_o")
39         }
40     } else {
41         prior_files_beta = NULL
42         gr_prior_files_beta = NULL
43     }
```

```

44
45 # Indicator of prior distribution for calibration parameters
46 iCalPrior = FALSE
47
48 if( calp && iCalPrior ){
49     # location of code for computing log-prior densities and gradients
50     prior_files_calp = NULL
51     if( igrad ){
52         gr_prior_files_calp = NULL
53     } else { gr_prior_files_calp = NULL }
54
55     # prior distribution for calibration parameters (calp)
56     p_cal$lp_calp$f = "lp_c"
57     if( igrad ){ p_cal$lp_calp$g = "lq_c" }
58
59     # parameters for calibration parameter prior distribution
60     #p_cal$pi_c_mu =
61     #p_cal$pi_c_sd =
62 } else {
63     prior_files_calp = NULL
64     gr_prior_files_calp = NULL
65 }
66
67 # Indicator of prior distribution for theta0
68 iTheta0Prior = FALSE
69
70 if( nev && iTheta0Prior ){
71     # location of code for computing log-prior densities and gradients
72     prior_files_theta0 = NULL
73     if( igrad ){
74         gr_prior_files_theta0 = NULL
75     } else { gr_prior_files_theta0 = NULL }
76
77     # prior distribution for new event parameters (theta0)
78     p_cal$lp_theta0$f = "lp_0"
79     if( igrad ){ p_cal$lp_theta0$g = "lq_0" }
80
81     # parameters for log yield parameter prior distribution
82     #p_cal$pi_w_mu =
83     #p_cal$pi_w_sd =
84     # parameters for HOB/DOB parameter prior distribution
85     #p_cal$pi_h_mu =
86     #p_cal$pi_h_sd =
87 } else {
88     prior_files_theta0 = NULL

```

```

89     gr_prior_files_theta0 = NULL
90 }
91
92 # fixed scale parameters for variance component prior
93 # comment out if these parameters should vary
94 p_cal$A = 20
95
96 # eta parameter in Lewandowski-Kurowicka-Joe (LKJ) prior
97 # distribution for correlation parameters
98 p_cal$lp_corr$eta = 1
99
100 # FGSN parameters for errors-in-variables yields prior
101 # number of components
102 p_cal$K = 0
103 # total number of FGSN parameters
104 p_cal$p_fgsn = 0
105 if( eiv ){
106     p_cal$K = 2
107     p_cal$p_fgsn = p_cal$K + 2
108 }
109
110 # specify Markov chain Monte Carlo (MCMC) algorithm
111 # options: "RAM", "FME", or "NUTS"
112 iMCMC = "RAM"
113
114 # burn-in
115 nburn = 10000
116
117 # production
118 nmcmc = 20000
119
120 # posterior sample thinning rate
121 nthin = 20
122
123 # number of cores to use for optimization
124 ncores_map = 30
125
126 # number of cores to use for generating parallel MCMC chains
127 ncores_mc = 30
128
129 # Indicator of prior gradient check
130 prior_grad_ck = TRUE
131
132 # Indicator of posterior gradient check
133 post_grad_ck = TRUE

```

```

134
135 # Bayesian calculations
136 p_cal = calc_bayes(p_cal,gen_dir,app_dir,nst=nstart,nburn=nburn,
137                   nmc=nmcmc,nthin=nthin,ncor_map=ncores_map,
138                   ncor_mc=ncores_mc,igrad=igrad,
139                   igrck_pr=prior_grad_ck,igrck_po=post_grad_ck,
140                   bfgs=bfgs,ibpr=iBetaPrior,icpr=iCalPrior,
141                   itpr=iTheta0Prior,fpr_b=prior_files_beta,
142                   fgpr_b=gr_prior_files_beta,fpr_c=prior_files_calp,
143                   fgpr_c=gr_prior_files_calp,
144                   fpr_t=prior_files_theta0,
145                   fgpr_t=gr_prior_files_theta0,Xnom=NULL,
146                   imcmc=iMCMC,pl=parallel_plan,t_cal=t_cal)
147 save.image()
148 }
149
150 #
151 # END BAYESIAN ANALYSIS
152 #

```

B.4 Output File

```
1 > # Preprocessing for statistical analysis routines
2 > tmp = prepro(gen_dir,app_dir,dat_dir,dat_cal,Rh,pbeta,bopt=opt_B,
3 +             nev=nev,itr=itransform,izmat=calc_Z,ieiv=ieiv,seiv=seiv,
4 +             ewsd=eiv_w_sd,Th=Th,pbetat=pbetat,ibetar=ibetar,
5 +             pvc_1=pvc_1,pvc_2=pvc_2,ptype=ptype,tnames=theta_names,
6 +             cnames=cal_par_names,fp_tr=tPars,tlb=lb_theta0,
7 +             tub=ub_theta0,ndir=dat_new,tsub=tsub)
8 [1] "Warning: Insufficient number of observations per Source for Variance
9      Component models with Phenomenology 3 and Response 1."
10 [1] "Warning: Insufficient number of observations per Source for Variance
11      Component models with Phenomenology 3 and Response 2."
12 [1] "Warning: Insufficient number of observations per Source for Variance
13      Component models with Phenomenology 4 and Response 1."
14 [1] "Warning: Insufficient number of observations per Source for Variance
15      Component models with Phenomenology 4 and Response 2."
16
17 > # MLE calculations
18 > p_cal = calc_mle(p_cal,gen_dir,app_dir,fm,nst=nstart,ncor=ncores_mle,
19 +                 ci_lev=ci_lev,igrad=igrad,bfgs=bfgs,igrck=mle_grad_ck,
20 +                 t_cal=t_cal,g=gfm,iresp=iResponse,fp_fm=fPars,
21 +                 fopt_in=opt_files_in,Xst=NULL,tst=tst,cst=cst,
22 +                 fopt_out=opt_files_out,phen=Phen,pl=parallel_plan)
23 [1] "MLE CONVERGENCE STATUS"
24
25 [1] 0
26 [1] 2
27 [1] "MAXIMUM LIKELIHOOD SUMMARY"
28
29 [1] "NEW EVENT INFERENCE PARAMETERS"
30
31 [1] "ESTIMATE: "
32
33      W    HOB
34 13.75  1.09
35
36 [1] "STANDARD DEVIATION FIXED MODEL PARAMETERS: "
37
38      W    HOB
39 0.18 0.43
40
41 [1] "CORRELATION MATRIX FIXED MODEL PARAMETERS: "
42
43      W    HOB
```

```

44 W    1.00 0.22
45 HOB 0.22 1.00
46
47 [1] "STANDARD DEVIATION: "
48
49     W  HOB
50 0.24 0.61
51
52 [1] "CORRELATION MATRIX: "
53
54     W  HOB
55 W    1.0 0.2
56 HOB 0.2 1.0
57
58 [1] "95%: CONFIDENCE INTERVAL FIXED MODEL PARAMETERS:"
59
60     W  HOB
61 lb_0 13.41 0.57
62 ub_0 14.10 2.86
63
64 [1] "95%: CONFIDENCE INTERVAL:"
65
66     W  HOB
67 lb 13.28 0.45
68 ub 14.23 5.23
69
70
71 [1] "ERRORS-IN-VARIABLES YIELDS"
72
73     7      8      9      10      11      13      14      16      17      20      21      22      23
74 16.31 16.21 16.50 16.60 16.99 12.26 17.57 17.28 16.54 14.50 15.75 17.59 15.19
75     24     25     28     29     30     31     33     34     35     36     37     38     39
76 15.89 16.45 14.50 12.14 17.66 23.11 23.48 17.45 21.90 22.33 16.74 21.04 18.52
77
78 [1] "COMMON COEFFICIENTS"
79
80     [1] "Phenomenology: 2; Response: 1"
81
82     [1] 6.13 -1.13
83
84     [1] "Phenomenology: 2; Response: 2"
85
86     [1] -5.26 0.23
87
88     [1] "Phenomenology: 3; Response: 1"

```



```

89
90      [1] -10.55   0.37  -0.47  -0.86
91
92      [1] "Phenomenology: 3; Response: 2"
93
94      [1] -8.26   0.38 -0.86 -6.83
95
96      [1] "Phenomenology: 4; Response: 1"
97
98      [1] -3.21   0.43
99
100     [1] "Phenomenology: 4; Response: 2"
101
102     [1] -1.64   0.24
103
104     [1] "EMPLACEMENT CONDITION DEPENDENT COEFFICIENTS"
105
106     [1] "Phenomenology: 1; Emplacement: 1; Response: 1"
107
108     [1] -9.70 -1.40 -1.14 22.04  4.03
109
110     [1] "Phenomenology: 1; Emplacement: 1; Response: 2"
111
112     [1]   0.25  -1.78  -0.95 252.86  36.60
113
114     [1] "Phenomenology: 1; Emplacement: 2; Response: 1"
115
116     [1] -10.54  -1.25 -128.05   1.74  -4.23
117
118     [1] "Phenomenology: 1; Emplacement: 2; Response: 2"
119
120     [1]   0.75  -1.84 -346.93   2.75  -5.90
121
122     [1] "Phenomenology: 1; Emplacement: 3; Response: 1"
123
124     [1] -6.61 -1.64 -4.16  7.34  0.76
125
126     [1] "Phenomenology: 1; Emplacement: 3; Response: 2"
127
128     [1]   3.88  -1.87  -2.43 105.59   3.62
129
130     [1] "Phenomenology: 2; Emplacement: 1; Response: 1"
131
132     [1] 4.81
133

```

134 [1] "Phenomenology: 2; Emplacement: 1; Response: 2"
 135
 136 [1] -0.2
 137
 138 [1] "Phenomenology: 2; Emplacement: 2; Response: 1"
 139
 140 [1] 3.86
 141
 142 [1] "Phenomenology: 2; Emplacement: 2; Response: 2"
 143
 144 [1] -1.14
 145
 146 [1] "Phenomenology: 2; Emplacement: 3; Response: 1"
 147
 148 [1] 2.45
 149
 150 [1] "Phenomenology: 2; Emplacement: 3; Response: 2"
 151
 152 [1] -1.42
 153
 154 [1] "SOURCE VARIANCE COMPONENTS"
 155
 156 [1] "Phenomenology: 1; Response: 1"
 157
 158 [1] 0.0379
 159
 160 [1] "Phenomenology: 1; Response: 2"
 161
 162 [1] 0.1026
 163
 164 [1] "Phenomenology: 2; Response: 1"
 165
 166 [1] 0.1539
 167
 168 [1] "Phenomenology: 2; Response: 2"
 169
 170 [1] 0.0314
 171
 172 [1] "PATH VARIANCE COMPONENTS"
 173
 174 [1] "Phenomenology: 1; Response: 1"
 175
 176 [1] 0.1344
 177
 178 [1] "Phenomenology: 1; Response: 2"

```

179
180         [1] 0.147
181
182         [1] "Phenomenology: 2; Response: 1"
183
184         [1] 0.0209
185
186         [1] "Phenomenology: 2; Response: 2"
187
188         [1] 0.0128
189
190 [1] "OBSERVATIONAL ERROR COVARIANCE PARAMETERS"
191
192 [1] "Phenomenology 1"
193
194 [1] "Variances"
195
196 [1] 0.0836 0.1851
197
198 [1] "Correlations"
199
200         [,1] [,2]
201 [1,]      1 0.41
202 [2,]      0 1.00
203
204 [1] "Phenomenology 2"
205
206 [1] "Variances"
207
208 [1] 0.0722 0.0168
209
210 [1] "Correlations"
211
212         [,1] [,2]
213 [1,]      1 -0.13
214 [2,]      0 1.00
215
216 [1] "Phenomenology 3"
217
218 [1] "Variances"
219
220 [1] 0.1794 0.1635
221
222 [1] "Correlations"
223

```

```

224      [,1] [,2]
225 [1,]    1 0.98
226 [2,]    0 1.00
227
228 [1] "Phenomenology 4"
229
230 [1] "Variances"
231
232 [1] 0.0124 0.0313
233
234 [1] "Correlations"
235
236      [,1] [,2]
237 [1,]    1 -0.21
238 [2,]    0 1.00
239
240 [1] "AIC = 1175.77"
241
242 [1] "BIC = 1539.03"
243
244 Loading required package: numDeriv
245 [1] "CHECK LOG-LIKELIHOOD GRADIENTS"
246
247 [1] "Analytic gradient"
248 [1] 1.889636e-04 1.979534e-05 9.327815e-05 1.186896e-05 7.037498e-07
249 [6] 3.156326e-05 -9.992847e-06 -2.352286e-04 3.373750e-05 1.632864e-04
250 [11] -2.621857e-04 3.442464e-05 -1.328103e-04 8.578609e-05 -1.154799e-04
251 [16] -3.325700e-05 1.005030e-04 2.426265e-04 -3.263952e-05 -1.048252e-05
252 [21] -8.495686e-04 -2.998059e-04 -1.788661e-04 -6.583879e-04 -1.557321e-06
253 [26] -3.829906e-04 -1.729691e-04 -1.039386e-03 -2.146308e-04 -4.559856e-04
254 [31] 1.811903e-04 -2.273453e-04 5.812326e-03 1.050803e-01 1.567457e-03
255 [36] 1.341854e-03 -6.142810e-03 -1.107020e-01 -5.752910e-04 -4.294192e-05
256 [41] -6.418059e-03 -1.308105e-01 -1.752086e-03 -3.755851e-02 -3.662864e-05
257 [46] -7.755280e-04 -1.452078e-03 -3.125450e-05 2.122689e-04 1.115359e-04
258 [51] 9.059122e-04 -8.573932e-05 -4.699025e-05 3.211258e-04 -3.329960e-02
259 [56] -1.627690e-01 2.095775e-02 2.677818e-02 9.657676e-02 -2.096955e-02
260 [61] -1.018079e-01 5.626800e-03 1.284405e-02 4.330289e-02 -1.216947e-04
261 [66] 1.544840e-04 2.384485e-04 1.482537e-05 3.346663e-05 -9.169129e-04
262 [71] -4.158536e-03 -6.889016e-04 2.040453e-04 1.270556e-04 -5.812449e-05
263 [76] -7.872415e-05 9.130954e-06 -2.800253e-05 9.783931e-05 -5.412616e-05
264 [81] 5.924874e-05 5.422608e-04 2.615568e-05 5.600618e-05 7.234010e-04
265 [86] -2.790801e-04 -6.360674e-05 3.193798e-04 -4.573840e-04 -3.912355e-04
266 [91] -1.239258e-03 -4.234052e-05 9.313203e-04 1.452440e-04 -7.311773e-04
267 [96] -7.359814e-05 1.592972e-03 4.851710e-05 3.033597e-04 6.840987e-04
268 [1] "Numerical gradient"

```

```

269 [1] 1.889672e-04 1.979524e-05 9.327871e-05 1.186899e-05 7.040755e-07
270 [6] 3.156248e-05 -9.991593e-06 -2.352281e-04 3.373693e-05 1.632866e-04
271 [11] -2.621872e-04 3.442490e-05 -1.328114e-04 8.578583e-05 -1.154789e-04
272 [16] -3.325648e-05 1.005034e-04 2.426258e-04 -3.264006e-05 -1.048215e-05
273 [21] -8.495685e-04 -2.998059e-04 -1.788661e-04 -6.583884e-04 -1.557373e-06
274 [26] -3.829912e-04 -1.729692e-04 -1.039386e-03 -2.146311e-04 -4.559512e-04
275 [31] 1.811876e-04 -2.273795e-04 5.812327e-03 1.050804e-01 1.567490e-03
276 [36] 1.341848e-03 -6.142814e-03 -1.107019e-01 -5.753007e-04 -4.294167e-05
277 [41] -6.418064e-03 -1.308105e-01 -1.752080e-03 -3.755855e-02 -3.662927e-05
278 [46] -7.755610e-04 -1.451958e-03 -3.125302e-05 2.122823e-04 1.115070e-04
279 [51] 9.059280e-04 -8.575542e-05 -4.699030e-05 3.211258e-04 -3.329960e-02
280 [56] -1.627690e-01 2.095775e-02 2.677818e-02 9.657676e-02 -2.096952e-02
281 [61] -1.018079e-01 5.626798e-03 1.284404e-02 4.330288e-02 -1.216961e-04
282 [66] 1.544934e-04 2.384560e-04 1.482035e-05 3.342587e-05 -9.169207e-04
283 [71] -4.158501e-03 -6.889003e-04 2.040453e-04 1.270596e-04 -5.813405e-05
284 [76] -7.872201e-05 9.128024e-06 -2.798809e-05 9.783516e-05 -5.413539e-05
285 [81] 5.923699e-05 5.422845e-04 2.616514e-05 5.602005e-05 7.233660e-04
286 [86] -2.790630e-04 -6.361130e-05 3.193673e-04 -4.573488e-04 -3.912442e-04
287 [91] -1.239248e-03 -4.229902e-05 9.313068e-04 1.477350e-04 -7.313718e-04
288 [96] -7.356558e-05 1.592941e-03 4.852152e-05 3.033360e-04 6.836520e-04
289 [1] "Difference"
290 [1] -2.491019e-06 4.467148e-07
291
292 + # Bayesian calculations
293 + p_cal = calc_bayes(p_cal,gen_dir,app_dir,nst=nstart,nburn=nburn,
294 + nmcmlc=nmcmlc,nthin=nthin,ncor_map=ncores_map,
295 + ncor_mc=ncores_mc,igrad=igrad,
296 + igrck_pr=prior_grad_ck,igrck_po=post_grad_ck,
297 + bfgs=bfgs,ibpr=iBetaPrior,icpr=iCalPrior,
298 + itpr=iTheta0Prior,fpr_b=prior_files_beta,
299 + fgpr_b=gr_prior_files_beta,fpr_c=prior_files_calp,
300 + fgpr_c=gr_prior_files_calp,
301 + fpr_t=prior_files_theta0,
302 + fgpr_t=gr_prior_files_theta0,Xnom=NULL,
303 + imcmlc=iMCMC,pl=parallel_plan,t_cal=t_cal)
304 + save.image()
305 + }
306 [1] "Perturbation added to Hessian diagonals: 1e-07"
307 [1] "MAP CONVERGENCE STATUS"
308
309 [1] 0
310 [1] 2
311 [1] "MAXIMUM A POSTERIORI SUMMARY"
312
313 [1] "NEW EVENT INFERENCE PARAMETERS"

```

```

314
315 [1] "ESTIMATE: "
316
317     W    HOB
318 13.65 36.77
319
320
321 [1] "ERRORS-IN-VARIABLES YIELDS"
322
323     7     8     9    10    11    13    14    16    17    20    21    22    23
324 16.31 16.21 16.51 16.60 16.99 12.26 17.57 17.28 16.54 14.51 15.75 17.58 15.19
325     24    25    28    29    30    31    33    34    35    36    37    38    39
326 15.89 16.45 14.50 12.14 17.66 23.10 23.46 17.44 21.91 22.33 16.73 21.05 18.51
327
328 [1] "COMMON COEFFICIENTS"
329
330     [1] "Phenomenology: 2; Response: 1"
331
332     [1] 6.10 -1.13
333
334     [1] "Phenomenology: 2; Response: 2"
335
336     [1] -5.25 0.23
337
338     [1] "Phenomenology: 3; Response: 1"
339
340     [1] -10.45 0.36 -0.43 -0.88
341
342     [1] "Phenomenology: 3; Response: 2"
343
344     [1] -8.17 0.38 -0.83 -6.56
345
346     [1] "Phenomenology: 4; Response: 1"
347
348     [1] -3.15 0.42
349
350     [1] "Phenomenology: 4; Response: 2"
351
352     [1] -1.60 0.24
353
354 [1] "EMPLACEMENT CONDITION DEPENDENT COEFFICIENTS"
355
356     [1] "Phenomenology: 1; Emplacement: 1; Response: 1"
357
358     [1]          -10.55          -1.37 -147674457.77          14.88          -46.24

```

359
360 [1] "Phenomenology: 1; Emplacement: 1; Response: 2"
361
362 [1] -0.46 -1.76 -154064746.72 -7.56 -44.20
363
364 [1] "Phenomenology: 1; Emplacement: 2; Response: 1"
365
366 [1] -1.07500e+01 -1.22000e+00 -5.32655e+08 1.97000e+00 -1.95800e+01
367
368 [1] "Phenomenology: 1; Emplacement: 2; Response: 2"
369
370 [1] 0.45 -1.79 -239762241.12 3.14 -19.50
371
372 [1] "Phenomenology: 1; Emplacement: 3; Response: 1"
373
374 [1] -5.62 -1.65 -763021708.24 1.45 -19.21
375
376 [1] "Phenomenology: 1; Emplacement: 3; Response: 2"
377
378 [1] 5.23 -1.85 -218052111.03 1.07 -18.06
379
380 [1] "Phenomenology: 2; Emplacement: 1; Response: 1"
381
382 [1] 4.78
383
384 [1] "Phenomenology: 2; Emplacement: 1; Response: 2"
385
386 [1] -0.19
387
388 [1] "Phenomenology: 2; Emplacement: 2; Response: 1"
389
390 [1] 3.85
391
392 [1] "Phenomenology: 2; Emplacement: 2; Response: 2"
393
394 [1] -1.14
395
396 [1] "Phenomenology: 2; Emplacement: 3; Response: 1"
397
398 [1] 2.43
399
400 [1] "Phenomenology: 2; Emplacement: 3; Response: 2"
401
402 [1] -1.42
403

```

404 [1] "SOURCE VARIANCE COMPONENTS"
405
406     [1] "Phenomenology: 1; Response: 1"
407
408     [1] 0.2394
409
410     [1] "Phenomenology: 1; Response: 2"
411
412     [1] 0.3172
413
414     [1] "Phenomenology: 2; Response: 1"
415
416     [1] 0.1641
417
418     [1] "Phenomenology: 2; Response: 2"
419
420     [1] 0.0328
421
422 [1] "PATH VARIANCE COMPONENTS"
423
424     [1] "Phenomenology: 1; Response: 1"
425
426     [1] 0.1295
427
428     [1] "Phenomenology: 1; Response: 2"
429
430     [1] 0.1501
431
432     [1] "Phenomenology: 2; Response: 1"
433
434     [1] 0.0217
435
436     [1] "Phenomenology: 2; Response: 2"
437
438     [1] 0.0131
439
440 [1] "OBSERVATIONAL ERROR COVARIANCE PARAMETERS"
441
442 [1] "Phenomenology 1"
443
444 [1] "Variances"
445
446 [1] 0.0835 0.1832
447
448 [1] "Correlations"

```



```

449
450      [,1] [,2]
451 [1,]    1 0.4
452 [2,]    0 1.0
453
454 [1] "Phenomenology 2"
455
456 [1] "Variances"
457
458 [1] 0.0718 0.0166
459
460 [1] "Correlations"
461
462      [,1] [,2]
463 [1,]    1 -0.12
464 [2,]    0 1.00
465
466 [1] "Phenomenology 3"
467
468 [1] "Variances"
469
470 [1] 0.1626 0.1477
471
472 [1] "Correlations"
473
474      [,1] [,2]
475 [1,]    1 0.98
476 [2,]    0 1.00
477
478 [1] "Phenomenology 4"
479
480 [1] "Variances"
481
482 [1] 0.0125 0.0286
483
484 [1] "Correlations"
485
486      [,1] [,2]
487 [1,]    1 -0.1
488 [2,]    0 1.0
489
490 [1] "FGSN PRIOR PARAMETERS"
491
492 [1] "Alpha = 20.77"
493 [1] "Lambda squared = 17.43"

```

```

494 [1] "Omega = 3.86" "Omega = -9.84"
495
496 [1] "CHECK LOG-PRIOR GRADIENTS"
497
498 [1] "Analytic gradient"
499 [1] 0.000000e+00 6.866078e-01 2.557242e-01 2.611331e-01 2.442877e-01
500 [6] 2.391311e-01 2.150300e-01 4.876328e-01 -2.950367e-01 1.316473e-01
501 [11] 2.422645e-01 3.590385e-01 2.876670e-01 -3.245607e-01 3.195985e-01
502 [16] 2.795230e-01 2.475521e-01 3.591038e-01 4.947293e-01 -4.889744e-01
503 [21] -8.365377e-01 -1.966228e+00 -5.858119e-02 7.114208e-02 -1.121370e-01
504 [26] 2.315801e-01 5.568784e-01 -1.201595e+00 0.000000e+00 0.000000e+00
505 [31] 0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00
506 [36] 0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00 1.455604e-01
507 [41] 0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00
508 [46] 0.000000e+00 9.593555e-05 0.000000e+00 0.000000e+00 0.000000e+00
509 [51] 0.000000e+00 9.392487e-05 0.000000e+00 0.000000e+00 0.000000e+00
510 [56] 0.000000e+00 5.051371e-05 0.000000e+00 0.000000e+00 0.000000e+00
511 [61] 0.000000e+00 7.529079e-05 0.000000e+00 0.000000e+00 0.000000e+00
512 [66] 0.000000e+00 4.220499e-05 0.000000e+00 0.000000e+00 0.000000e+00
513 [71] 0.000000e+00 7.894999e-05 0.000000e+00 0.000000e+00 0.000000e+00
514 [76] 0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00
515 [81] 4.994018e-01 4.992077e-01 4.995898e-01 4.999180e-01 4.996763e-01
516 [86] 4.996249e-01 4.999458e-01 4.999673e-01 0.000000e+00 -5.094154e-01
517 [91] -2.834417e+00 0.000000e+00 -9.532280e-01 2.905942e+00 -4.440892e-16
518 [96] 1.859669e+00 -7.620194e+00 0.000000e+00 -9.675227e-01 1.845838e+00
519 [101] -1.269265e-05 7.668126e-05 4.768596e-06 1.397674e-06
520 [1] "Numerical gradient"
521 [1] 0.000000e+00 6.866078e-01 2.557242e-01 2.611331e-01 2.442877e-01
522 [6] 2.391311e-01 2.150300e-01 4.876328e-01 -2.950367e-01 1.316473e-01
523 [11] 2.422645e-01 3.590385e-01 2.876670e-01 -3.245607e-01 3.195985e-01
524 [16] 2.795230e-01 2.475521e-01 3.591038e-01 4.947293e-01 -4.889744e-01
525 [21] -8.365377e-01 -1.966228e+00 -5.858119e-02 7.114208e-02 -1.121370e-01
526 [26] 2.315801e-01 5.568784e-01 -1.201595e+00 0.000000e+00 0.000000e+00
527 [31] 0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00
528 [36] 0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00 1.455604e-01
529 [41] 0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00
530 [46] 0.000000e+00 9.593555e-05 0.000000e+00 0.000000e+00 0.000000e+00
531 [51] 0.000000e+00 9.392487e-05 0.000000e+00 0.000000e+00 0.000000e+00
532 [56] 0.000000e+00 5.051371e-05 0.000000e+00 0.000000e+00 0.000000e+00
533 [61] 0.000000e+00 7.529079e-05 0.000000e+00 0.000000e+00 0.000000e+00
534 [66] 0.000000e+00 4.220499e-05 0.000000e+00 0.000000e+00 0.000000e+00
535 [71] 0.000000e+00 7.894999e-05 0.000000e+00 0.000000e+00 0.000000e+00
536 [76] 0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00
537 [81] 4.994018e-01 4.992077e-01 4.995898e-01 4.999180e-01 4.996763e-01
538 [86] 4.996249e-01 4.999458e-01 4.999673e-01 0.000000e+00 -5.094154e-01

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539 [91] -2.834417e+00 0.000000e+00 -9.532280e-01 2.905942e+00 1.057769e-19
540 [96] 1.859669e+00 -7.620194e+00 0.000000e+00 -9.675227e-01 1.845838e+00
541 [101] -1.269263e-05 7.668105e-05 4.768979e-06 1.397674e-06
542 [1] "Difference"
543 [1] -5.942113e-09 4.385131e-08
544
545 [1] "CHECK LOG-POSTERIOR GRADIENTS"
546
547 [1] "Analytic gradient"
548 [1] -1.017061e-04 -6.198745e-06 -2.195747e-04 -1.084250e-04 1.706170e-05
549 [6] -8.544150e-05 8.660167e-05 -4.038308e-05 -1.934465e-05 -1.534313e-05
550 [11] 7.326406e-05 8.004617e-05 3.549186e-05 -3.918374e-05 9.904637e-06
551 [16] -7.836092e-05 7.661220e-05 1.921951e-05 -7.011434e-06 -9.905870e-06
552 [21] -1.442063e-04 -5.705642e-05 -4.236935e-05 -2.924564e-05 2.436260e-05
553 [26] -8.611112e-07 -8.998067e-05 -4.869752e-06 -3.840009e-05 -8.107434e-05
554 [31] 7.762245e-05 -4.260097e-05 1.087814e-04 2.103136e-03 1.186941e-04
555 [36] 1.571887e-04 -1.622334e-04 -2.185773e-03 1.160528e-04 1.850237e-04
556 [41] -2.483704e-04 -4.339237e-03 -3.364853e-05 -6.955617e-04 -6.251959e-04
557 [46] -3.518368e-03 9.593555e-05 1.394843e-08 2.255195e-08 -9.408589e-05
558 [51] -3.482771e-06 9.392481e-05 4.076915e-07 -3.425394e-07 1.688876e-04
559 [56] 9.293371e-04 5.048304e-05 -1.012064e-04 -3.036084e-04 1.887201e-05
560 [61] -1.180778e-04 7.527802e-05 1.409681e-05 -8.481309e-05 2.201795e-05
561 [66] 2.735661e-04 4.216416e-05 1.961556e-04 -4.836977e-04 4.422984e-04
562 [71] 2.017716e-03 7.884983e-05 -8.333723e-04 -6.343277e-04 2.144477e-05
563 [76] -1.358425e-05 -3.789958e-05 6.578306e-06 5.496561e-05 1.197309e-05
564 [81] 5.765107e-05 1.782721e-04 4.717755e-06 1.800562e-05 2.411407e-05
565 [86] 1.817448e-05 -1.194070e-05 1.250188e-05 -2.387841e-04 5.537983e-05
566 [91] -3.503936e-05 1.328072e-05 -5.317068e-05 -6.734721e-05 -8.241768e-05
567 [96] -8.531565e-05 6.700947e-05 1.763811e-06 1.801670e-06 -1.052975e-04
568 [101] -1.269265e-05 7.668126e-05 4.768596e-06 1.397674e-06
569 [1] "Numerical gradient"
570 [1] -1.017055e-04 -6.208960e-06 -2.195756e-04 -1.084239e-04 1.706136e-05
571 [6] -8.544209e-05 8.660072e-05 -4.038400e-05 -1.934414e-05 -1.534332e-05
572 [11] 7.326330e-05 8.004583e-05 3.548906e-05 -3.918489e-05 9.903710e-06
573 [16] -7.836123e-05 7.661098e-05 1.922200e-05 -7.012369e-06 -9.905426e-06
574 [21] -1.442083e-04 -5.705689e-05 -4.236974e-05 -2.924474e-05 2.436096e-05
575 [26] -8.596070e-07 -8.998048e-05 -4.867341e-06 -3.839853e-05 -8.103171e-05
576 [31] 7.762432e-05 -4.253559e-05 1.087721e-04 2.103174e-03 1.185531e-04
577 [36] 1.571119e-04 -1.622209e-04 -2.185311e-03 1.160947e-04 1.850282e-04
578 [41] -2.483650e-04 -4.339188e-03 -3.364749e-05 -6.956318e-04 -6.251930e-04
579 [46] -3.518362e-03 9.593555e-05 1.494315e-08 2.233542e-08 -9.402663e-05
580 [51] -3.486597e-06 9.392481e-05 4.095422e-07 -3.424171e-07 1.688882e-04
581 [56] 9.293524e-04 5.048304e-05 -1.012252e-04 -3.036092e-04 1.891406e-05
582 [61] -1.180729e-04 7.527802e-05 1.409442e-05 -8.481255e-05 2.201730e-05
583 [66] 2.735433e-04 4.216416e-05 1.961515e-04 -4.836963e-04 4.422999e-04

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584 [71] 2.017718e-03 7.884983e-05 -8.333728e-04 -6.343274e-04 2.144276e-05
585 [76] -1.349009e-05 -3.790013e-05 6.560494e-06 5.498022e-05 1.197686e-05
586 [81] 5.761285e-05 1.782545e-04 4.703847e-06 1.799020e-05 2.408198e-05
587 [86] 1.815746e-05 -1.192233e-05 1.249933e-05 -2.387806e-04 5.529707e-05
588 [91] -3.491855e-05 1.333231e-05 -5.320085e-05 -7.160914e-05 -8.236477e-05
589 [96] -8.526496e-05 6.688301e-05 1.802031e-06 1.830750e-06 -1.039566e-04
590 [101] -1.269184e-05 7.668236e-05 4.771679e-06 1.397438e-06
591 [1] "Difference"
592 [1] -1.340853e-06 4.261934e-06
593
594 [1] "ACCEPTANCE RATES:"
595
596 [1] "Core 1: 0.237"
597 [1] "Core 2: 0.235"
598 [1] "Core 3: 0.243"
599 [1] "Core 4: 0.24"
600 [1] "Core 5: 0.247"
601 [1] "Core 6: 0.251"
602 [1] "Core 7: 0.246"
603 [1] "Core 8: 0.236"
604 [1] "Core 9: 0.239"
605 [1] "Core 10: 0.239"
606 [1] "Core 11: 0.247"
607 [1] "Core 12: 0.235"
608 [1] "Core 13: 0.23"
609 [1] "Core 14: 0.241"
610 [1] "Core 15: 0.244"
611 [1] "Core 16: 0.239"
612 [1] "Core 17: 0.246"
613 [1] "Core 18: 0.245"
614 [1] "Core 19: 0.25"
615 [1] "Core 20: 0.246"
616 [1] "Core 21: 0.232"
617 [1] "Core 22: 0.242"
618 [1] "Core 23: 0.246"
619 [1] "Core 24: 0.242"
620 [1] "Core 25: 0.251"
621 [1] "Core 26: 0.249"
622 [1] "Core 27: 0.248"
623 [1] "Core 28: 0.245"
624 [1] "Core 29: 0.236"
625 [1] "Core 30: 0.239"
626
627 [1] "POSTERIOR SUMMARY"
628

```

```

629 [1] "NEW EVENT INFERENCE PARAMETERS"
630
631 [1] "POSTERIOR MEAN: 13.6" "POSTERIOR MEAN: 35.67"
632
633 [1] "POSTERIOR SD: 0.33" "POSTERIOR SD: 19.55"
634
635 [1] "LEVEL 2.5%: 12.88" "LEVEL 2.5%: 8.16"
636
637 [1] "LEVEL 5%: 13.02" "LEVEL 5%: 10.09"
638
639 [1] "LEVEL 50%: 13.61" "LEVEL 50%: 32.38"
640
641 [1] "LEVEL 95%: 14.1" "LEVEL 95%: 75.85"
642
643 [1] "LEVEL 97.5%: 14.18" "LEVEL 97.5%: 88.43"
644
645 [1] "CORRELATION MATRIX:"
646
647           W    HOB
648 W      1.00 -0.15
649 HOB -0.15  1.00
650
651 [1] "ERRORS-IN-VARIABLES YIELDS"
652
653 [1] "POSTERIOR MEAN: 16.3" "POSTERIOR MEAN: 16.22" "POSTERIOR MEAN: 16.53"
654 [4] "POSTERIOR MEAN: 16.58" "POSTERIOR MEAN: 16.98" "POSTERIOR MEAN: 12.27"
655 [7] "POSTERIOR MEAN: 17.57" "POSTERIOR MEAN: 17.25" "POSTERIOR MEAN: 16.55"
656 [10] "POSTERIOR MEAN: 14.5" "POSTERIOR MEAN: 15.74" "POSTERIOR MEAN: 17.59"
657 [13] "POSTERIOR MEAN: 15.19" "POSTERIOR MEAN: 15.89" "POSTERIOR MEAN: 16.44"
658 [16] "POSTERIOR MEAN: 14.52" "POSTERIOR MEAN: 12.16" "POSTERIOR MEAN: 17.65"
659 [19] "POSTERIOR MEAN: 23.08" "POSTERIOR MEAN: 23.45" "POSTERIOR MEAN: 17.47"
660 [22] "POSTERIOR MEAN: 21.92" "POSTERIOR MEAN: 22.35" "POSTERIOR MEAN: 16.71"
661 [25] "POSTERIOR MEAN: 21.06" "POSTERIOR MEAN: 18.5"
662
663 [1] "LEVEL 2.5%: 16.13" "LEVEL 2.5%: 16.03" "LEVEL 2.5%: 16.34"
664 [4] "LEVEL 2.5%: 16.38" "LEVEL 2.5%: 16.77" "LEVEL 2.5%: 12.08"
665 [7] "LEVEL 2.5%: 17.37" "LEVEL 2.5%: 17.05" "LEVEL 2.5%: 16.4"
666 [10] "LEVEL 2.5%: 14.29" "LEVEL 2.5%: 15.52" "LEVEL 2.5%: 17.38"
667 [13] "LEVEL 2.5%: 15" "LEVEL 2.5%: 15.69" "LEVEL 2.5%: 16.26"
668 [16] "LEVEL 2.5%: 14.33" "LEVEL 2.5%: 11.94" "LEVEL 2.5%: 17.46"
669 [19] "LEVEL 2.5%: 22.91" "LEVEL 2.5%: 23.27" "LEVEL 2.5%: 17.27"
670 [22] "LEVEL 2.5%: 21.75" "LEVEL 2.5%: 22.18" "LEVEL 2.5%: 16.52"
671 [25] "LEVEL 2.5%: 20.84" "LEVEL 2.5%: 18.31"
672
673 [1] "LEVEL 5%: 16.15" "LEVEL 5%: 16.06" "LEVEL 5%: 16.37" "LEVEL 5%: 16.42"

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674 [5] "LEVEL 5%: 16.81" "LEVEL 5%: 12.12" "LEVEL 5%: 17.4" "LEVEL 5%: 17.1"
 675 [9] "LEVEL 5%: 16.42" "LEVEL 5%: 14.33" "LEVEL 5%: 15.56" "LEVEL 5%: 17.4"
 676 [13] "LEVEL 5%: 15.03" "LEVEL 5%: 15.73" "LEVEL 5%: 16.28" "LEVEL 5%: 14.35"
 677 [17] "LEVEL 5%: 11.97" "LEVEL 5%: 17.48" "LEVEL 5%: 22.94" "LEVEL 5%: 23.29"
 678 [21] "LEVEL 5%: 17.3" "LEVEL 5%: 21.77" "LEVEL 5%: 22.21" "LEVEL 5%: 16.55"
 679 [25] "LEVEL 5%: 20.87" "LEVEL 5%: 18.34"
 680
 681 [1] "LEVEL 50%: 16.3" "LEVEL 50%: 16.22" "LEVEL 50%: 16.53" "LEVEL 50%: 16.58"
 682 [5] "LEVEL 50%: 16.98" "LEVEL 50%: 12.27" "LEVEL 50%: 17.58" "LEVEL 50%: 17.26"
 683 [9] "LEVEL 50%: 16.55" "LEVEL 50%: 14.5" "LEVEL 50%: 15.74" "LEVEL 50%: 17.59"
 684 [13] "LEVEL 50%: 15.19" "LEVEL 50%: 15.9" "LEVEL 50%: 16.44" "LEVEL 50%: 14.52"
 685 [17] "LEVEL 50%: 12.16" "LEVEL 50%: 17.65" "LEVEL 50%: 23.08" "LEVEL 50%: 23.45"
 686 [21] "LEVEL 50%: 17.47" "LEVEL 50%: 21.92" "LEVEL 50%: 22.34" "LEVEL 50%: 16.72"
 687 [25] "LEVEL 50%: 21.06" "LEVEL 50%: 18.5"
 688
 689 [1] "LEVEL 95%: 16.44" "LEVEL 95%: 16.37" "LEVEL 95%: 16.67" "LEVEL 95%: 16.74"
 690 [5] "LEVEL 95%: 17.14" "LEVEL 95%: 12.43" "LEVEL 95%: 17.74" "LEVEL 95%: 17.39"
 691 [9] "LEVEL 95%: 16.7" "LEVEL 95%: 14.67" "LEVEL 95%: 15.9" "LEVEL 95%: 17.8"
 692 [13] "LEVEL 95%: 15.35" "LEVEL 95%: 16.07" "LEVEL 95%: 16.6" "LEVEL 95%: 14.67"
 693 [17] "LEVEL 95%: 12.33" "LEVEL 95%: 17.81" "LEVEL 95%: 23.23" "LEVEL 95%: 23.61"
 694 [21] "LEVEL 95%: 17.64" "LEVEL 95%: 22.07" "LEVEL 95%: 22.51" "LEVEL 95%: 16.86"
 695 [25] "LEVEL 95%: 21.22" "LEVEL 95%: 18.66"
 696
 697 [1] "LEVEL 97.5%: 16.47" "LEVEL 97.5%: 16.4" "LEVEL 97.5%: 16.7"
 698 [4] "LEVEL 97.5%: 16.76" "LEVEL 97.5%: 17.17" "LEVEL 97.5%: 12.45"
 699 [7] "LEVEL 97.5%: 17.77" "LEVEL 97.5%: 17.41" "LEVEL 97.5%: 16.72"
 700 [10] "LEVEL 97.5%: 14.7" "LEVEL 97.5%: 15.91" "LEVEL 97.5%: 17.83"
 701 [13] "LEVEL 97.5%: 15.37" "LEVEL 97.5%: 16.09" "LEVEL 97.5%: 16.65"
 702 [16] "LEVEL 97.5%: 14.7" "LEVEL 97.5%: 12.37" "LEVEL 97.5%: 17.83"
 703 [19] "LEVEL 97.5%: 23.25" "LEVEL 97.5%: 23.65" "LEVEL 97.5%: 17.67"
 704 [22] "LEVEL 97.5%: 22.1" "LEVEL 97.5%: 22.58" "LEVEL 97.5%: 16.88"
 705 [25] "LEVEL 97.5%: 21.25" "LEVEL 97.5%: 18.69"
 706
 707 [1] "COMMON COEFFICIENTS"
 708
 709 [1] "Phenomenology: 2; Response: 1"
 710
 711 [1] "POSTERIOR MEAN: 6.13" "POSTERIOR MEAN: -1.13"
 712
 713 [1] "LEVEL 2.5%: 5.89" "LEVEL 2.5%: -1.18"
 714
 715 [1] "LEVEL 5%: 5.92" "LEVEL 5%: -1.17"
 716
 717 [1] "LEVEL 50%: 6.13" "LEVEL 50%: -1.13"
 718

719 [1] "LEVEL 95%: 6.33" "LEVEL 95%: -1.1"

720

721 [1] "LEVEL 97.5%: 6.37" "LEVEL 97.5%: -1.1"

722

723 [1] "Phenomenology: 2; Response: 2"

724

725 [1] "POSTERIOR MEAN: -5.25" "POSTERIOR MEAN: 0.23"

726

727 [1] "LEVEL 2.5%: -5.42" "LEVEL 2.5%: 0.21"

728

729 [1] "LEVEL 5%: -5.38" "LEVEL 5%: 0.21"

730

731 [1] "LEVEL 50%: -5.25" "LEVEL 50%: 0.23"

732

733 [1] "LEVEL 95%: -5.13" "LEVEL 95%: 0.26"

734

735 [1] "LEVEL 97.5%: -5.1" "LEVEL 97.5%: 0.26"

736

737 [1] "Phenomenology: 3; Response: 1"

738

739 [1] "POSTERIOR MEAN: -10.45" "POSTERIOR MEAN: 0.36" "POSTERIOR MEAN: -0.59"

740 [4] "POSTERIOR MEAN: -0.79"

741

742 [1] "LEVEL 2.5%: -11.51" "LEVEL 2.5%: 0.3" "LEVEL 2.5%: -1.38"

743 [4] "LEVEL 2.5%: -1.27"

744

745 [1] "LEVEL 5%: -11.39" "LEVEL 5%: 0.31" "LEVEL 5%: -1.26" "LEVEL 5%: -1.18"

746

747 [1] "LEVEL 50%: -10.43" "LEVEL 50%: 0.36" "LEVEL 50%: -0.58"

748 [4] "LEVEL 50%: -0.78"

749

750 [1] "LEVEL 95%: -9.41" "LEVEL 95%: 0.42" "LEVEL 95%: 0.01" "LEVEL 95%: -0.45"

751

752 [1] "LEVEL 97.5%: -9.25" "LEVEL 97.5%: 0.43" "LEVEL 97.5%: 0.3"

753 [4] "LEVEL 97.5%: -0.4"

754

755 [1] "Phenomenology: 3; Response: 2"

756

757 [1] "POSTERIOR MEAN: -8.19" "POSTERIOR MEAN: 0.38" "POSTERIOR MEAN: -1.17"

758 [4] "POSTERIOR MEAN: -6.22"

759

760 [1] "LEVEL 2.5%: -9.16" "LEVEL 2.5%: 0.31" "LEVEL 2.5%: -2.07"

761 [4] "LEVEL 2.5%: -12.66"

762

763 [1] "LEVEL 5%: -9.01" "LEVEL 5%: 0.33" "LEVEL 5%: -1.98" "LEVEL 5%: -12.17"

764
765 [1] "LEVEL 50%: -8.19" "LEVEL 50%: 0.38" "LEVEL 50%: -1.16" "LEVEL 50%: -5.94"
766
767 [1] "LEVEL 95%: -7.38" "LEVEL 95%: 0.43" "LEVEL 95%: -0.31" "LEVEL 95%: -1.54"
768
769 [1] "LEVEL 97.5%: -7.04" "LEVEL 97.5%: 0.44" "LEVEL 97.5%: -0.03"
770 [4] "LEVEL 97.5%: -1.38"
771
772 [1] "Phenomenology: 4; Response: 1"
773
774 [1] "POSTERIOR MEAN: -3.06" "POSTERIOR MEAN: 0.42"
775
776 [1] "LEVEL 2.5%: -3.85" "LEVEL 2.5%: 0.39"
777
778 [1] "LEVEL 5%: -3.67" "LEVEL 5%: 0.39"
779
780 [1] "LEVEL 50%: -3.04" "LEVEL 50%: 0.42"
781
782 [1] "LEVEL 95%: -2.5" "LEVEL 95%: 0.45"
783
784 [1] "LEVEL 97.5%: -2.42" "LEVEL 97.5%: 0.46"
785
786 [1] "Phenomenology: 4; Response: 2"
787
788 [1] "POSTERIOR MEAN: -1.74" "POSTERIOR MEAN: 0.25"
789
790 [1] "LEVEL 2.5%: -3.54" "LEVEL 2.5%: 0.19"
791
792 [1] "LEVEL 5%: -2.44" "LEVEL 5%: 0.2"
793
794 [1] "LEVEL 50%: -1.75" "LEVEL 50%: 0.25"
795
796 [1] "LEVEL 95%: -0.82" "LEVEL 95%: 0.28"
797
798 [1] "LEVEL 97.5%: -0.6" "LEVEL 97.5%: 0.35"
799
800 [1] "EMPLACEMENT CONDITION DEPENDENT COEFFICIENTS"
801
802 [1] "Phenomenology: 1; Emplacement: 1; Response: 1"
803
804 [1] "POSTERIOR MEAN: -10.45" "POSTERIOR MEAN: -1.38"
805 [3] "POSTERIOR MEAN: -921836513.41" "POSTERIOR MEAN: 5016.44"
806 [5] "POSTERIOR MEAN: -21066.47"
807
808 [1] "LEVEL 2.5%: -11.33" "LEVEL 2.5%: -1.54"

809 [3] "LEVEL 2.5%: -3248024320.67" "LEVEL 2.5%: -20806.44"
 810 [5] "LEVEL 2.5%: -48365.66"
 811
 812 [1] "LEVEL 5%: -11.19" "LEVEL 5%: -1.52"
 813 [3] "LEVEL 5%: -2939730943.81" "LEVEL 5%: -13278.49"
 814 [5] "LEVEL 5%: -40898.25"
 815
 816 [1] "LEVEL 50%: -10.47" "LEVEL 50%: -1.38"
 817 [3] "LEVEL 50%: -593848596.07" "LEVEL 50%: 4955.43"
 818 [5] "LEVEL 50%: -17928.44"
 819
 820 [1] "LEVEL 95%: -9.64" "LEVEL 95%: -1.25"
 821 [3] "LEVEL 95%: -75738559.09" "LEVEL 95%: 22390.4"
 822 [5] "LEVEL 95%: -6350.23"
 823
 824 [1] "LEVEL 97.5%: -9.47" "LEVEL 97.5%: -1.22"
 825 [3] "LEVEL 97.5%: -52550562.37" "LEVEL 97.5%: 25026.47"
 826 [5] "LEVEL 97.5%: -5059.12"
 827
 828 [1] "Phenomenology: 1; Emplacement: 1; Response: 2"
 829
 830 [1] "POSTERIOR MEAN: -0.35" "POSTERIOR MEAN: -1.78"
 831 [3] "POSTERIOR MEAN: -1101601766.01" "POSTERIOR MEAN: -5606.99"
 832 [5] "POSTERIOR MEAN: -14364.57"
 833
 834 [1] "LEVEL 2.5%: -1.41" "LEVEL 2.5%: -1.96"
 835 [3] "LEVEL 2.5%: -4782355486.68" "LEVEL 2.5%: -24814.25"
 836 [5] "LEVEL 2.5%: -44067.04"
 837
 838 [1] "LEVEL 5%: -1.19" "LEVEL 5%: -1.93"
 839 [3] "LEVEL 5%: -4423190842.86" "LEVEL 5%: -21571.61"
 840 [5] "LEVEL 5%: -32714.41"
 841
 842 [1] "LEVEL 50%: -0.32" "LEVEL 50%: -1.78"
 843 [3] "LEVEL 50%: -432816811.32" "LEVEL 50%: -4553.58"
 844 [5] "LEVEL 50%: -11118.7"
 845
 846 [1] "LEVEL 95%: 0.55" "LEVEL 95%: -1.63"
 847 [3] "LEVEL 95%: -21350203.37" "LEVEL 95%: 4691"
 848 [5] "LEVEL 95%: -3253.55"
 849
 850 [1] "LEVEL 97.5%: 0.77" "LEVEL 97.5%: -1.59"
 851 [3] "LEVEL 97.5%: -8210026.41" "LEVEL 97.5%: 6454.02"
 852 [5] "LEVEL 97.5%: -2353.58"
 853

854 [1] "Phenomenology: 1; Emplacement: 2; Response: 1"

855

856 [1] "POSTERIOR MEAN: -11.01" "POSTERIOR MEAN: -1.22"

857 [3] "POSTERIOR MEAN: -1119969809.77" "POSTERIOR MEAN: 3.25"

858 [5] "POSTERIOR MEAN: -20.62"

859

860 [1] "LEVEL 2.5%: -12.62" "LEVEL 2.5%: -1.37"

861 [3] "LEVEL 2.5%: -2857319457.58" "LEVEL 2.5%: 0.51"

862 [5] "LEVEL 2.5%: -23.92"

863

864 [1] "LEVEL 5%: -12.43" "LEVEL 5%: -1.35"

865 [3] "LEVEL 5%: -2640501804.65" "LEVEL 5%: 0.68"

866 [5] "LEVEL 5%: -23.46"

867

868 [1] "LEVEL 50%: -11.22" "LEVEL 50%: -1.22"

869 [3] "LEVEL 50%: -953306334.5" "LEVEL 50%: 2.67"

870 [5] "LEVEL 50%: -20.49"

871

872 [1] "LEVEL 95%: -8.7" "LEVEL 95%: -1.07"

873 [3] "LEVEL 95%: -129238181.99" "LEVEL 95%: 8.18"

874 [5] "LEVEL 95%: -18.18"

875

876 [1] "LEVEL 97.5%: -8.38" "LEVEL 97.5%: -1.05"

877 [3] "LEVEL 97.5%: -110724130" "LEVEL 97.5%: 8.79"

878 [5] "LEVEL 97.5%: -17.79"

879

880 [1] "Phenomenology: 1; Emplacement: 2; Response: 2"

881

882 [1] "POSTERIOR MEAN: -0.14" "POSTERIOR MEAN: -1.79"

883 [3] "POSTERIOR MEAN: -814365953.32" "POSTERIOR MEAN: 8.59"

884 [5] "POSTERIOR MEAN: -22.63"

885

886 [1] "LEVEL 2.5%: -1.58" "LEVEL 2.5%: -2"

887 [3] "LEVEL 2.5%: -2227401669.99" "LEVEL 2.5%: 1.18"

888 [5] "LEVEL 2.5%: -27.13"

889

890 [1] "LEVEL 5%: -1.34" "LEVEL 5%: -1.97"

891 [3] "LEVEL 5%: -2036623857.57" "LEVEL 5%: 1.62"

892 [5] "LEVEL 5%: -26.78"

893

894 [1] "LEVEL 50%: -0.15" "LEVEL 50%: -1.79"

895 [3] "LEVEL 50%: -727297974.98" "LEVEL 50%: 6.93"

896 [5] "LEVEL 50%: -22.93"

897

898 [1] "LEVEL 95%: 1.11" "LEVEL 95%: -1.62"

899 [3] "LEVEL 95%: -29297892.17" "LEVEL 95%: 18.59"
900 [5] "LEVEL 95%: -18.73"
901
902 [1] "LEVEL 97.5%: 1.36" "LEVEL 97.5%: -1.59" "LEVEL 97.5%: -12450736"
903 [4] "LEVEL 97.5%: 20.27" "LEVEL 97.5%: -18.26"
904
905 [1] "Phenomenology: 1; Emplacement: 3; Response: 1"
906
907 [1] "POSTERIOR MEAN: -5.57" "POSTERIOR MEAN: -1.67"
908 [3] "POSTERIOR MEAN: -1061191538.16" "POSTERIOR MEAN: 1.6"
909 [5] "POSTERIOR MEAN: -19.43"
910
911 [1] "LEVEL 2.5%: -7.7" "LEVEL 2.5%: -1.84"
912 [3] "LEVEL 2.5%: -2343477675.4" "LEVEL 2.5%: 0.61"
913 [5] "LEVEL 2.5%: -20.93"
914
915 [1] "LEVEL 5%: -7.45" "LEVEL 5%: -1.81"
916 [3] "LEVEL 5%: -2226050598.28" "LEVEL 5%: 0.7"
917 [5] "LEVEL 5%: -20.79"
918
919 [1] "LEVEL 50%: -5.74" "LEVEL 50%: -1.67"
920 [3] "LEVEL 50%: -1056115409.12" "LEVEL 50%: 1.56"
921 [5] "LEVEL 50%: -19.59"
922
923 [1] "LEVEL 95%: -3.1" "LEVEL 95%: -1.53"
924 [3] "LEVEL 95%: -229903815.28" "LEVEL 95%: 2.58"
925 [5] "LEVEL 95%: -17.93"
926
927 [1] "LEVEL 97.5%: -2.84" "LEVEL 97.5%: -1.51"
928 [3] "LEVEL 97.5%: -164495097.53" "LEVEL 97.5%: 3.04"
929 [5] "LEVEL 97.5%: -17.82"
930
931 [1] "Phenomenology: 1; Emplacement: 3; Response: 2"
932
933 [1] "POSTERIOR MEAN: 5.06" "POSTERIOR MEAN: -1.83"
934 [3] "POSTERIOR MEAN: -400736185.19" "POSTERIOR MEAN: 1.44"
935 [5] "POSTERIOR MEAN: -18.63"
936
937 [1] "LEVEL 2.5%: 2.37" "LEVEL 2.5%: -2.03"
938 [3] "LEVEL 2.5%: -974868968.61" "LEVEL 2.5%: 0.34"
939 [5] "LEVEL 2.5%: -20.62"
940
941 [1] "LEVEL 5%: 2.69" "LEVEL 5%: -1.99"
942 [3] "LEVEL 5%: -903833273.27" "LEVEL 5%: 0.4"
943 [5] "LEVEL 5%: -20.29"

944
 945 [1] "LEVEL 50%: 4.84" "LEVEL 50%: -1.83"
 946 [3] "LEVEL 50%: -357907575.38" "LEVEL 50%: 1.34"
 947 [5] "LEVEL 50%: -18.68"
 948
 949 [1] "LEVEL 95%: 8.49" "LEVEL 95%: -1.66"
 950 [3] "LEVEL 95%: -74797271.12" "LEVEL 95%: 2.85"
 951 [5] "LEVEL 95%: -16.75"
 952
 953 [1] "LEVEL 97.5%: 8.93" "LEVEL 97.5%: -1.63"
 954 [3] "LEVEL 97.5%: -52972650.74" "LEVEL 97.5%: 3.09"
 955 [5] "LEVEL 97.5%: -16.61"
 956
 957 [1] "Phenomenology: 2; Emplacement: 1; Response: 1"
 958
 959 [1] "POSTERIOR MEAN: 4.83"
 960
 961 [1] "LEVEL 2.5%: 4.03"
 962
 963 [1] "LEVEL 5%: 4.12"
 964
 965 [1] "LEVEL 50%: 4.83"
 966
 967 [1] "LEVEL 95%: 5.53"
 968
 969 [1] "LEVEL 97.5%: 5.65"
 970
 971 [1] "Phenomenology: 2; Emplacement: 1; Response: 2"
 972
 973 [1] "POSTERIOR MEAN: -0.18"
 974
 975 [1] "LEVEL 2.5%: -0.62"
 976
 977 [1] "LEVEL 5%: -0.55"
 978
 979 [1] "LEVEL 50%: -0.18"
 980
 981 [1] "LEVEL 95%: 0.18"
 982
 983 [1] "LEVEL 97.5%: 0.27"
 984
 985 [1] "Phenomenology: 2; Emplacement: 2; Response: 1"
 986
 987 [1] "POSTERIOR MEAN: 3.91"
 988

989 [1] "LEVEL 2.5%: 3.16"

990

991 [1] "LEVEL 5%: 3.26"

992

993 [1] "LEVEL 50%: 3.9"

994

995 [1] "LEVEL 95%: 4.65"

996

997 [1] "LEVEL 97.5%: 4.84"

998

999 [1] "Phenomenology: 2; Emplacement: 2; Response: 2"

1000

1001 [1] "POSTERIOR MEAN: -1.52"

1002

1003 [1] "LEVEL 2.5%: -3.54"

1004

1005 [1] "LEVEL 5%: -2.94"

1006

1007 [1] "LEVEL 50%: -1.43"

1008

1009 [1] "LEVEL 95%: -0.51"

1010

1011 [1] "LEVEL 97.5%: -0.41"

1012

1013 [1] "Phenomenology: 2; Emplacement: 3; Response: 1"

1014

1015 [1] "POSTERIOR MEAN: 2.32"

1016

1017 [1] "LEVEL 2.5%: 1.29"

1018

1019 [1] "LEVEL 5%: 1.41"

1020

1021 [1] "LEVEL 50%: 2.34"

1022

1023 [1] "LEVEL 95%: 3.07"

1024

1025 [1] "LEVEL 97.5%: 3.14"

1026

1027 [1] "Phenomenology: 2; Emplacement: 3; Response: 2"

1028

1029 [1] "POSTERIOR MEAN: -1.69"

1030

1031 [1] "LEVEL 2.5%: -3.24"

1032

1033 [1] "LEVEL 5%: -2.94"

```

1034
1035         [1] "LEVEL 50%: -1.62"
1036
1037         [1] "LEVEL 95%: -0.69"
1038
1039         [1] "LEVEL 97.5%: -0.57"
1040
1041 [1] "SOURCE VARIANCE COMPONENTS"
1042
1043         [1] "Phenomenology: 1; Response: 1"
1044
1045         [1] "POSTERIOR MEAN: 0.3476"
1046
1047         [1] "LEVEL 2.5%: 0.1893"
1048
1049         [1] "LEVEL 5%: 0.2011"
1050
1051         [1] "LEVEL 50%: 0.3144"
1052
1053         [1] "LEVEL 95%: 0.5998"
1054
1055         [1] "LEVEL 97.5%: 0.6669"
1056
1057         [1] "Phenomenology: 1; Response: 2"
1058
1059         [1] "POSTERIOR MEAN: 0.4476"
1060
1061         [1] "LEVEL 2.5%: 0.252"
1062
1063         [1] "LEVEL 5%: 0.2754"
1064
1065         [1] "LEVEL 50%: 0.4206"
1066
1067         [1] "LEVEL 95%: 0.7252"
1068
1069         [1] "LEVEL 97.5%: 0.7725"
1070
1071         [1] "Phenomenology: 2; Response: 1"
1072
1073         [1] "POSTERIOR MEAN: 0.2313"
1074
1075         [1] "LEVEL 2.5%: 0.1179"
1076
1077         [1] "LEVEL 5%: 0.126"
1078

```

1079 [1] "LEVEL 50%: 0.2179"

1080

1081 [1] "LEVEL 95%: 0.3959"

1082

1083 [1] "LEVEL 97.5%: 0.4572"

1084

1085 [1] "Phenomenology: 2; Response: 2"

1086

1087 [1] "POSTERIOR MEAN: 0.0437"

1088

1089 [1] "LEVEL 2.5%: 0.0239"

1090

1091 [1] "LEVEL 5%: 0.0264"

1092

1093 [1] "LEVEL 50%: 0.0415"

1094

1095 [1] "LEVEL 95%: 0.0682"

1096

1097 [1] "LEVEL 97.5%: 0.0743"

1098

1099 [1] "PATH VARIANCE COMPONENTS"

1100

1101 [1] "Phenomenology: 1; Response: 1"

1102

1103 [1] "POSTERIOR MEAN: 0.1323"

1104

1105 [1] "LEVEL 2.5%: 0.0918"

1106

1107 [1] "LEVEL 5%: 0.0963"

1108

1109 [1] "LEVEL 50%: 0.1315"

1110

1111 [1] "LEVEL 95%: 0.1692"

1112

1113 [1] "LEVEL 97.5%: 0.1779"

1114

1115 [1] "Phenomenology: 1; Response: 2"

1116

1117 [1] "POSTERIOR MEAN: 0.1441"

1118

1119 [1] "LEVEL 2.5%: 0.0928"

1120

1121 [1] "LEVEL 5%: 0.1008"

1122

1123 [1] "LEVEL 50%: 0.1416"

```

1124
1125         [1] "LEVEL 95%: 0.1995"
1126
1127         [1] "LEVEL 97.5%: 0.2143"
1128
1129         [1] "Phenomenology: 2; Response: 1"
1130
1131         [1] "POSTERIOR MEAN: 0.0212"
1132
1133         [1] "LEVEL 2.5%: 0.013"
1134
1135         [1] "LEVEL 5%: 0.014"
1136
1137         [1] "LEVEL 50%: 0.0209"
1138
1139         [1] "LEVEL 95%: 0.0285"
1140
1141         [1] "LEVEL 97.5%: 0.0302"
1142
1143         [1] "Phenomenology: 2; Response: 2"
1144
1145         [1] "POSTERIOR MEAN: 0.0132"
1146
1147         [1] "LEVEL 2.5%: 0.0093"
1148
1149         [1] "LEVEL 5%: 0.0098"
1150
1151         [1] "LEVEL 50%: 0.013"
1152
1153         [1] "LEVEL 95%: 0.0174"
1154
1155         [1] "LEVEL 97.5%: 0.0186"
1156
1157 [1] "OBSERVATIONAL ERROR COVARIANCE PARAMETERS"
1158
1159 [1] "Phenomenology 1"
1160
1161 [1] "POSTERIOR MEAN:"
1162
1163 [1] "Variances"
1164
1165 [1] 0.0863 0.1895
1166
1167 [1] "Correlations"
1168

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1169      [,1] [,2]
1170 [1,]    1 0.41
1171 [2,]    0 1.00
1172
1173 [1] "Variances"
1174
1175      [1] "LEVEL 2.5%: 0.0715" "LEVEL 2.5%: 0.16"
1176
1177 [1] "Correlations"
1178
1179      [1] "LEVEL 2.5%:"
1180      [,1] [,2]
1181 [1,]    1 0.32
1182 [2,]    0 1.00
1183
1184 [1] "Variances"
1185
1186      [1] "LEVEL 5%: 0.0733" "LEVEL 5%: 0.1652"
1187
1188 [1] "Correlations"
1189
1190      [1] "LEVEL 5%:"
1191      [,1] [,2]
1192 [1,]    1 0.33
1193 [2,]    0 1.00
1194
1195 [1] "Variances"
1196
1197      [1] "LEVEL 50%: 0.0858" "LEVEL 50%: 0.189"
1198
1199 [1] "Correlations"
1200
1201      [1] "LEVEL 50%:"
1202      [,1] [,2]
1203 [1,]    1 0.41
1204 [2,]    0 1.00
1205
1206 [1] "Variances"
1207
1208      [1] "LEVEL 95%: 0.0998" "LEVEL 95%: 0.2157"
1209
1210 [1] "Correlations"
1211
1212      [1] "LEVEL 95%:"
1213      [,1] [,2]

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1214 [1,]    1 0.49
1215 [2,]    0 1.00
1216
1217 [1] "Variances"
1218
1219 [1] "LEVEL 97.5%: 0.107" "LEVEL 97.5%: 0.2206"
1220
1221 [1] "Correlations"
1222
1223 [1] "LEVEL 97.5%:"
1224 [1] [,1] [,2]
1225 [1,]    1 0.51
1226 [2,]    0 1.00
1227
1228 [1] "Phenomenology 2"
1229
1230 [1] "POSTERIOR MEAN:"
1231
1232 [1] "Variances"
1233
1234 [1] 0.0730 0.0168
1235
1236 [1] "Correlations"
1237
1238 [1] [,1] [,2]
1239 [1,]    1 -0.12
1240 [2,]    0 1.00
1241
1242 [1] "Variances"
1243
1244 [1] "LEVEL 2.5%: 0.0623" "LEVEL 2.5%: 0.014"
1245
1246 [1] "Correlations"
1247
1248 [1] "LEVEL 2.5%:"
1249 [1] [,1] [,2]
1250 [1,]    1 -0.24
1251 [2,]    0 1.00
1252
1253 [1] "Variances"
1254
1255 [1] "LEVEL 5%: 0.0641" "LEVEL 5%: 0.0143"
1256
1257 [1] "Correlations"
1258

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1259         [1] "LEVEL 5%:"
1260             [,1] [,2]
1261 [1,]      1 -0.22
1262 [2,]      0  1.00
1263
1264 [1] "Variances"
1265
1266         [1] "LEVEL 50%: 0.0724" "LEVEL 50%: 0.0168"
1267
1268 [1] "Correlations"
1269
1270         [1] "LEVEL 50%:"
1271             [,1] [,2]
1272 [1,]      1 -0.12
1273 [2,]      0  1.00
1274
1275 [1] "Variances"
1276
1277         [1] "LEVEL 95%: 0.0838" "LEVEL 95%: 0.0198"
1278
1279 [1] "Correlations"
1280
1281         [1] "LEVEL 95%:"
1282             [,1] [,2]
1283 [1,]      1      0
1284 [2,]      0      1
1285
1286 [1] "Variances"
1287
1288         [1] "LEVEL 97.5%: 0.0871" "LEVEL 97.5%: 0.0205"
1289
1290 [1] "Correlations"
1291
1292         [1] "LEVEL 97.5%:"
1293             [,1] [,2]
1294 [1,]      1 0.02
1295 [2,]      0 1.00
1296
1297 [1] "Phenomenology 3"
1298
1299 [1] "POSTERIOR MEAN:"
1300
1301 [1] "Variances"
1302
1303 [1] 0.1900 0.1781

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1304
1305 [1] "Correlations"
1306
1307      [,1] [,2]
1308 [1,]    1 0.97
1309 [2,]    0 1.00
1310
1311 [1] "Variances"
1312
1313      [1] "LEVEL 2.5%: 0.0995" "LEVEL 2.5%: 0.0953"
1314
1315 [1] "Correlations"
1316
1317      [1] "LEVEL 2.5%:"
1318      [,1] [,2]
1319 [1,]    1 0.94
1320 [2,]    0 1.00
1321
1322 [1] "Variances"
1323
1324      [1] "LEVEL 5%: 0.1063" "LEVEL 5%: 0.1028"
1325
1326 [1] "Correlations"
1327
1328      [1] "LEVEL 5%:"
1329      [,1] [,2]
1330 [1,]    1 0.94
1331 [2,]    0 1.00
1332
1333 [1] "Variances"
1334
1335      [1] "LEVEL 50%: 0.1852" "LEVEL 50%: 0.1714"
1336
1337 [1] "Correlations"
1338
1339      [1] "LEVEL 50%:"
1340      [,1] [,2]
1341 [1,]    1 0.97
1342 [2,]    0 1.00
1343
1344 [1] "Variances"
1345
1346      [1] "LEVEL 95%: 0.2862" "LEVEL 95%: 0.2665"
1347
1348 [1] "Correlations"

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1349
1350         [1] "LEVEL 95%:"
1351             [,1] [,2]
1352 [1,]      1 0.98
1353 [2,]      0 1.00
1354
1355 [1] "Variances"
1356
1357         [1] "LEVEL 97.5%: 0.3134" "LEVEL 97.5%: 0.3079"
1358
1359 [1] "Correlations"
1360
1361         [1] "LEVEL 97.5%:"
1362             [,1] [,2]
1363 [1,]      1 0.99
1364 [2,]      0 1.00
1365
1366 [1] "Phenomenology 4"
1367
1368 [1] "POSTERIOR MEAN:"
1369
1370 [1] "Variances"
1371
1372 [1] 0.0343 0.0915
1373
1374 [1] "Correlations"
1375
1376         [,1] [,2]
1377 [1,]      1 -0.12
1378 [2,]      0 1.00
1379
1380 [1] "Variances"
1381
1382         [1] "LEVEL 2.5%: 0.0063" "LEVEL 2.5%: 0.018"
1383
1384 [1] "Correlations"
1385
1386         [1] "LEVEL 2.5%:"
1387             [,1] [,2]
1388 [1,]      1 -0.76
1389 [2,]      0 1.00
1390
1391 [1] "Variances"
1392
1393         [1] "LEVEL 5%: 0.0076" "LEVEL 5%: 0.0204"

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1394
1395 [1] "Correlations"
1396
1397         [1] "LEVEL 5%:"
1398             [,1] [,2]
1399 [1,]      1 -0.71
1400 [2,]      0  1.00
1401
1402 [1] "Variances"
1403
1404         [1] "LEVEL 50%: 0.0219" "LEVEL 50%: 0.0569"
1405
1406 [1] "Correlations"
1407
1408         [1] "LEVEL 50%:"
1409             [,1] [,2]
1410 [1,]      1 -0.15
1411 [2,]      0  1.00
1412
1413 [1] "Variances"
1414
1415         [1] "LEVEL 95%: 0.0966" "LEVEL 95%: 0.3034"
1416
1417 [1] "Correlations"
1418
1419         [1] "LEVEL 95%:"
1420             [,1] [,2]
1421 [1,]      1  0.55
1422 [2,]      0  1.00
1423
1424 [1] "Variances"
1425
1426         [1] "LEVEL 97.5%: 0.1304" "LEVEL 97.5%: 0.4019"
1427
1428 [1] "Correlations"
1429
1430         [1] "LEVEL 97.5%:"
1431             [,1] [,2]
1432 [1,]      1  0.66
1433 [2,]      0  1.00
1434
1435 [1] "FGSN PRIOR PARAMETERS"
1436
1437 [1] "POSTERIOR MEAN:"
1438

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```

1439 [1] "Alpha = 20.94"
1440 [1] "Lambda squared = 20.74"
1441 [1] "Omega = 3.4"      "Omega = -10.98"
1442
1443 [1] "Alpha:"
1444 [1] "LEVEL 2.5%: 20.05"
1445
1446 [1] "LEVEL 5%: 20.2"
1447
1448 [1] "LEVEL 50%: 20.92"
1449
1450 [1] "LEVEL 95%: 21.58"
1451
1452 [1] "LEVEL 97.5%: 22.58"
1453
1454 [1] "Lambda squared:"
1455 [1] "LEVEL 2.5%: 11.08"
1456
1457 [1] "LEVEL 5%: 12.51"
1458
1459 [1] "LEVEL 50%: 19.76"
1460
1461 [1] "LEVEL 95%: 31.31"
1462
1463 [1] "LEVEL 97.5%: 34.5"
1464
1465 [1] "Omega:"
1466 [1] "LEVEL 2.5%: -2.94" "LEVEL 2.5%: -20.7"
1467
1468 [1] "LEVEL 5%: -1.23" "LEVEL 5%: -19.31"
1469
1470 [1] "LEVEL 50%: 3.46" "LEVEL 50%: -10.99"
1471
1472 [1] "LEVEL 95%: 8.17" "LEVEL 95%: -2.38"
1473
1474 [1] "LEVEL 97.5%: 9.13" "LEVEL 97.5%: -1.23"
1475
1476 [1] "DIC = 1241.51"
1477
1478 [1] "PIC = 1325.2"

```